

MuleSoft Interview Questions

Q1. What is MuleSoft?

Ans. MuleSoft is a solution based cluster that integrates Data, Applications, and APIs on-premises and over the cloud platform. MuleSoft works on the **AnyPoint Connectivity Model** which helps in connecting any existing SaaS-based applications or set of APIs through one single API interface. The architecture is service-oriented which gives the flexibility to access all the required applications through this integration. MuleSoft owns the Programmable Web portal that is widely used by application developers to help them build web, mobile and other user applications. So Mule is the most popularly used application integration and communication platform for the business needs.

Q2. What is Mule ESB?

Ans. ESB stands for enterprise service bus, Mule is a runtime engine of the Java-based ESB integration platform. It allows the development teams to connect and access flexibly and easily exchange data. It helps to enable easy integrations between application platforms without creating fuzz about different technologies being used by the applications (such as HTTP, SaaS, Payment Gateways, Web services, JMS). EBS can be implemented and deployed anywhere in the application network. It triggers events in real-time or in the batch system providing global connectivity.

Q3. Why Mule ESB is so popular?

Ans. Mule Soft is a lightweight technology that supports a high level of scalability, which means it allows you to start small and scale as per your requirement. You can connect N number of application within a single environment which has zero restriction for Apps communication. Mule Sot manages N number of application interactions with each other without restricting that application into the same VM interaction or other VM interaction. It means even though the application is running into different VMs, that can interact with each other. Its flexibility with transportation protocol enables the interactions easily.

Though there are many ESBs are available in the market, Mule Soft gives you the most flexible and reliable services. Another EBs comes with the restriction of limited functionalities or access to an existing application server or a specific messaging server, they put restrictions for the use of a specific vendor. Whereas, Mule Soft comes with a high level of accessibility and adaptability with vendor-neutral methods. You can easily plug and play any application through any vendor.

Q4. What are the Features of Mule ESB?

Ans. An ESB is used for integration using a service-oriented approach. Its main features are:

- Set of Service Container
- Message Transformation Service
- Message Routing Service
- Web Service Security

Q5. What are different types of Primitives used in Mediation?

Ans. These are the different types of primitives in mediation:

- Message Filter

- Type Filter
- Endpoint Lookup
- Service Invoke
- Fan-out
- Fan-in
- XSLT
- BO Map
- Message Element Setter
- DB lookup
- Data Handler
- Custom Mediation
- Header Setters
- Message Logger
- Even Emitter
- Stop
- Fail
- Sub Flow

Q6.What are various types of Exception Handling?

Ans.

- Choice Exception Handling.
- Catch Exception Handling.
- Rollback Exception Handling.
- Global Exception Handling.
- Default Exception Handling.

Q7. What is Shared Resource in Mule and how they have been used?

Ans. We can make connectors as a reusable component by defining them as common resources and expose them to all applications deployed under the same domain, these resources are known as shared resources. These shared resource needs to be defined inside Mule Domain Project and then referred to each of the projects that are meant to use the elements in it.

Q8. What are the different ESBs in the market?

Ans. There are various ESB's available in the market. Some are opensource, some licenced:

- Talend
- Mule ESB
- JBoss Fuse ESB

Q9. Various types of variables in mule?

- Record Variable.
- Flow Variable.
- Session Variable.

Q10. What is the definition of Web Services?

Ans. Web service is a function or program in any language that can be accessed over HTTP. Message format can be XML or JSON or any other program as long as the other programs can understand and communicate. Any web service has a server-client relationship. Web services can be synchronous or asynchronous. Any web service can have multiple clients.

Q11. How to find when the project needs ESB?

Ans. ESB implementation is not suitable for all projects. Proper analysis should be done if the use of ESB will really benefit the project.

Some of the points to be considered while analyzing the need of ESB are as follows:

- If the project requires integrating 3 or more applications/services. If the need is to communicate between two applications, using point-to-point integration would suffice.
- If the project would need to be scaled in future where it might be needed to interact with more services in future. Not all projects need this as they may perform not that big a task.
- If the project needs message routing capabilities such as forking and aggregating message flows. Such features are not required by all projects.
- Is the architecture of what is to be achieved clear. It's much better to do simple POCs integrating small parts to evaluate the benefits.
- Most ESBs are a costly affair. Does the project budget allow the use of ESB

Q12. What are the different types of Flow Processing Strategies?

Ans. There are six different types of Flow Processing Strategies. They are

- Asynchronous Flow Processing Strategy.
- Custom Processing Strategy.
- Thread Per Processing Strategy.
- Queued Asynchronous Flow Processing Strategy.
- Synchronous Flow Processing Strategy.
- Non-blocking Flow Processing Strategy.
- Queued Flow Processing Strategy.

Q13. How to create and consume SOAP service in Mule?

Ans. Creating SOAP Service – We can create a SOAP service same as we create Mule Project With RAML, the only change is instead of RAML we need to import Concert WSDL.

Consuming SOAP Service – We can use Web Service Consumer or CXF component in our mule flow to access/consume SOAP service.

Q14. Explain ESB Integration core principles?

Ans. Transformation - Data transformation between canonical data formats and specific data formats required by each ESB connector.

Transportation - Transport protocol negotiation between multiple formats. Such as HTTP, JMS, JDBC.

Mediation - Providing multiple interfaces for the purpose of a) supporting multiple versions of a service for backwards compatibility or alternatively, b) to allow for multiple channels to the same underlying component implementation. This second requirement may involve providing multiple interfaces to the same component, one legacy interface (flat file) and one standards compliant (SOAP/XML) interface.

Non-functional consistency - For a typical ESB initiative, this can include consistency around the way security and monitoring policies are applied and implemented.

Mule ESB Interview Questions For Experienced:

Q15. How to select an ESB?

Ans. Usability: How complicated is the installation process? The learning curve of the ESB should not be long.

Maintainability: How to monitor the product? If any GUI terminal is available for monitoring the services.

Community: Is there an active community for the ESB. Various discussion forums, tutorials etc.

Enterprise Support- Is the product support reliable. What services are available.

Flexibility: Can the ESB be configured to meet the business requirements.

Reliability: The reliability of the ESB. Its current users and various case studies if available.

Cost: The cost of the ESB. Its Licence policies. Support costs. etc.

Q16. What does the statement “Any Point Platform” mean to Mule Soft?

Ans. MuleSoft is a secure, robust and highly scalable communication network that allows the application to do self-services, it helps the organizations to integrate applications, data and multiple devices in a flexible environment. It is a hybrid application integration platform that includes ESB, unified solutions for APIs management, iPaaS, application design, and publishing.

MuleSoft allows using N number of software and tools which includes the below list

- API Designer- Any Time Design Center
- Any Point Studio- Any Point Design Center
- API Manager-Any Point Management Center
- API Analytics-Any Point Management Center
- API Portal-Any Time Exchange
- MCQs- Runtime Services
- Mule ESB-Mule Runtime Engine
- Any point Connectors
- CloudHub- Hybrid Cloud
- Mule Enterprise Management

Q17. What are the advantages of using an ESB?

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Ans.

- High level of operational control from the central web-based portal.
- Wide range of connectivity through more than 120 leading SaaS-based applications on-premises.
- Ensuring zero message loss reliability and gives you high availability.
- Custom code out of data mapping with graphical integration and transformation
- Analytics and API Management
- Equipped with thousands of automated tests and bug fixing techniques.
- EDI/B2B Integration
- Batch Integration with Real-Time Integration techniques
- Premium connectors for HL7, Oracle, Epic, and SAP.

Q18. What is Transient Context?

Ans. Transient Context is used to pass the required values within the existing flow either the requesting flow or the responding flow. The transient flow could not make it link requests or responses together, so it is not used across. It is not being used if you want to save an input message before service gets invoked into the request or response flow. Transient is made to act as temporary storage of messages in general after the service invokes a call, the next primitive creates another message by combining the two, first the invoked response and second the original message that is stored in the Transient Context.

Q19. What are the different types of messages in MuleSoft?

Ans. Echo and Log message- log messages and move them from inbound to outbound routers.

Bridge Message – Basically, a passed message from inbound to outbound routers.

Build Message- Messages that are created from fixed or dynamic values.

Q20. What is API? Define the most prominent features of API.

Ans. API (Application Programming Interface) is a software interface that allows two different applications to interact in one platform.

- API has N number of benefits in terms of user's usability
- APIs have become modern and advanced within time, it adheres to many standards such as HTTP and REST.
- APIs are developer-friendly, easy to understand and broadly acceptable.
- APIs are mostly used as a product rather than a set of codes. They are specifically designed for a particular group of audiences such as application developers, and Mobile App developers.
- Every API comes with detailed documentation with version-specific information and up-gradation details.
- APIs are highly secure application interfaces that allow you to operate within a robust environment.
- APIs as similar to any software product has their own SDLC (software development lifecycle).

Q21. What is REST?

Ans. REST stands for Representational State Transfer or RESTful web service. REST is a client-server architecture which means each unique URL is a representation of some object or resource. Any REST API developed uses HTTP methods explicitly and in a way that's consistent with the protocol definition. This basic REST design principle establishes a one-to-

one mapping between creating, read, update, and delete (CRUD) operations and HTTP methods. According to this mapping:

1. To create a resource on the server, use POST.
2. To retrieve a resource, use GET.
3. To change the state of a resource or to update it, use PUT.
4. To remove or delete a resource, use DELETE.

Q22. What is RAML and why we use it?

Ans. RAML – RESTful API Modeling Language

RAML is similar to WSDL, it contains endpoint URL, request/response schema, HTTP methods and query and URI parameter.

RAML helps client (a consumer of the service) know, what the service is and what/how all operations can be invoked.

RAML helps the developer in creating the initial structure of this API. RAML can also be used for documentation purpose.

Q23. What are the Message Sources in Mule ESB?

Ans. Message sources in Mule are usually Anypoint Connectors, elements which provide connectivity to a specific external source, either via a standard protocol (such as HTTP, FTP, SMTP) or a third-party API (such as Salesforce.com, Twitter, or MongoDB).

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Q24. What is the difference between ESB and JMS?

Ans. ESB provides the middleware and interfaces that allow businesses to connect their applications without writing code.

JMS provides the messaging capability and facilitates communication between the modules/applications.

Q25. Why the Mulesoft is preferred than other ESB implementations?

Ans. Mule is lightweight but highly scalable, allowing you to start small and connect more applications over time. The ESB manages all the interactions between applications and components transparently, regardless of whether they exist in the same virtual machine or over the Internet, and regardless of the underlying transport protocol used.

Several commercial ESB implementation provides limited functionality or built on top of an existing application server or messaging server, locking you into that specific vendor. Mule is vendor-neutral, so different vendor implementations can plug in to it. You are never locked into a specific vendor when you use Mule.

Q26. Define Payload In MuleSoft?

Ans. The payload is a Mule Runtime Variable that stores arrays or objects. It is wrapped under `org.mule.api.MuleMessage` which helps you get the different means of accessing the payload under different forms. Mule messages are similar to any other SOAP message JMS message it also contains properties, headers and multiple names attached to it. The main content of the message is called Payload.

Q27. Explain the concept of endpoint?

Ans. An endpoint is a destination shared by many other routers in the group. It also helps to create a global endpoint. A global endpoint is not similar for inbound and outbound routing service, but it makes it useful in many different places in the configuration file. The entire endpoint destination must be named to be specified during the services. These names

identify the global endpoint in the group of routers. The global endpoint also offers to clarify the usage of a specific destination.

Q28. Explain Mule Transformer?

Ans. Mule transfer is an event instance that belongs to org.mule.api.MuleEvent library. This object carries the actual context of the message with the context of the event. The main purpose of the transformer is to translate the message from one form to another. It is also possible to create a chain of transformers. It transits the message from one medium to another while staging into different services.

Q29. What is the Transport Service Descriptor in Mule?

Ans. The transport service description is a technical configuration of the connector. This is a hidden configuration that is used in each instance of the connector. It gives definitions to the parameters such as the use of particular parameters, what classes are required for that particular message receiver, dispatchers and requesters. The definition is about default transformation to use inbound or outbound and utilizing the response of the router.

Q30. What is a Router in MuleSoft?

Ans. The router is one of the most critical services of MuleSoft. The Router finalizes and assigns the running territory for the messages to move from one service to another. Basically, routing is a process of controlling the transitory decided by the router for the moving message to transit from one source to another. It can be also called a gatekeeper of the endpoint services. It keeps tracks of targeted successions to ensure the message gets to deliver on the right intended destination. Routers can also act as a bundle of classified tasks, such as it can split, sort, group or regroup, messages based on specified conditions or certain mappings.

Q31. Define the use of Filter in Mule

Ans. Filters are the most powerful capability given to the routers to make smart decisions on the message delivery or request and response environment. It also gives sight to the router to decide what to do with the messages in the transit stage. Some filters go through intense analysis of the message to find out the actual value for the desired outputs.

Q32. Define SDO and SMO.

Ans. SDO stands for service data objects that are the representation of variables and objects.

SMO stands for Service Model Object and it is a pattern for using SDO for messages.

Q33. Define the purpose of Endpoint in Mule

Ans. Endpoint defines the specific usage of the transport protocol. Either you are reading the message, writing it, listening or polling to a targeted destination the endpoint directs the usability of the protocol. Endpoint controls the underlined entity which ensures the usability of the connectors. The targeted destination is defined as URI. It depends on the connectors that whether the destination will be treated as URI or URL or JMS destination itself.

Q34. What is the difference between service invoke and Callout?

Ans. Service Invoke:

Service invoke is an initial process of creating a service for either request or response flow. The service can be request, response or both and One-way implementation. Multiple instances of a service can be created and permitted into a flow, that allows a series of services to operate within the flow.

CallOut:

The purpose of the CallOut method is to receive messages and request the required service and operation in a flow. There is a CallOut node always presented into the median flow for connected target operation.

If the call attempt is successful the callout node in the responses flows median will start receiving the message.

If the call attempt is unsuccessful the callout node will be set to retry service in the flow depending on the type of fault that occurred.

Q35. What is Mule Context?

Ans. In general, the message context defines the overall purpose of the message but here the context defines the temporary area which is created along with the SMO(server message object) in the median flow while the transition of the message.SMO contains the shared context in the message flow. This shared context usually gets used at the time of aggregation, which means if you are aggregating and need to iterate the BO for the specified time. Aggregative context maintains data between FanOut and FanIn primitives. The data(context) present in the request flow cannot be persistent in the throughout request and responses flow because it only belongs to request BO.

Q36. How the messages in Mule are composed?

Ans. The four different part of the composed message in Mule

- Payload – is the main data context carried by a particular message
- Properties- it contains the Meta information or header similar to the SOAP message.
- And Optionally
- Multiple Name Attachments- To provide support for multipart messages. Or an exceptional payload for holding errors that occur during the event processing.

Q37. How to judge the need for ESB in a project ?

Ans. Project application needs need to be analyzing very carefully to avoid unnecessary arrangements. ESB benefits in several ways for projects which need to operate in a huge setup of multifunctional application support.

Q38. The analysis depends on many factors such as

- If the project requires a group of actions with additional applications and servers where the interaction is happening between two or more complex resources.
- If the project needs a broad scale up in the future and requires multiple application communication over a single setup.
- If the project requires to route the messages between one end to such as forking and aggregation of the routed messages.
- If the project allows this huge investment in need of application support.

Q39. Define Fileage properties in file connector in MuleSoft?

Ans. Fileage defines the waiting duration of the endpoint before starting to read the file again. Such as file age 60000 indicates that the endpoint should wait about 1 minute before starting the next processing.

Q40. What is streaming property in the file connector in MuleSoft?

Ans. Streaming properties in connectors represent the values in true and false format. If the streaming value is true, it means you are working on streaming data on the connector otherwise if it's false you are working on the file system.

Q41. What is polling frequency in the file connector in MuleSoft?

Ans. If you want a new file inbound endpoints to poll direction to read the new content then you require to set the polling frequency to a few milliseconds value to achieve this. Here polling frequency defines the value of the poll.

Q42. Explain the Configuration builder in MuleSoft?

Ans. Configuration builder in Mule helps to transfer the human authorized configuration file into a complex graph of objects that substitute a running node in ESB. Configuration builder can be defined into two types:

- Spring Driven Builder: it works with XML files
- Script Builder: it accepts scripting language files.

Q43. Explain the concept of the Auto-Delete feature in the file connector.

Ans. The auto-delete value of this feature is zero by default which means it inbound endpoint will automatically delete the file from the source directory. In case the source directory file does not require to be auto-deleted then you need to set the value to false.

Q44. Difference between ESB and JMS

Ans. ESB solves the business problem by providing middleware support for application integration and communication issues without writing code.

JMS help you get equipped with messaging capability and solve the communication issues between module and instances.

Q45. Explain the working functionality of the service layer.

Ans. Mule service is a set of all Mule entity which requires to support the processing request in an arranged manner. This service is defined by a particular configuration that defines the different elements from a different layer of services. It mobilizes the request which is open to receive a particular request. It depends on the service layer's input channels whether a service can or cannot be accessible outside in public ESB.

Q46. List of flow processing strategies in MuleSoft.

Ans. It represents six types of strategy for Flow Processing

- Asynchronous Flow Processing
- Custom Flow Processing
- Tread Per Processing
- Queued Asynchronous Flow Processing
- Synchronous Flow Processing
- Non-blocking Flow Processing
- Queued Flow Processing

Q47. Define the available resources used for modularizing configuration in mule?

Ans. There are many resources available for modularizing configuration in Mule as:

Independent Configuration

A Mule instance is capable of loading multiple independent configuration files one after another.

Inherited Configuration

As the term inheritance referred to represent the parent and child relationships between the instances this configuration also showcases the parent-child relationship between two configurations. It strongly provides this dependency with the guarantee that no configuration file will be omitted at the time of application booting. It used the same name for the parent and child models.

For ex:

```
1 <model name="myConfig"><model name="myConfig" inherit="true">
```

Imported Configuration

It supports spring configuration so that you can easily use your spring config file to Mule configuration files.

For ex: you need to extend instance.xml then it will import spring context file:

```
1 <spring:beans>,<spring:import resources="instance-beans.xml"/></spring:beans>
```

Heterogeneous configuration

As the term, heterogeneous refers you can configure many types of files using this configuration method. The instances can be created using the Groovy script method and Spring XML Configuration builders.

Q48. Transport layer functionality in MuleSoft.

Ans. The main operating task for the transport layer is to sending and receiving messages through the inbound and outbound communication network. The transport layer gets configured with connectors, endpoints, and transformers.

Transport is also responsible for message adaptor where the responsibility of the adaptor is to extract all the information from the particular request such as data, Meta information or header info, and attachments. It also stores the information available in it.

Q49. Explain the functionality of FanIn and FanOut

Ans. Fan-Out: you can use Fan-Out primitive to trigger the output terminal initially with the input message or many times as per need. Fan-Out can be used individually or as a combination of Fan-In and Fan-Out.

Fan-In: FanIn in a combination with Fan-Out works as a decision point to identify when to continue flow execution in the connector. It accepts n number of messages until the decision point is made. The last message received in the flow is represented as an output terminal.

Q50. What configuration is required for JDBC Adaptor implementation?

Ans. JDBC adaptor configuration is not a complex task for those who have experience in connecting DB. You need a data source to be connected and configured with a database. Also if your DB is having secure access then you need to create a security authentication program.

Q51. Name the technique of implementing a loop in mediation?

Ans. Fan-In and FanOut is the method to implement a loop in the mediation.

Q52. Explain the concept of Correlation Context.

Ans. When the mediation primitive requires passing the value form the request flow to response flow the correlation context plays the role.

Q53. List out the primitives used in Mediation.

Ans. Mediation has many primitive as below

- Type Filter
- Message Filter
- Service Invoke
- Fan-out
- Fan-in
- XSLT
- BO Map
- Header Setters
- Message Logger
- Even Emitter

- Endpoint Lookup
- Message Element Setter
- DB lookup
- Data Handler
- Custom Mediation
- Stop
- Fail
- Sub Flow

Q54. Define Mule UMO

Ans. UMO is Universal Message Object; UMO components are also referred to as service components.

Q55. Explain Mule Data Integrator.

Ans. Data Integrator tool is launched by Mule that is a data visualization mapping tool having support for flat files, XML Mapping and Java objects. It was a tedious task for the developer to code complex mapping functionalities so the Mule Data Integrator tool provides drag and drop features to make the coding process easier.

The mapping process gets support from eclipse (where plug-in has to be done before process) to run the Data Integrator which is part of top layer applications in Mule Architecture.

Q56. Find out the path for class Abstract Mule TestCase?

Ans. The path for Abstract Mule TestCase is /lib/mule/mule-core-.jar

Q57. Advantages of Mulesoft

Ans. There are many advantages of MuleSoft integration platform

- Very lightweight but highly scalable.
- It allows you to connect over a large scale of applications so that you can expand large over time.
- ESB ensures that the communication stays uninterrupted and consistent between the applications.
- Despite using the same protocols for Virtual Machine it allows you to build and maintain communication between applications.
- Ability to integrate and connect over the vast range of application network.
- POJO integration support to connect other components in the network.
- Reusability of Components
- You can use existing components without writing MuleSoft specific codes.
- A wide variety of vendors can be plugged in as it is a vendor neutral solution.
- Eliminates the vendor dependencies by reducing vendor restrictions.
- Can be deployed into a number of topologies despite sticking to one ESB as per your business need.
- The great adaptability of change for higher business support.

1) What is MuleSoft?

MuleSoft is an integration platform for connecting enterprise and SaaS applications in the cloud and on-premise.

2) List types of variables in MuleSoft

Types of variables in MuleSoft are:

- **Flow Variable:** It is used to either set or removes variables tied to a particular message in the current flow.
- **Record Variable:** It is used for batch processing flows.
- **Session Variable:** This variable is used to either set or remove variables tied to a particular message for the complete lifecycle.

3) What are the various types of messages in MuleSoft?

Various types of messages in MuleSoft are: 1) echo and log message, 2) bridge message, and 3) build message.

4) Explain Mule ESB

Mule ESB is an architecture developed for programmers. They can integrate a range of applications together using the bus-like infrastructure. Mule ESB can be integrated with HTTP, web service, JMS, etc.

5) What is fan-in?

Fan-in helps in taking a decision to continue flow execution. It be used in combination with fan out.

6) What is a fan-out?

Fan out is primitive that can be used to input messages through the output terminal once or more than one time. It can also be used as a combination of both fan-out and fan-in.

7) Mention the features of Mule ESB

The features of Mule ESB are:

- Offer SLA (Service Level Agreement) monitoring and API management facility.
- It has easy to use and drag and drop graphical design.
- Mule ESB provides high scalability.
- It enables developers to deploy in one click cloud or on-premise deployments.

8) Mention the basic principles of ESB Integration

The basic principles of ESB integration are:

- **Transportation:** It negotiates between different formats like JDBC, HTTP, JMS, etc.
- **Transformation:** It deals with the transportation of data between data formats needed by the ESP connector.
- **Non-functional consistency:** It is the way of how monitoring and security policies are applied and should be consistent.
- **Mediation:** It involves offering different interfaces to:
 1. Enables different channels to the same component implementation.
 2. Support various service versions for backward compatibility.

9) What is Mule Expression Language?

MEL or Mule Expression Language is a light-weight mule specific language that can be used to access and evaluate data in the payload.

10) List various types of endpoints in Mule ESB

Various types of Endpoints in Mule are 1) JMS, 2) HTTP, 3) SMTP, 4) IMAP, and 5) AJAX.

11) What is the full form of SDO?

The full form of SDO is a Service Data Object.

12) Mention different types of Exception Handling

Different types of exception handlings are:

- Rollback exception handling.
- Default exception handling.
- Catch exception handling.
- Global exception handling.
- Choice exception handling.

13) What are the advantages of using ESB?

The advantages of using ESB are:

- It offers a high level of the operational controlling facility from the portal that is based on the web.
- ESB provides numerous connectivity options using SaaS-based applications.
- It provides API and analytics management.
- ESB is equipped with numerous bug fixing and automated testing facilities.
- EDI (Electronic Data Interchange)/B2B (business to business) integration.
- Batch integration feature using real-time integration methods.

14) What is a transient context?

Transient context is used to pass the values within the existing flow, either requesting flow or the responding flow.

15) What is Mule Transformer?

Mule transformer is an event instance that refers to a library, org.mule.api.MuleEvent. This object carries the message with the event. The main aim of the Mule transformer is to create a chain of transformers.

16) What is API?

API is the acronym for Application Programming Interface. It is a software interface that allows two applications to interact with each other without any user intervention.

APIs provides product or service to communicate with other products and services without having to know how they're implemented.

17) What are the features of API?

Here are some essential features of API:

- Efficiency
- Wider reach
- Customizable
- Personalization

- Data ownership
- Easy integration with GUI
- Time effective
- Language-independent

18) What is the payload in MuleSoft?

The payload is a mule runtime variable that stores objects or arrays. It helps developers to access payload under different forms.

19) What are the various parts of composing a message in Mule?

Different parts of composing a message in Mule are:

- **Properties:** it contains the header or meta-information or header similar to the SOAP (Simple Object Access Protocol) message.
- **Payload:** It is the main data context carried by a particular message.
- **Multiple name attachments:** It provides support for multiple messages or payload that occurs during event processing.

20) Mention flow processing strategies in MuleSoft.

MuleSoft has six types of strategy for flow processing:

- A queued asynchronous flow processing
- Custom flow processing
- Thread per processing
- Queued flow processing
- Non-blocking flow processing
- Synchronous flow processing
- Asynchronous flow processing

21) Explain the concept of Correlation Context.

It is a primitive that is used to pass values from request flow to response flow.

22) Mention different types of primitives used in Mediation

Different types of primitives used in Mediation are:

- Endpoint lookup
- Data handler
- Message element setter
- DB lookup
- Service Invoke
- Type filter
- Stop
- Sub Flow
- Custom mediation fan-out
- Fan-in
- Even emitter
- Header setters
- Message logger
- XSLT

- BO MapMessage filter
- Fail

23) Name different types of ESPs used in the market

Different types of ESPs used in the market are:

- Mule ESB
- JBoss fuse ESB
- Talend

24) Define the model layer in mule

The model layer is the first logical layer. It represents the runtime environment that hosts services. This layer describes the behaviour of Mule when processing requests that are handled by services. It offers services with default values to simplify configuration.

25) Explain connector in MuleSoft

A connector in Mule controls how a particular protocol used. It can be configured with parameters which are specific to that protocol. The connector holds any state context which can be shared with any entity in charge of actual communications.

26) What is Endpoint in Mule?

An endpoint in MuleSoft indicates a particular usage of a protocol. It is for polling, reading from, or writing to a destination.

Therefore, it controls what underlying entities would be used with a dependent connector.

27) Define component in Mule

Components perform an important role in MuleSoft services. Every service is organized with core components and core and inbound and outbound routers.

They are used to implement behavior in service. It can be very simple, like logging messages or invoking other services.

28) What is the use of Outbound Endpoint in MuleSoft?

Outbound Endpoint in MuleSoft is used to perform the following things:

- Send SOAP messages
- Write to file streams
- Send email messages

29) Define configuration builders In MuleSoft

MuleSoft is a configuration builder to translate a configuration file into the graph of the object that makes the running node of ESB.

30) List the types of configuration builders in MuleSoft

Types of configuration builders In MuleSoft are 1) Spring-driven builder and 2) script builder.

31) What Is TSD in Mule?

TSD or transport service descriptor is a connector used for technical configuration. It defines classes name used for message receivers, dispatchers, and requesters. This default value can vary to grasp the behaviour of transport.

32) Define multicasting router in MuleSoft

Multicasting router in MuleSoft sends messages to more than one endpoints over different transports. It allows the user to move the same messages across different endpoints.

33) What are the characteristics of Global Endpoint?

Characteristics of global endpoint are:

- The global endpoint is not typified or outbound routing.
- It can be usable in different places of configuration files.
- The global service name must be applied so that it can reference the endpoint.
- It helps to clarify the usage of a particular destination.

34) Explain VM transport in MuleSoft

The VM (Virtual Machine) transport is a special type of transport that can be used to send a message via memory. These messages never leave the Java Virtual Machine, and the Mule instance is running in.

35) Name different types of web services

Different types of web services: 1) RESTful web services and 2) SOAP web services.

36) What are Web Services?

Web services are a standardized way or medium to propagate communication between the client and server applications on the World Wide Web.

37) What is Restful Web Service?

Restful Web Service is a light-weight, maintainable, and scalable service that is built on the REST architecture. Restful Web Service, expose API from your application in a secure, uniform, stateless manner to the calling client. The calling client can perform predefined operations using the Restful service.

38) Mention the difference between SOAP and REST

The difference between SOAP and REST is:

SOAP	REST
SOAP stands for Simple Object Access Protocol	REST stands for Representational State Transfer
SOAP cannot make use of REST since SOAP is a protocol, and REST is an architectural pattern.	REST can make use of SOAP as the underlying protocol for web services because, in the end, it is just an architectural

pattern.

SOAP can only work with XML format. As seen from SOAP messages, all data passed is in XML format.

REST offers various data formats such as JSON, plain text, HTML, XML, etc. But the most preferred format for transferring data is JSON.

39) What is RAML?

The full form of RAML is the RESTful API Modeling Language. It is a YAML-based language that describes RESTful APIs.

RAML is best for the information needed to describe RESTful APIs. It is similar to WSDL (Web Services Description Language).

RAML contains request/response schema, URI parameter, endpoint URL, HTTP methods, and query.

40) What is caching?

The cache concept is a way of storing the copy of the file in the cache, or any temporary storage location to access it quickly.

41) What are the Models?

Model is a grouping of services that are created in MuleSoft studio. User has the liberty to start and stop all the services inside a particular model.

42) Name supported languages by MuleSoft

Supported languages of MuleSoft are 1) Ruby, 2) Python, 3) Groovy, and 4) JavaScript.

43) List various the categories of Mule Processors

Categories of Mule Processors are: 1) Components, 2) Exception strategies, 3) Business events, 4) Routers, 5) Connectors, and 6) Transformers.

44) What are the configuration patterns provided by MuleSoft?

Configuration patterns provided by of MuleSoft are:

- Bridge
- Validator
- WS proxy
- Simple service pattern
- HTTP proxy

45) What are the advantages of the logger component?

The advantages of the logger component are:

- Users can add this core component anywhere in the workflow.
- It can be configured to any combination of strings and expressions.

46) What is scheduler Endpoint?

Scheduler Endpoint is a MuleSoft component or middleware are working on time-based conditions. It allows the user to trigger whenever this condition is met.

47) Explain the parameters to configure a scheduler

Parameters related to configuring a scheduler are:

- **Frequency:** It is a frequency used by Scheduler to triggers flows.
- **Start Delay:** It is the time to wait before triggering any flow.
- **Time Unit:** The time unit for frequency and Start Delay.

48) What is Choice Router?

Choice Router dynamically routes messages using a flow. It is based on a set of DataWeave expressions to evaluate the message content.

49) What is a Scatter-Gather Router?

Scatter-Gather Router is the most used routing event processor. It can send a request message to more than one target concurrently. This router then collects responses from all routes and aggregate back into one response.

50) What are error types in MuleSoft?

Following are the effort types:

- Transformation
- Expression
- Routing
- Duplicate_Message
- Source_Response
- Timeout
- Security
- Connectivity
- Validation

51) What are the features of MUnit?

The features of MUnit are:

- In MUnit framework, a developer can create a Mule test by using Java code as well as Mule code.
- The programmer can design and test Mule APIs and apps, either in XML or graphically within Anypoint studio or platform.
- MUnit allows integrating the testing into the current CI/CD process.
- MUnit offers auto-generated tests and coverage reports to reduce manual work.
- Developers can also use local FTP/DB/mail servers to make the testing process more portable through the Continuous Integration.
- It allows enabling/disable tests.
- Programmers can extend the MUnit framework using plugins.
- Features to verify message processor calls.
- It provides error reports with a Mule stack trace.

52) What is Exchange?

Exchange is a hub for the development team. It is used to store and access API, connectors, templates, documentation, and more.

53) What are the advantages of SOAP?

The advantages of SOAP are:

- SOAP is the perfect medium that is developed for web service to talk with client applications.
- SOAP is a light-weight protocol which can be used for data interchange between applications.
- SOAP protocol can work any programming language based applications on Windows and Linux platforms.
- It does not require customization to run the web services built on the SOAP protocol to work on the WWW.

54) Define Batch Jobs in Mule ESB

A batch job is an element in Mule that split large size messages into records that process asynchronously in a batch job.

55) Explain the Mule data integrator.

A mule data integrator is a tool that is used for mapping data by visualizing it. It offers drag and drop feature to make a developer's coding process easier.

Define Mule ESB

It is an integration platform and a lightweight enterprise service bus (ESB) based on Java. It enables developers to easily and quickly connect applications and exchange data. With Mule ESB, developers can easily integrate existing systems irrespective of how different are the technologies that the application use such as Web Services, HTTP, JDBC, JMS and the like.

What is the benefit of using Mule ESB?

Mule ESB is an integration framework that is lightweight and highly scalable. It enables developers in starting small applications and also in connecting different applications. With Mule managing the exchanges between components and applications transparently and ESB taking care of a variety of applications, it is easy to integrate third-party applications with the help of Mule.

What are the various types of Exception Handling?

- Global Exception Handling
- Catch Exception Handling
- Choice Exception Handling
- Default Exception Handling
- Rollback Exception Handling

What are the characteristics of Mule ESB?

An ESB is used for the purpose of integration with an approach that is service-oriented. Its features include:

- Message Routing Service
- Message Transformation Service
- Set of Service Container
- Web Service Security

In Mule, how do you develop and consume SOAP services?

SOAP services can be created just like how we create a Mule project by using RAML. The difference here is that we need Concert WSDL importing rather than RAML. And SOAP services can be consumed by using our Mule flow CXF component or Web Service Consumer.

How can you find out whether your project requires ESB?

As every project might not require an ESB, you should analyze first to see if your project might benefit from ESB implementation. Certain things that should be at the front of your mind while you analyze the need for ESB are:

- If the project requires integration of more than 3 applications or services and if communication between two application is needed, it would be enough to use point to point integration
- Sometimes there will be a need for you to scale the project in the future where there might arise a need to interact with multiple services. This is required only by a few projects that perform heavy tasks
- If the project requires message routing abilities such as aggregating and forking message flows. This feature is not necessary for all projects
- You should have clarity on the architecture of the thing that needs to be achieved. A simple POC integration of small parts to find out the benefits is much better
- As most of the ESBs are on the expensive side, first evaluate whether your project budget permits ESB use

Name the various kinds of Primitives that are used in Mediation.

The following are the various kinds of primitives in mediation

- Endpoint Lookup
- Service Invoke
- DB lookup
- Data Handler
- Type Filter
- Message Element Setter
- Custom MediationFan-out
- Fan-in
- Header Setters
- Message Logger
- Even Emitter
- XSLT
- BO MapMessage Filter
- Fail
- Stop
- Sub Flow

Name the various ESBs that are in the market

There are different ESBs in the market, both licensed and open source. They are:

- JBoss Fuse ESB
- Mule ESB
- Talend

What do you know about Mule's shared resources?

Connectors can be made into reusable components. For that, you first need to define them as common resources and then expose them to all applications that have been deployed under the same domain. Such resources are called shared resources. Shared resources need to be defined inside Mule Domain Project and need to be referred to every one of the projects that are intended to incorporate the elements in it.

What are the core principles of ESB Integration?

Transportation — Transport protocol negotiation between different formats including JMS, JDBC, HTTP, etc.

Transformation — Transformation of data between certain data formats as needed by every ESB connector individually

Non-functional consistency — The way monitoring and security policies are applied as well as implemented should be consistent

Mediation — This involves offering many different interfaces to:

- Enable different channels to the same unlying component implementation
- Support different service versions for backwards compatibility

What are the different types of variables in Mule?

The different types of Mule variables are:

- Record variable
- Session variable
- Flow variable

What are the different kinds of Flow Processing Strategies?

The following are the six kinds of Flow Processing Strategies:

- Thread Per Processing Strategy
- Custom Processing Strategy
- Queued Asynchronous Flow Processing Strategy.
- Asynchronous Flow Processing Strategy
- Synchronous Flow Processing Strategy
- Queued Flow Processing Strategy
- Non-blocking Flow Processing Strategy

1. What is Mulesoft? For what Mulesoft is used for?

Answer:

MuleSoft is the most widely used integration platform. Here we will find 2 types Mule ESB and Cloud Hub for connecting enterprise and SAAS applications in the on-premises and cloud. Mulesoft allows developers to connect applications together quickly and easily and it helps in exchanging the data.

2. Explain the types of variables in mule?

Answer:

There are 3 types of variables available in a mule.

Flow Variable

- Session Variable
- Record Variable

3. What is Mule ESB?

Answer:

Mule ESB is a Java-based ESB (enterprise service bus) and it is the integration platform all the developers can connect to their respective application directly with ESB. Mule ESB uses service oriented architecture. The main use of Mule ESB is it enables easy integration of existing systems. It can integrate with different technologies that the applications use- Including JMS, **Web Services**, HTTP, etc.

Let us move to the next Mulesoft Interview Questions.

4. What is ESB?

Answer:

ESB stands for Enterprise Storage bus. It's software architecture for middleware and it provides fundamental services for more complex architectures. This is the common Mulesoft Interview Questions which is frequently asked in an interview.

Example: In ESB incorporates with the features required to implement an SOA(service-oriented architecture).

5. What is the MuleSoft Anypoint platform and where it will be used?

Answer:

MuleSoft Anypoint Platform of integration products is designed to tie both software as a service (SaaS) and on-premises software.

Part 2 – Mulesoft Interview Questions (Advanced)

Let us now have a look at the advanced Mulesoft Interview Questions.

6. What are the main features of Mule ESB? What are the different ESBs in the market?

Answer:

1)The main Features of Mule ESB are::

- It is very simple and easy to use-Drag and drops Graphical design
- SLA monitoring and API management
- High scalability
- We can deploy in a One-click cloud or on-premise deployments

2)Different ESB's in Market are::

- Talend
- Mule ESB
- JBoss Fuse ESB

7. How will we identify ESB is needed in a project?

Answer:

Implementation of ESB is not suitable for all the projects. We should analyze is really ESB is required here or not. You need to analyze by taking below points into consideration:

- In the project, require 3 or more applications and services to be integrated and there must be a need to communicate between the applications.
- If there is plain of interacting with more applications and Services in the future then we can go with Mule ESB because it is highly scalable.
- We need to keep cost also in mind before going to ESB implementation

Let us move to the next Mulesoft Interview Questions

8. Explain the difference between Callout and Service Invoke?

Answer:

Callout: We can call the service using callout or with service invoke. Use the Callout if we need to mediate a message (without calling an intermediate service) and then call a service provider.

- The Callout provides the simplest model for this configuration.
Service Invoke: You need to interact with multiple services, and produce an output that combines service responses. The Service Invoke primitive does not switch from request flow to response flow.
- Use the Service invoke if we need to call an intermediate service.
Example: We can use an intermediate service to adjust a message or to validate a message externally. The mediation flow contains Service Invoke mediation primitive, and a Callout node that is connected to the service provider there will be no intermediate service.

9. What is the full form of SDO and SMO?

Answer:

- **SDO:** Service Data Object and it represent the Object.
- **SMO:** Service Message object and it is used to represent messages

10. Explain about Fan-in and Fan-out?

Answer:

Fan-In: Fan-In is always in the flow with Fan-Out and helps in taking a decision to continue flow execution. The Fan In may only be used in combination with Fan Out.

Fan-out: We can use the Fan Out-primitive to fire the output terminal once with the input message or to fire the output terminal multiple times. Fan-out can be used as a combination of Fan Out and Fan In.

1. What Is Mule?

Answer: Mule is a lightweight event-driven enterprise service bus (ESB) and an integration platform. It is a lightweight and modular solution that could scale from an application-level messaging framework to an enterprise-wide highly ([Mulesoft Training](#)) distributable object broker.

2. What is Shared Context?

Answer:

Shared Context: Context is a temporary area which is created along with Service Message Object (SMO) in the Mediation Flows. Shared Context is a type of context which is present in the SMO. Shared Context is mainly used when we are using Aggregation process where we need to Iterate the BO for Certain times. Shared Context maintains Aggregation data between Aggregation (FanOut and FanIn) primitives. The Content (data) which is present in the shared context BO does not persist across Request and Response flows i.e The Data in the Shared Context which is used in Request flow can not be used again in Response flow. ([Selenium Training](#))

3. What is the functionality of Fan-in and Fan-out?

Answer: Fan-out: We can use the Fan Out primitive to fire the output terminal once (with the input message) or fire the output terminal multiple times. You can use Fan Out in isolation or as part of a Fan-Out and Fan In combination. ([Sap Fico Online Training](#))

Fan-In: Fan In is always partnered with a Fan-Out in the same flow and acts as a decision point for when to continue flow execution. It receives a number of messages until a decision point is reached, at which point the last message to be received is propagated to the output terminal. The Fan In primitive may only be used in combination with Fan-Out. ([Azure Training](#))

4. What is the Difference between SDO and SMO?

Answer: SDO: Service Data Object is the representation of the variable or Object.

SMO: The SMO model is a pattern for using SDO Data Objects to represent messages

5. Why The Name Mule?

Answer: There is a lot of infrastructure work to be done before we can really start thinking about implementing any logic. So this infrastructure work is regarded as “donkey work” as it needs doing for every project. A Mule is also commonly referred to as a carrier of load, moving it from one place to another. The load specializes in moving is our enterprise information.

6. What are Batch Jobs in Mule ESB?

Answer: A batch job is a top-level element in Mule which exists outside all Mule flows. Batch jobs split large messages into records which Mule processes asynchronously in a batch job; just as flows process messages, batch jobs process records. ([Mulesoft Training](#))

A batch job contains one or more batch steps which, in turn, contain any number of message processors that act upon records as they move through the batch job. During batch processing, you can use record-level variables (recorders) and MEL expressions to enrich, route or otherwise act upon records.

7. What are the different types of Flow Processing Strategies?

Answer: There are six different types of Flow Processing Strategies. They are

1. Asynchronous Flow Processing Strategy.
2. Custom Processing Strategy.
3. Thread Per Processing Strategy.
4. Queued Asynchronous Flow Processing Strategy.
5. Synchronous Flow Processing Strategy.
6. Non-blocking Flow Processing Strategy.
7. Queued Flow Processing Strategy.

8. What Are Available Esbs Apart From Mule?

Answer: All major JEE vendors (BEA, IBM, Oracle, Sun) have an ESB in their catalog. It is unremarkably based on their middleware technologies and is usually at the core of a much broader SOA product suite. There are also some commercial ESBs that have been built by vendors not in the field of JEE application servers, like the ones from Progress Software, IONA Technologies, and Software AG.

9. Why the Mulesoft is preferred than other ESB implementations?

Answer: Mule is lightweight but highly scalable, allowing you to start small and connect more applications over time. The ESB manages all the interactions between applications and components transparently, regardless of whether they exist in the same virtual machine or over the Internet, and regardless of the underlying transport protocol used.

Several commercial ESB implementation provides limited functionality or built on top of an existing application server or messaging server, locking you into that specific vendor. Mule is vendor-neutral, so different vendor implementations can plug into it. You are never locked into a specific vendor when you use Mule. ([Sql Server Training](#))

10. What Is Filter In Mule?

Answer: Filters are a powerful complement to the routers. Filters provide the brains routers need to make smart decisions about what to do with messages in transit. Some filters go as far as deeply analyzing the content of a message for a particular value on which their outcome will be based.

11. What are the different types of Web services?

Answer: There are mainly two types of web services.

- SOAP web services.
- RESTful web services.

12. What is the difference between Stop and fail?

Answer: Stop: Stops a particular path in the flow, without generating an exception.

Fail: Generates a failure in the flow.

13. How to add classes/jars to the Mule classpath?

Answer: Use the MULE_LIB variable (generally set in the run script)

To include JAR file(s) in a mule classpath, declare each dependent jar file in the MULE_LIB entry.

For spring resource, if the XML bean declaration is placed within a project, including the project JAR file in the classpath too (i.e., if not included, Mule will throw a *.xml not found on the classpath)

14. What is Mule Data Integrator?

Answer: Mule has released a data integrator tool, it is a visual mapping tool which supports flat-file, java object, XML mappings, etc. Coding complex mappings can be very tedious and additionally difficult to maintain, the mule data integrator with drag and drop facilities makes building and maintaining mappings very simple.

The mapping is done in eclipse (plugins required) and executed on a data integrator runtime which sits on top of Mule ESB – this requires a license.

15. Where to find the class Abstract Mule TestCase?

Answer: This is in 1.4/1.4.1 distributions but was missing from the 1.3.3 distribution – the class is defined in /lib/mule/mule-core-jar.

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16. What is Flow Processing Strategies?

Answer: Flow processing strategies in Mulesoft are:

- Synchronous Flow Processing Strategy
- Queued Flow Processing Strategy
- Asynchronous Flow Processing Strategy
- Thread Per Processing Strategy
- Queued Asynchronous Flow Processing Strategy
- Non-blocking Flow Processing Strategy
- Custom Processing Strategy.

17. Why am I getting 'Unable to get resource from repository' while building the Mule examples?

Answer: If you are getting this error message while building the Mule examples:

18. How to use Mule HQ with Mule?

Answer: Mule Configuration

Add the following code snippet to your Mule configuration:

```
<agentname="RMI" className="org.mule.management.agents.RmiRegistryAgent"/>
<agentname="JMX" className="org.mule.management.agents.JmxAgent">
<propertyname="connectorServerUrl" value="service:jmx:rmi:///jndi/rmi://localhost:1099/server"/>
<mapname="connectorServerProperties"> <propertyname="jmx.remote.jndi.rebind" value="true"/>
```

Start your Mule instance.

Ensure the HQ agent is running on the server the Mule instance is configured on and is pointing to the desired HQ server. ([Sql Server Training](#))

Check the Mule HQ server page to see if information about the Mule instance is being received.

19. What is a Mule UMO?

Answer: A Mule UMO is a Universal Message Object

UMO is now a legacy term. What was once referred to as UMO Components are now referred to as Service Components? (What is MuleSoft Used for)

20. A mule won't start? Check your Environment?

Answer: The java and mule environment variables must be set up correctly for a mule to start. If you are experiencing problems check the following variables:

MULE_HOME – should be the location of the mule install

JAVA_HOME – should be the location of the JDK

PATH – should have both JAVA_HOME\bin and MULE_HOME\bin in the path.

Check all of the above carefully. Some systems with multiple JDK's installed can end up with incorrect mappings between the PATH and the JAVA_HOME, which will stop mule from loading.

21. Is MuleSoft open source?

Answer: MuleSoft is open source technology and it provides open source SOA middleware and related products. ([python training online](#))

22. Why do we need ESB?

Answer: ESB, a middleware technology, is a Bus-like architecture used to integrate heterogeneous systems. In ESB, each application is independent and yet able to communicate with other systems. It, thus, prevents scalability issues and ensures that communication happens only through it.

23. What are Web Services?

Answer: Web service is functionality or a small piece of program in any programming language that can be accessed over HTTP. Message format can be XML or JSON or any other program as long as the other programs can understand and communicate.

A Web service can be of synchronous or asynchronous type. Any web service has a server-client relationship and it can have multiple clients.

24. How can we create and consume SOAP service in Mule?

Answer: Creating SOAP Service – We can create a SOAP service same as we create Mule Project With RAML, the only [change](#) is instead of RAML we need to import Concert WSDL. Consuming SOAP Service – We can use Web Service Consumer or CXF component in our mule flow to access/consume SOAP service.

25. What would you consider your technical specialty?

Answer: Professional Answers Preview “Computing and Network Communications has always been a really strong area for me. I have a big interest in how network communication improves our daily lives and our overall business efficiency. I see that you are looking for team members who are experts in network communications which is what really caught my eye when I first saw your job posting.” “I have been focusing my career on becoming a cloud integration specialist although I do have cross-platform app development expertise as well.” ([oracle apex training](#))

26. What is the MuleSoft Anypoint platform used for?

Answer: MuleSoft’s Anypoint Platform of integration products is designed to tie together software as a service (SaaS) and on-premises software.

27. What is the use of Web service?

Answer: Web services are XML-based information exchange systems that use the Internet for direct application-to-application interaction. These systems can include programs, objects, messages, or documents.

28. What is Mulesoft used for?

Answer: Mulesoft is an integration platform that allows developers to connect applications together quickly and easily, enabling them to exchange data.

29. Is Mulesoft a middleware?

Answer: Mule Enterprise Service Bus is a middleware technology that quickly, easily, and securely connects the enterprise. Unlike typical middleware software, Mule as an ESB is a Java-based middleware solution that is easy to use and easy to scale. ([devops training videos](#))

30. A web service is a collection of open protocols and stand

ards used for exchanging data between applications or systems. Mule ESB?

Answer: The different types of variables in Mule ESB are:

- Flow Variable
- Session Variable
- Record Variable

31. What are the various types of Exception Handling in Mule ESB?

Answer: The types of exception handling in Mule ESB are:

- Default Exception Handling
- Global Exception Handling
- Catch Exception Handling
- Choice Exception Handling

32. What are the advantages of Soap Web Services?

Answer: WS Security: SOAP defines its own security known as WS Security.

Language and Platform independent: SOAP web services can be written in any programming language and executed in any platform.

Disadvantages of Soap Web Services:

Slow: SOAP uses XML format that must be parsed to be read. It defines many standards that must be followed while developing SOAP applications. So it is slow and consumes more bandwidth and resource.

WSDL dependent: SOAP uses WSDL and doesn’t have any other mechanism to discover the service.

33. What are the Features of Mule ESB?

Answer: An ESB is used for integration using a service-oriented approach. Its main features are:

1. Set of Service Container
2. Message Transformation Service
3. Message Routing Service
4. Web Service Security

34. What Are Differences Between Mule And Other Commercial Tabs?

Answer: Prescriptive deployment model, whereas Mule supports a wide variety of deployment strategies. Prescriptive SOA methodology, whereas Mule can embrace the [architectural](#) style and SOA practices in place where it is deployed. Mainly focused on higher-level concerns, whereas Mule deals extensively with all the details of integration. Strict full-stack web service orientation, whereas Mule’s capacities as an integration

framework open it to all sorts of other protocols. Comprehensive documentation, a subject on which MuleSource has made huge progress recently.

35. What Is Connector In Mule?

Answer: A connector is in charge of controlling the usage of a particular protocol. It is configured with parameters that are specific to this protocol and holds any state that can be shared with the underlying entities in charge of the actual communications.

For example, a JMS connector is configured with a Connection, which is shared by the different entities in charge of the actual communication. ([big data online training](#))

36. What Is Model Layer In Mule?

Answer: The first logical layer is the model layer. A Mule model represents the runtime environment that hosts services. It defines the behavior of Mule when processing requests handled by services. The model provides services with supporting features, such as exception strategies. It also provides services with default values that simplify their configuration.

37. What Is Service Layer In Mule?

Answer: A Mule service is composed of all the Mule entities involved in processing particular requests in predefined manners. A service is defined by a specific configuration. This configuration determines the different elements, from the different layers of responsibility, that will be mobilized to process the requests that it will be open to receive. Depending on the type of input channel it uses, a service may or may not be publicly accessible outside of the ESB.

38. Where would you like your career with Mulesoft to take you?

Answer: Professional Answers Preview “One of the reasons I was so excited to interview with Mulesoft is because of the endless growth opportunities you appear to offer your high performing employees. My biggest ambition is to be awarded a management role in the next 3-5 years.” “What interests me most about your company is that there are so many options when it comes to career growth. Ideally, I would like to earn my way into a team lead role, then a division manager position in the digital management department. Eventually, a CDO (Chief Data Officer) level role would be ideal. Of course, I understand that this will take many years of dedication and hard work.”

39. Have you researched Mulesoft enough to know how their internal hierarchy works? Do they have a variety of departments and management levels, offering you choices when it comes to carving out your career path?

Answer: the interviewer about your career ambitions specifically related to this role and their organization. Professional Answers Preview “There have been so many great advancements in technology this year. Recently, I was reading an article by MIT Technology Review and was amazed at the incredible advancements this year. The one that stood out most to me was the autopilot software update released by Tesla this year. We truly will have the software for vehicles to be controlled by tech rather than humans.” “The most impressive advancement in technology, in my opinion, is the continued advancement of mobile integration. Most of the modern world is connected by tech and we can do absolutely everything from our phones these days.” Display to the interviewer that you have an invested interest in the technology industry by discussing your thoughts on recent advancements. Be sure to include the ways in which you stay up to date on industry changes and new technology.([hadoop online training](#))

40. What is the definition of Web Services?

Answer: Web service is a function or program in any language that can be accessed over HTTP. Message format can be XML or JSON or any other program as long as the other programs can understand and communicate. Any web service has a server-client relationship. Web services can be synchronous or asynchronous. Any web service can have multiple clients.

41. What is your understanding of this position and the responsibilities that come with it?

Answer: Professional Answers Preview “From our previous conversation, understanding the job description, and researching the role online I feel that the job is best described as a project leader focused on X, Y, and Z. The technical requirements that you would like to see are A, B, and C. I am well versed in the tasks and programs you are asking for and am confident that I meet the strict requirements to successfully perform in this role.” “My understanding is that you seek a candidate with expert level skills in X, Y, and Z, who can train junior employees.

42. Do you have any questions about my technical expertise?

Answer: Assure the interviewer that you bring a full understanding of the technical requirements attached to this particular job. It is very important that you research the role and thoroughly consume the information provided in the company’s job posting/job description. Keep your answer to the point. ([data science online training](#))

43. What is the difference between SOAP and REST?

Answer:

SOAP: REST

1. SOAP is a protocol. -> REST is an architectural style.
2. SOAP stands for Simple Object Access Protocol. -> REST stands for REpresentational State Transfer.
3. SOAP can't use REST because it is a protocol. -> REST can use SOAP web services because it is a concept and can use any protocol like HTTP, SOAP.
4. SOAP uses services interfaces to expose business logic. -> REST uses URI to expose business logic.
5. SOAP defines standards to be strictly followed. -> REST does not define too many standards like SOAP.
6. SOAP defines standards to be strictly followed. -> REST does not define too many standards like SOAP.
7. SOAP requires more bandwidth and resource than REST. -> REST requires less bandwidth and resource than SOAP.
8. SOAP defines its own security. -> RESTful web services inherit security measures from the underlying transport.
9. SOAP permits XML data format only. -> REST permits different data format such as Plain text, HTML, XML, JSON etc.
10. SOAP is less preferred than REST. -> REST more preferred than SOAP.

44. What Is Component In Mule?

Answer: Components are the centerpiece of Mule's services. Each service is organized with a component at its core and inbound and outbound routers around it. Components are used to implement a specific behavior in service. This behavior can be as simple as logging messages or can go as far as invoking other services. Components can also have no behavior at all; in that case, they are pass-through and make the service act as a bridge between its inbound and outbound routers. ([tableau online training](#))

45. What is a SOAP Web Service?

Answer: SOAP stands for Simple Object Access Protocol. It is an XML-based protocol for accessing web services. SOAP is a W3C recommendation for communication between two applications. SOAP is XML based protocol. It is platform-independent and language independent. By using SOAP, you will be able to interact with other programming language applications.

46. What Is Transport Layer In Mule?

Answer: The transport layer is in charge of receiving or sending messages. This is why it is involved with both inbound and outbound communications. A transport manifests itself in the configuration by the following elements: connectors, endpoints, and transformers.

A transport also defines one message adapter. A message adapter is responsible for extracting all the information available in a particular request (data, meta information, attachments, and so on) and storing them in transport-agnostic fashion in a Mule message.

sewer :

A Mule message is composed of different parts: The payload, which is the main data content carried by the message.

The properties, which contain the meta information much like the header of a SOAP envelope or the properties of a JMS message. Optionally, multiple named attachments, to support the notion of multipart messages.

Optionally, an exception payload, which holds any error that occurred during the processing of the event.

47. How to find when the project needs ESB?

Answer: ESB implementation is not suitable for all projects. Proper analysis should be done if the use of ESB will really benefit the project.

Some of the points to be considered while analyzing the need of ESB are as follows:

- If the project requires integrating 3 or more applications/services. If the need is to communicate between two applications, using point-to-point integration would suffice.
 - If the project would need to be scaled in the future where it might be needed to interact with more services in the future. Not all projects need this as they may perform not that big a task.
 - If the project needs message routing capabilities such as forking and aggregating message flows.
-

How to achieve the parallel processing in MuleSoft?

To achieve the parallel processing you can use [scatter gather](#) or batch job and with mule 4.2 onwards there is also parallel for each has been added to process the message simultaneously, Please visit [scatter gather](#) and [parallel for each](#) urls for more information

How to invoke the Custom Java Class from inside a dataweave

```
%dw 2.0
Import java!com.mycompany::classname

output application/json
---
{
  a: classname::methodName("myString")
}
```

How to access secure property inside dataweave?

By using `p('secure::<property name>')` you can access the secure property defined as part of secure property file in mule application

How can you raise error explicitly in Mule flow/subflow?

By using the Raise Error component you can explicitly raise the error in between the flow

How can you handle errors in middle of the processing in MuleSoft?

You can use the Try block to handle the errors in between the flows, try block also support the transaction. By using try block you can use the on-error continue or on-error propagate to handle the error

How to call Flow from inside a dataweave?

You can use dataweave lookup function to call flow from inside a dataweave It works in Mule apps that are running on Mule Runtime version 4.1.4 and later It takes the flow's name and an input payload as parameters. For example, `lookup("flowname", payload)`

What is the difference between map, mapObject and pluck?

- map is used to iterate over array
- mapObject is used to iterate over object and return object
- pluck iterates over an object and returns an array of keys, values, or indices in that object.

How will you combine 2 arrays into single array in MuleSoft?

You can use flatten dataweave function to combine two arrays in to single array

What are target variable in MuleSoft

When creating a flow in mule app, you can store data in variable so that any component in a flow can use it. In Mule 4 Target Variable replaces the Message Enricher of Mule 3, target variable is available with most of the connectors

How many phases are there for batch job in Mule 4

Each [batch job](#) contains 3 different Phase

-
- Load and Dispatch.
- Process
- On Complete.

When we should use batch processing

You can use batch processing when:

-
- Synchronizing data sets between business applications, such as syncing contacts between two different ERPs
- Extracting, transforming and loading (ETL) information into a target system, such as uploading data from a flat file (CSV) to Hadoop.
- Handling large quantities of incoming data from an API into a different systems

Why we use Batch Aggregator scope

Batch Aggregator scope is used to accumulate a subset of records from a batch step and process them to external source or service for example rather than processing single record to target system you can use batch aggregator to process all the records at once. You can also configure batch aggregator scope to process fixed size groups of records inside a batch aggregator scope

What is the difference between For each and Batch Process

[For each](#) do the processing in single thread while [Batch Process](#) performs multi-threaded processing [more information](#)

How to manage the Transaction in MuleSoft

MuleSoft supports two types of transactions

- Single resource(Local)
- XA transactions (extended architecture)

You can configure the transaction depends on the requirement

You can start a transaction from a message source. In this case, the entire flow takes part in transaction This is useful when working with messaging connectors to prevent the consumption of the message if a problem occurs when processing it, allowing you to retry later (because of the rollback).

You can also use the Try block which support the transaction when you want to handle the transaction in middle of the flow

There are below transaction Action define the type of action that operations take regarding transactions

-
- Always_Begin
- Always_Join
- Begin-or-join
- Join-if-possible
- Indifferent
- None
- Not Supported

What are various types of error handling in Mule 4?

Mule 4 has redesigned [error handling](#) by introducing the error-handler component, which can contain any number of internal handlers and can route an error to the first one matching it. Such handlers are on-error-continue and on-error-propagate.

How can you process message asynchronously in MuleSoft?

You can use the [Async](#) Scope to process the message asynchronously. – [more information](#)

Async can be used to do the processing in parallel with the main flow, it can be used to process the time consuming operations that does not expect a response back

How can you achieve the Reliability in MuleSoft?

If your application uses a transactional transport such as JMS, VM, DB etc. [Reliable messaging](#) is ensured by the built in support of transactions in the transport. Reliable messaging pattern is important when dealing with non-transactional endpoints

How can you improve performance of Mule Application in MuleSoft?

There are multiple ways to improve the performance

-
- Try to save the result and reuse them later
- If processing is expensive do it only once and save result and use them later
- Try to put the validation at start of the flow
- Use Streaming to process data
- Try to process the data asynchronously wherever is possible

How can you optimize memory efficiency of the Code in MuleSoft?

-
- Do not save the Payload in to the flow Variable as it is more memory consuming element
- Do not load unnecessary part of the document
- Use better Database polling for highly concurrent scenarios

What is distributed file polling?

In mule 4 distributed file polling is used to poll files in cluster nodes

There are few connectors where this feature is enabled by default

-
- [File](#) Connector
- FTP Connector
- [SFTP](#) Connector

You can configure connectors to poll only from primary node @primaryNode Only which ignores the default setting set by mule runtime engine

How will you implement caching in MuleSoft?

There are 2 ways to implement the caching in MuleSoft

1.
 - Use Object Store [– more information](#)
 - Use Cache scope to cache the Response – [more information](#)

If we wanted to process only one message at a time in mule flow how can we achieve that?

we can make a flow to process in single thread, there is an option to define the Max Concurrency you can configure the same as 1 to process only one message at a time

What is Anypoint Platform?

- Anypoint platform is highly productive, hybrid integration platform that creates an application network of app, data and devices by producing and consuming reusable assets.
- It is collection of Tools and Framework (building application) and runtime for running application

What are Anypoint Connectors in MuleSoft?

Anypoint connectors are used to integrate with SaaS and Cloud API e.g. Salesforce, AWS etc. also you can connect your on-premise applications using connectors, there are various connectors available like Database, Email, File, Salesforce etc.

What is API Portal in MuleSoft?

API Portal allows providers to expose and publicize their APIs, educate developers communities about them, provision user access generate client keys and more

What is API Notebook in MuleSoft?

API Notebook is web based tool for building interactive tutorials and examples in a Java Script workspace

What are deployment options available in MuleSoft?

- Cloud hub
- On-premise
- Runtime Fabric(RTF)
- Anypoint PCE(Anypoint Private Cloud Edition)
- Anypoint PCF(Anypoint Platform for Pivotal Cloud Foundry)

What is Cloudhub?

Cloudhub is an (iPasS) Integrated platform as a service which is multi-tenant, secure, highly available service where we can deploy our integration application on cloud also integrate on-premise application with cloud services

How many ways you can deploy application on Cloudhub?

- [Anypoint Studio](#)
- [Runtime Manager](#)
- Anypoint CLI
- Cloudhub API

What is PCE (Platform Cloud Edition in MuleSoft)?

- With Anypoint Platform Private Cloud Edition (Anypoint Platform PCE), you run and manage Mule applications on your local servers
- Anypoint Platform PCE uses Docker and Kubernetes to provide built-in high availability and scalability.
- You can install the platform without understanding Docker or Kubernetes

What is RTF (Run time Fabric)?

Anypoint Runtime Fabric is a container service that automates the deployment and orchestration of your Mule applications Runtime Fabric runs on customer-managed infrastructure on AWS, Azure, virtual machines (VMs)

Features of RTF

-
- You can scale your mule application
- There is option to run multiple version of mule runtime on same set of resources
- Isolation between application by running a separate Mule runtime per application

Why we use Mule Maven Plugin?

The [Mule Maven plugin](#) allows you to integrate the packaging and deployment of your Mule applications with your Maven lifecycle.

By using mule maven plugin you can automate the deployment of mule application on Cloudhub, on-premise or on RTF

What is the difference between op-premise deployment and cloud hub deployment strategy?

- With Cloud hub deployment strategy you can deploy maximum 10 Application on 1 vCore machine but in on-premise you can deploy 100 application on 1 vCore depends on the
- In Cloudhub mule runtime will be managed by the MuleSoft, in on-premise mule runtime will be managed by the Customer itself

What is persistent queue?

Persistent queues ensure zero message loss and allow you to distribute non-HTTP workloads across a set of workers, Persistent queues also guarantee delivery of your messages; even if one or more workers or data-centers go down, persistent queues facilitate disaster recovery and provide resilience to hardware or application failures.

Why we used Anypoint VPN?

We use Anypoint VPN to create a secure connection between MuleSoft VPC and on-premise network

What is static IP and in what scenarios static IP gets changed?

Static IP is the IP Address of the worker displayed on the cloud hub console; by default you are allocated a number of static IP equal to twice the number of production vCores. Static IP can be changed in below scenarios

-
- Deleting an application will also remove its Static IP. If you create a new application with the same name, it will have a new dynamically assigned IP address.
- Adding a new application by moving it from Dev to QA. This requires a new application name and a new Static IP is generated
- When application is re-deployed to a different geographic region

In which time zone MuleSoft scheduler runs?

Mulesoft Schedulers always run in UTC/GMT timezone

What happens to your logs when you delete mule application from Cloudhub?

After you delete your application your log data is no longer accessible through the console, Cloudhub archive old log data for a limited period of time before being purged, if you want to recover data you have to raise support ticket with MuleSoft.

What is object store V2?

- [Object Store v2](#) lets Cloudhub applications store data and states across batch processes, Mule components and applications, from within an application.
- Persists keys for 30 days unless updated. If a key is updated, the TTL (time-to-live) is extended by another 30 days from the current expiration date
- Allows for an unlimited number of entries. There is no limit on the total size of v2 object-stores.
- Stores values up to 10 MB

How can one mule app access the object store of another mule app?

To share data between Mule 4 apps create a flow to retrieve the data using the Retrieve operation in the Object Store Connector

In Mule 3, use the [Object Store v2 REST API](#) to read from one app and store to another.

What is Anypoint MQ?

- Anypoint MQ is a multi-tenant, cloud messaging service that offers customers to perform asynchronous messaging scenarios between their applications.
- Anypoint MQ is fully integrated with Anypoint Platform
- Anypoint MQ requires a separate Enterprise subscription

How MuleSoft scheduler runs in multi worker cloud hub environment

Mule runtime cluster or multi-worker Cloudhub deployment, the Scheduler runs (or triggers) only on the primary node (that is, only in one Mule runtime).

How you can prevent your application hosted in Cloudhub from being publicly accessible

Remove the Anypoint VPC Firewall rules with source Anywhere (0.0.0.0/0) and port 8081/82 or create a whitelist in your dedicated load balancer with the IP Address you want to authorize.

What is Control Plane?

In Anypoint Platform, the control plane consists of Anypoint Design Center, Anypoint Management Center, and Anypoint Exchange, API Manager Etc. Control plane is used to deploy, monitor and manage the running application in Mule runtime

What is Anypoint runtime Plane?

Anypoint runtime plane is used where we deploy the mule runtime application also where the mule runtime engine runs, in case of Cloudhub runtime plane will be managed by MuleSoft and for on-premise and RTF Runtime plane will be managed by Customer

What is EU Control Plane?

MuleSoft hosts a version of the Anypoint Platform control plane entirely within the EU (Frankfurt or Dublin data centers). This enables you to deploy and manage your APIs, applications, and related data entirely within the European Union. This includes data and metadata about your MuleSoft implementation. The EU control plane ensures that all MuleSoft data is stored within the EU

What is Anypoint Visualizer?

- Anypoint Visualizer displays views of different aspects of an application network graph.
- Anypoint Visualizer provides a real-time, graphical representation of the APIs and mule application that are running.
- It also displays third party system that are invoked by a mule API, proxy or application within your application network
- Anypoint Visualizer collects data from Mule applications, APIs, and proxies deployed to Cloudhub; from standalone Mule runtime engines (Mule instances); or from Runtime Fabric, to discover all incoming and outgoing connections through an embedded plugin.

What is Anypoint VPC?

The Anypoint Virtual Private Cloud (VPC) offering allows you to create a virtual, private, and isolated network segment in the cloud to host your Cloudhub workers

Connecting to your Anypoint VPC extends your corporate network and allows Cloudhub workers to access resources behind your corporate firewall. You can connect on-premises data centers through a secured VPN tunnel, or a private AWS VPC through VPC peering, or by using AWS Direct Connect.

What are the methods to connect on-premise network with Anypoint VPC

You can use an IPsec tunnel with network-to-network configuration to connect your on-premises data centers to your Anypoint VPC. An IPsec VPN tunnel is generally the recommended solution for VPC to on-premises connectivity

What is VPC Peering?

VPC peering provides connectivity between two VPC, this enables you to route traffic between 2 VPCs so that they can communicate as though they are in same network.

To use VPC peering your AWS and Anypoint VPCs must be located in the same AWS region

Why we use Shared Load Balancer (SLB)?

Shared load balancer is a default load balancer provides basic load balancing functionality, with Shared load balancer you cannot configure customer SSL certificate proxy rule. Shared load balancer has lower rate limits

Why we used Dedicated Load Balancer?

Cloudhub dedicated load balancers (DLBs) are component of Anypoint Platform that enable you to route external HTTP and HTTPS traffic to multiple Mule applications deployed to Cloudhub workers in a Virtual Private Cloud (VPC).

Features of DLB:

-
- It handles the load balancing among different cloud hub workers

- You can also configure the custom SSL certificate and enforce 2-way SSL client authentication

What are the prerequisite to create DLB?

- Your profile is authorized to perform this action by adding the **Cloudhub Network Administrator** permission to the profile of the organization where you are creating the load balancer.
- Create an Anypoint Virtual Private Cloud (Anypoint VPC) in the organization where you want to create a load balancer.
- Create at least one certificate and private key for your certificate.

How many ways you can configure the Dedicated Load Balancer

There are three ways to create and configure a dedicated load balancer for your Anypoint VPC

- [Runtime Manager](#) from Anypoint Platform
- The [Cloudhub load-balancer create](#) command from the Anypoint Platform CLI
- The [Cloudhub API](#)

How can you achieve the High Availability in MuleSoft?

- HA is to keep overall system operational when a system component fails, usually it is achieved with multiple level of fault tolerance or load balancing.
- In case of on-premise you can deploy the mule application on multiple Clusters
- In case of Cloudhub you can deploy the mule application on multiple cloud hub workers i.e horizontal scaling to multiple mule run time

What is API Manager?

- API Manager is a component of Anypoint Platform for managing APIs.
- API Manager manages APIs that reside in Exchange
- API Manger is used to apply the policies on APIs

Please visit [this url](#) for more information

What is API Autodiscovery?

- [API Autodiscovery](#) is used to pair the API in API Manger to it deployed mule application
- When autodiscovery is correctly configured in your Mule application you can say that your application's API is *tracked by or paired to* API Manager.
- An API in a Mule runtime can only be associated with one autodiscovery instance. No two Autodiscoveries can be associated to the same API in a Mule Runtime.

Please visit [this url](#) to for more information

What is API Gateway?

API Gateway is embedded inside mule runtime, using this gateway any user can apply basic authentication policy on top of Mule application, enrich and incoming outgoing message or any other complex capability to an API without writing code. Basically it is an orchestration layer on top of your back end API to help you separate orchestration from implementation.

What are API security policies are available in MuleSoft?

- [XML](#) or [JSON threat](#) protection – this will protect against the oversized XML or JSON payload
- [Client ID enforcement](#) – authentication is need for proper use of an API, only client authorized can use the API and no one else
- [SLA-based Rate Limiting](#) – this is more need in case we want to monetize an API otherwise ignored e.g.
 - Free – 20 request per minute
 - Unlimited – 100K request per minute

- [IP blacklisting](#) – this can be used if we want to limit the consumption of an API to particular IPs e.g. IPs know for hacking etc.
- Tokenization – to tokenize any element which can be sensitive e.g. credit cards etc.
- [HTTP Caching](#) – in case we think the response don't change frequently and it's ok to send the same response for particular request
- Cross-Origin resource sharing – is a standard mechanism that allows JavaScript XMLHttpRequest (XHR) calls executed in a web page to interact with resources from non-origin domains. CORS is a commonly implemented solution to the "same-origin policy" that is enforced by all browsers.
- [Spike Control API](#) – it will protect the system against burst requests
- [Header removal](#) or [injection](#) – add remove HTTP header
- [Message logging](#) – Logs custom messages between policies and flow. the payload will be consumed by the policy if it's a non-repeatable streams

What is the difference between Throttling and Rate Limiting?

Rate Limiting and Throttling are the API Policies available in MuleSoft, you can apply these policies through API Manager.

Rate Limiting: The Rate Limiting policy limits the number of requests an API accepts within a window of time. The API rejects requests that exceed the limit

Throttling: The Throttling policy queues requests that exceed limits for possible processing in a subsequent window. The API eventually rejects the request if processing cannot occur after a certain number of attempts. You can configure a delay between retries, as well as limit the number of retries

What is RAML?

- RAML stands for Restful API Modelling Language,
- RAML is a well-defined contract in a human readable format to actually exist as your source code
- it is open-specification language built on [YAML 1.2](#) and JSON for describing Restful APIs.
- Restful API Modelling Language ([RAML](#)) makes it easy to manage the [API lifecycle](#) from design to deployment to sharing. It's concise and reusable; you only have to write what you need to define and you can use it again and again.

What are API Fragments in RAML

[API Fragment](#) is part of RAML specification; it is not in itself a complete RAML specification. These are the small reusable component used in API specification. You can create fragment and publish it to exchange to make it reuse in the organization. Example of Fragments: Library, Data Type etc.

Please visit [this url](#) to get more information

Why we use Design Center

- Design center can be used to write API definitions with RAML using API Designer
- You can also create API Fragment in RAML, after you create a specification or fragment you can publish it to Anypoint Exchange so that it can be reused in the organization
- Design center can also be used to create mule applications using flow designer

What is runtime manager?

- Runtime manager is to deploy and manage mule application on mule runtime engine, where mule runtime is running on Cloudhub or on-premise or on RTF.
- By using runtime manager you can deploy/undeploy the mule app
- You can change the runtime version at any time
- You can increase/decrease the worker size

What is vCore?

It is unit of compute capacity for processing on Cloudhub. In 1 vCore maximum 10 applications can be deployed where 0.1 vCore will be consumed by each mule application

What is worker in MuleSoft?

Worker is an dedicated Instance of Mule hosted on AWS is used to run your integration application

Features of workers:

-
- Capacity: Each worker has a specific amount of capacity to process data.
- Isolation: Each worker run in a separate container from every other application
- Manageability: Each worker is deployed and Managed independently
- Locality: Each worker runs in specific worker cloud such as US,EU or Asia-Pacific

How many mule applications you can deployed in one worker in Cloudhub

Only one Mule Application can be deployed in MuleSoft in one worker

What is minimum and maximum Worker size available in Cloudhub?

0.1 vCores is the Minimum and 16 vCores is the maximum worker size available in Cloudhub

What is horizontal scaling and vertical scaling?

Vertical scaling is to increase the worker size, when you want to process the CPU Intensive API's or process large payload with small number of request increase the vCore size.

Horizontal scaling is to increase the number of worker, if you want to increase the throughput of high frequency small payload

How to achieve clustering in MuleSoft

You can achieve this by adding multiple workers to your application to make it horizontally scale, Cloudhub automatically distribute multiple workers for same application across 2 or more data centres for maximum reliability When deploying your application to two or more workers, the HTTP load balancing service distributes requests across these workers, allowing you to scale your services horizontally. Requests are distributed on a round-robin basis.

Can you disable the Cloudhub Logs?

Yes, in that case only the system logs are available in Runtime Manager. System logs provide the status of your worker deployment and whether your application started correctly, but do not provide application logs

What is Mule Runtime?

- A Mule runtime is a runtime engine used to host and run Mule applications – Similar to an application server
- Mule runtimes can be provisioned on-premises and/or in the cloud
- One Mule runtime can host several Mule applications
- **How to invoke the Custom Java Class from inside a Dataweave?**
- Below is the example

```
%dw 2.0
Import java!com.mycompany::classname
output application/json
---
{
  a: classname::methodName("myString")
}
```

- More info – <https://mulesy.com/invoke-custom-java-class-using-dataweave>
-
- **How to access secure property inside Dataweave?**
- By using `p('secure::<property name>')` you can access the secure property defined as part of secure property file in mule application
- More info – <https://mulesy.com/read-properties-in-mule-4>

-
- **How to call Flow from inside a Dataweave?**
- You can use Dataweave lookup function to call flow from inside a Dataweave It works in Mule apps that are running on Mule Runtime version 4.1.4 and later. It takes the flow's name and an input payload as parameters. For example, **lookup("flowname", payload)**
-
- **How to Log message in Dataweave?**
- Use **log(" Current Time Stamp ",now())** function in Dataweave
- More info – <https://mulesy.com/logging-inside-dataweave-transformation>
-
- **How to use comment line in Dataweave?**

```
// single line comment.
```

```
/*
 * This is.
 * multi-line.
 * comments in.
 * DataWeave.
 */
```

-
- **What is the latest version available for Dataweave?**
- Dataweave latest version is 4.3, please see the new features in latest Dataweave version <https://mulesy.com/latest-features-dataweave-2-3-0>
-
- **How to use map operator in Dataweave?**
- Please see – <https://mulesy.com/map>
-
- **How to use filters in Dataweave?**
- Please see – <https://mulesy.com/filter-in-dataweave-mule-4>
-
- **How to use the Custom Dataweave functions in Mule?**
- Please see – <https://mulesy.com/custom-dwl-function-using-custom-module>
-
- **How you can define a variable in Dataweave?**
- Please see – <https://mulesy.com/variables-in-dataweave>
-
- **How to convert an Object into Array using Data weave?**
- Use Pluck operator which iterate over an Object and return an array of keys, values, or indices from the object. More info – <https://docs.mulesoft.com/mule-runtime/4.3/dw-core-functions-pluck>
-
- **How you will parse an XML with or without namespace in Dataweave? How you will retrieve the attribute value**
- Please see – <https://mulesy.com/xml-processing-in-dataweave>
-
- **Difference between map and mapObject?**
- **map** – Iterates over items in an array and outputs the results into a new array.
- **mapObject** – Iterates over an object using a mapper that acts on keys, values, or indices of that object.
- From the functionality wise both are same but both work on different input and return different output, where array and object are Java array and Java object.
-
- **How you can merge two Arrays in Dataweave?**
- Use flatten operator to merge two array into one – more info – <https://docs.mulesoft.com/mule-runtime/4.3/dw-core-functions-flatten>
-
- **How you will convert string into Array and vice versa?**
- **String to Array** – <https://mulesy.com/convert-string-to-array-in-dataweave-2-0>

- **Array to String** – <https://mulesy.com/convert-array-to-string>
 -
 - **What is filter and how we will use it in Dataweave?**
 - Please see – <https://mulesy.com/filter-in-dataweave-mule-4>
 -
 - **How you will do null check in Dataweave?**
 - Please see – <https://mulesy.com/perform-null-check-in-mule-4>
-

Q: What is Mule ?

A: Mule ESB (a.k.a. Mule) is a lightweight Java-based enterprise service bus (ESB) and integration platform that allows developers to connect applications together quickly and easily, enabling them to exchange data. Mule ESB enables easy integration of existing systems, regardless of the different technologies that the applications use, including JMS, Web Services, JDBC, HTTP, and more.

Q: What is ESB ?

A: An enterprise service bus (ESB) is software architecture for middleware that provides fundamental services for more complex architectures. For example, an ESB incorporates the features required to implement a service-oriented architecture (SOA). In a general sense, an ESB can be thought of as a mechanism that manages access to applications and services (especially legacy versions) to present a single, simple, and consistent interface to end-users via Web- or forms-based client-side front ends.

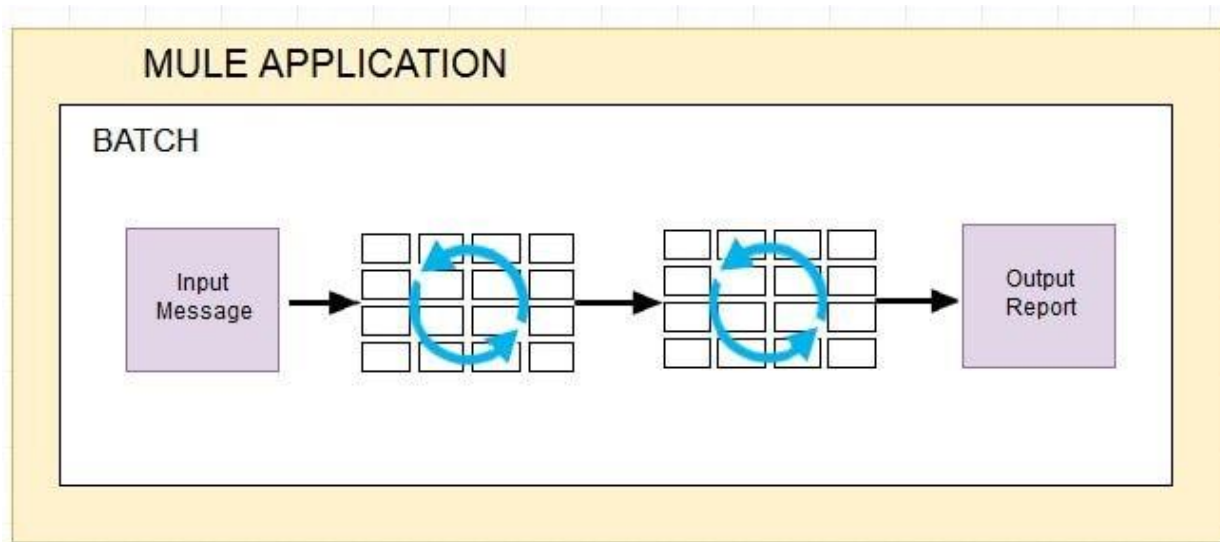
Q: What is Shared Context ?

A: Shared Context: Context is a temporary area which is created along with Service Message Object (SMO) in the Mediation Flows. Shared Context is a type of context which is present in the SMO. Shared Context is mainly used when we are using Aggregation process where we need to Iterate the BO for Certain times. Shared Context maintains Aggregation data between Aggregation (FanOut and FanIn) primitives. The Content (data) which is present in the shared context BO does not persist across Request and Response flows i.e The Data in the Shared Context which is used in Request flow can not be used again in Response flow.

Q: What are Batch Jobs in Mule ESB ?

A: A batch job is a top-level element in Mule which exists outside all Mule flows. Batch jobs split large messages into records which Mule processes asynchronously in a batch job; just as flows process messages, batch jobs process records.

A batch job contains one or more batch steps which, in turn, contain any number of message processors that act upon records as they move through the batch job. During batch processing, you can use record-level variables (recordVars) and MEL expressions to enrich, route or otherwise act upon records.



Q: What is Transient Context ?

A:Transient Context: Used for passing values between Mediation primitives within the current flow – either the request flow or the responses flow. The transient context cannot link requests and responses and hence cannot be used across. Used when you want to save an input message before a service invokes call (within a request or response flow). After the services invoke call, the next primitive can create another message by combining the service invoke response and the original message stored in the transient context.

Q: What are the different endpoints available in Mule ESB ?

A:Different types of Endpoints are available in Mule, such as HTTP, JMS, IMAP, SMTP, and AJAX.

Q: How to check if file exists in Mule ESB ?

A:This can be done using Mule Expression Language(MEL) as follows-

```
<choice>
  <when expression="#[xmlFileExists()]">
    ...
  </when>
  <otherwise>
    ...
  </otherwise>
</choice>
```

Q: What are Inbound endpoints and Outbound endpoints in Mule ESB ?

A:**Inbound endpoints** are message sources

(<http://www.mulesoft.org/documentation/display/current/Message+Sources>), which as the name suggests is where messages are created. They can be created based on external events (like an incoming HTTP request or JMS message) or by polling (like files in a directory).

Outbound endpoints and anything else you see in a flow (except exception strategies) are message processors (<http://www.mulesoft.org/documentation/display/current/Message+Processors>), which means they do something with the message in-flight the flow.

Q: What is difference between Mule Connectors and Transports Mule ESB ?

A: Transports are targeted towards a way of transporting data, i.e. a protocol like HTTP or reading/writing files. These are general concepts and the other party behind such a data channel can be anything, a pure data sink or a party with whom data can be exchanged, own company or other.

Connectors are made for using specific APIs, e.g. those from salesforce.com or facebook. Usually, choosing a connector also determines how the data will be transferred in the end, e.g. HTTP.

1. Question 1. What Is Mule ?

Answer :

Mule is a lightweight event-driven enterprise service bus (ESB) and an integration platform. It is a lightweight and modular solution that could scale from an application-level messaging framework to an enterprise-wide highly distributable object broker.

2. Question 2. What Difficulties Mule Does Encompass ?

Answer :

- **Transport:** applications can accept input from a variety of means, from the file system to the network.
- **Data format:** speaking the right protocol is only part of the solution, as applications can use almost any form of representation for the data they exchange.
- **Invocation styles:** synchronous, asynchronous, or batch call semantics entail very different integration strategies.
- **Lifecycles:** applications of different origins that serve varied purposes tend to have disparate development, maintenance, and operational lifecycles.

3. Question 3. Why Mule Was Designed ?

Answer :

Mule's core was designed as an event-driven framework combined with a unified representation of messages, expandable with pluggable modules. These modules would provide support for a wide range of transports or add extra features, such as distributed transactions, security, or management. Mule was also designed as a programmatic framework offering programmers the means to graft additional behavior such as specific message processing or custom data transformation.

4. Question 4. Why The Name Mule ?

Answer :

There is a lot of infrastructure work to be done before we can really start thinking about implementing any logic. So this infrastructure work is regarded as "donkey work" as it needs doing for every project. A Mule is also commonly referred to as a carrier of load, moving it from one place to another. The load it specializes in moving is our enterprise information.

5. Question 5. What Are Available Esbs Apart From Mule ?

Answer :

All major JEE vendors (BEA, IBM, Oracle, Sun) have an ESB in their catalog. It is unremarkably based on their middleware technologies and is usually at the core of a much broader SOA product suite. There are also some commercial ESBs that have been built by vendors not in the field of JEE application servers, like the ones from Progress Software, IONA Technologies, and Software AG.

6. Question 6. What Are Differences Between Mule And Other Commercial Esbs ?

Answer :

Prescriptive deployment model, whereas Mule supports a wide variety of deployment strategies. Prescriptive SOA methodology, whereas Mule can embrace the architectural style and SOA practices in place where it is deployed. Mainly focused on higher-level concerns, whereas Mule deals extensively with all the details of integration. Strict full-stack web service orientation, whereas Mule's capacities as an integration framework open it to all sorts of other protocols. Comprehensive documentation, a subject on which MuleSource has made huge progress recently.

7. Question 7. What Is Model Layer In Mule ?

Answer :

The first logical layer is the model layer. A Mule model represents the runtime environment that hosts services. It defines the behavior of Mule when processing requests handled by services. The model provides services with supporting features, such as exception strategies. It also provides services with default values that simplify their configuration.

8. Question 8. What Is Service Layer In Mule ?

Answer :

A Mule service is composed of all the Mule entities involved in processing particular requests in predefined manners. A service is defined by a specific configuration. This configuration determines the different elements, from the different layers of responsibility, that will be mobilized to process the requests that it will be open to receive. Depending on the type of input channel it uses, a service may or may not be publicly accessible outside of the ESB.

9. Question 9. What Is Transport Layer In Mule ?

Answer :

The transport layer is in charge of receiving or sending messages. This is why it is involved with both inbound and outbound communications. A transport manifests itself in the configuration by the following elements: connectors, endpoints and transformers.

A transport also defines one message adapter. A message adapter is responsible for extracting all the information available in a particular request (data, meta information, attachments, and so on) and storing them in transport-agnostic fashion in a Mule message.

10. Question 10. What Is Connector In Mule ?

Answer :

A connector is in charge of controlling the usage of a particular protocol. It is configured with parameters that are specific to this protocol and holds any state that can be shared with the underlying entities in charge of the actual communications.

For example: a JMS connector is configured with a Connection, which is shared by the different entities in charge of the actual communication.

11. Question 11. What Is Endpoint In Mule ?

Answer :

An endpoint represents the specific usage of a protocol, whether it is for listening/polling, reading from, or writing to a particular target destination. Hence it controls what underlying entities will be used with the connector they depend on. The target destination itself is defined as a URI. Depending on the connector, the URI will bear a different meaning; for example, it can represent a URL or a JMS destination.

12. Question 12. What Is Transformer In Mule ?

Answer :

A transformer takes care of translating the content of a message from one form to another. It is possible to chain transformers to cumulate their effects. Transformers can kick in at different stages while a message transits through a service.

13. Question 13. What Is Router In Mule ?

Answer :

Routers play a crucial role in controlling the trajectory a message will follow when it transits in Mule. They are the gatekeepers of the endpoints of a service, taking care of keeping messages on the right succession of tracks so they can reach their intended destinations. Certain routers act like the big classification yards: they can split, sort, or regroup messages based on certain conditions.

14. Question 14. What Is Filter In Mule ?

Answer :

Filters are a powerful complement to the routers. Filters provide the brains routers need to make smart decisions about what to do with messages in transit. Some filters go as far as deeply analyzing the content of a message for a particular value on which their outcome will be based.

15. Question 15. What Is Component In Mule ?

Answer :

Components are the centerpiece of Mule's services. Each service is organized with a component at its core and inbound and outbound routers around it. Components are used to implement a specific behavior in a service. This behavior can be as simple as logging messages or can go as far as invoking other services. Components can also have no behavior at all; in that case they are pass-through and make the service act as a bridge between its inbound and outbound routers.

16. Question 16. How Message In Mule Is Composed ?

Answer :

A Mule message is composed of different parts: The payload, which is the main data content carried by the message.

The properties, which contain the meta information much like the header of a SOAP envelope or the properties of a JMS message. Optionally, multiple named attachments, to support the notion of multipart messages.

Optionally, an exception payload, which holds any error that occurred during the processing of the event.

17. Question 17. What Are Configuration Builders In Mule ?

Answer :

Mule uses configuration builders that can translate a human-authored configuration file into the complex graph of objects that constitutes a running node of this ESB. The main builders are of two kinds: a Spring-driven builder, which works with XML files, and a script builder, which can accept scripting language files.

18. Question 18. Why Spring-driven Configuration Builder Is Important Than Script Builder ?

Answer :

The advantages of Spring-driven configuration builder

- It is the most popular — you are more likely to find examples using this syntax.
- It is the most user friendly — Spring takes care of wiring together all the moving parts of the ESB, something you must do by hand with a script builder.
- It is the most expressive — dedicated XML schemas define the domain-specific language of Mule, allowing you to handle higher-level concepts than the scripting approach does.

19. Question 19. What Is Bridge Component In Mule ?

Answer :

A bridge component is used to pass the messages from the inbound router to the outbound one. A bridge is a neutral component: it does not perform any action or modify messages that it processes.

20. Question 20. What Tags Are Used To Configure Spring Elements In Mule ?

Answer :

Tags like `<spring:bean/>` `<spring:entry/>` etc. are used to configure spring stuff.

21. Question 21. What Are Available Approaches Used For Modularizing Configurations In Mule ?

Answer :

There are different following approaches that can be used when modularizing a configuration.

- **Independent configurations** – a Mule instance can load several independent configuration files side by side.
- **Inherited configurations** – main idea is to express a formal parent-child dependency between two configurations. By strongly expressing this dependency, you will have the guarantee at boot time that no configuration file has been omitted. Simply by using the same name for the parent and child models and by flagging the child as being an heir, as shown here:
 - `<model name="myConfig">`
 - `<model name="myConfig" inherit="true">`
- **Imported configurations** – You can easily import external Spring application context files into your Mule configuration files. The following illustrates how instance.xml would import its Spring context file:
 - `<spring:beans>`
 - `<spring:import resource="instance-beans.xml" />`
 - `</spring:beans>`

- **Heterogeneous configurations** – It is possible to mix several styles of Mule configuration in an instance. An instance can be configured with a Groovy script and Spring XML configuration builders.

22. Question 22. Give An Example Of Stdio Connector In Mule ?

Answer :

```
<stdio:connector name="SystemStreamConnector" promptMessage="Please enter something: "
messageDelayTime="1000"/>
```

23. Question 23. Give An Example Of Http Connector In Mule ?

Answer :

```
<http:connector name="HttpConnector" proxyHostname="proxyHostname"
proxyPort="proxyPort"
proxyUsername="proxyUsername" proxyPassword="proxyPassword"/>
```

24. Question 24. When Does Mule Instantiates A Connector ?

Answer :

If Mule figures out that one of our endpoints needs a particular connector, it will automatically instantiate one for us, using all the default values for its different configuration parameters. This is a perfectly viable approach if we are satisfied with the behavior of the connector when it uses its default configuration. This is often the case for the VM or HTTP transports. Note that Mule will name these default connectors with monikers such as connector.http.0.

25. Question 25. What Is Transport Service Descriptor In Mule ?

Answer :

The connector has a technical configuration known as the Transport Service Descriptor (TSD). This hidden configuration is automatically used for each instance of the connector. It defines technical parameters such as what classes to use for the message receivers, requesters, and dispatchers; or the default transformers to use in inbound, outbound, and response routers. Knowing these default values is essential to grasping the behavior of a transport.

26. Question 26. How Many Endpoints Are There In Mule ?

Answer :

There are two endpoints :

- inbound
- outbound.

You will use inbound and outbound endpoints to communicate between components and services inside Mule as well as with the outside world.

27. Question 27. What Is An Outbound Endpoint In Mule ?

Answer :

Outbound endpoints are used to send data. An outbound endpoint is used to do things such as send SOAP messages, write to file streams, and send email messages.

28. Question 28. What Is Global Endpoint In Mule ?

Answer :

An endpoint destination that is shared by several routers, it is worth creating a global endpoint. A global endpoint is not typified for inbound or outbound routing, making it usable in many different places in a configuration file. It must be named so it can actually be used in a service, which will reference the global endpoint by its name. A global endpoint can also help clarify the usage of a particular destination.

29. Question 29. Why Does An Endpoint In Mule Offer An Address Attribute ?

Answer :

This allows us to configure a generic endpoint using the Mule 1.x style of URI-based destination addresses instead of the dedicated attributes of the specific endpoint element.

30. Question 30. Give An Example Of File Endpoint In Mule ?

Answer :

```
<file:endpoint name="tmpPoller" path="/tmp" fileAge="1000" pollingFrequency="2000"/>
```

31. Question 31. What Is Streaming Property In File Connector In Mule ?

Answer :

The value of this streaming property can be either true or false. If it is set to true then we are actually working on stream of file data otherwise we are working with file itself.

32. Question 32. What Is Pollingfrequency Property In File Connector In Mule ?

Answer :

When we want file inbound endpoints to poll their source directories for new content. This is accomplished by setting the pollingFrequency to some milliseconds value.

33. Question 33. What Is Autodelete Property In File Connector In Mule ?

Answer :

The default value of autoDelete is true. Therefore, a file inbound endpoint will, by default, remove the file from the source directory once it is read by the inbound endpoint. If you do not want to delete file automatically then you can set autoDelete property to false.

34. Question 34. What Is Fileage Property In File Connector In Mule ?

Answer :

The fileAge property specifies how long the endpoint should wait before reading the file again. For instance, a fileAge of 60000 indicates Mule should wait a minute before processing the file again.

35. Question 35. How To Send Only Certain Types Of File From One Directory To Another In Mule ?

Answer :

Use the below element in file inbound to filter certain types of files.

`<file:filename-wildcard-filter pattern="file*.xml"/></file:inbound-endpoint>` pattern indicates what pattern of file names should move from one directory to another directory.

36. Question 36. What Is Vm Transport In Mule ?

Answer :

The VM transport is a special kind of transport that you'll use to send messages via memory. These messages never leave the JVM the Mule instance is running in.

37. Question 37. What Is Multicasting Router In Mule ?

Answer :

The multicasting router can send messages to multiple endpoints over different transports. The multicasting router allows you to easily move the same messages across these different endpoints.

38. Question 38. What Is Mule Transformer ?

Answer :

It is an event, more specifically an instance of `org.mule.api.MuleEvent`. This object carries not only the actual content of the message but also the context of the event.

39. Question 39. What Is Mule Context ?

Answer :

The Mule context is composed of references to different objects, including security credentials, if any, the session in which the request is processed. All internals of the ESB are accessible through Mule context

40. Question 40. What Is Payload In Mule ?

Answer :

The content of a message, also known as payload. It is wrapped in an instance of `org.mule.api.MuleMessage`, which provides different means of accessing the payload under different forms. A `MuleMessage` also contains properties, much like the header of a SOAP envelope or the properties of a JMS message, and can also have multiple named attachments.

41. Question 41. What Are Different Type Of Messages In Mule ?

Answer :

- **Bridge messages:** Pass messages from inbound to outbound routers.
- **Echo and log messages:** Log messages and move them from inbound to outbound routers.
- **Build messages:** Create messages from fixed or dynamic values

42. Question 42. Do I Need An Esb?

Answer :

Mule and other ESBs offer real value in scenarios where there are at least a few integration points or at least 3 applications to integrate. They are also well suited to scenarios where loose coupling, scalability and robustness are required.

43. Question 43. Why Mule Esb?

Answer :

Mule ESB is lightweight but highly scalable, allowing you to start small and connect more applications over time. Mule manages all the interactions between applications and components transparently, regardless of whether they exist in the same virtual machine or over the Internet, and regardless of the underlying transport protocol used.

There are currently several commercial ESB implementations on the market. However, many of these provide limited functionality or are built on top of an existing application server or messaging server, locking you into that specific vendor. Mule is vendor-neutral, so different vendor implementations can plug in to it. You are never locked in to a specific vendor when you use Mule.

What are inbound and Outbound properties ?

Inbound properties are *immutable*, are automatically generated by the message source and cannot be set or manipulated by the user. They contain metadata specific to the message source. A message retains its inbound properties only for the duration of the flow; when a message passes out of a flow, its inbound properties do not follow it.

What are Web Services?

Web service is a function or program in any language that can be accessed over HTTP. Message format can be XML or JSON or any other program as long as the other programs can understand and communicate. Web services can be synchronous or asynchronous. Any web service has server-client relationship. Any web service can have multiple clients. Eg: When a travel portal is selling tickets of an airliner, Portal is client and the Airline is the server as it is selling its service.

How to use RAML in anypoint studio?

RAML can be imported while creating a new application from scratch either from local machine or from Anypoint Design Center. You can also create flows and error handlers using APIKit module. RAML can also be imported into existing applications by right clicking on the application and selecting "Anypoint Platform" -> "Import from Design Center" (in Anypoint Studio 7.* and Mule 4.*).

2. What are Traits in RAML?

Traits are functions in RAML which defines common properties for HTTP methods, can be declared once and used at multiple places using keyword "is". traits should be defined in a separate file and import it into the main RAML to follow Best practices.

3. Explain Request/Response lifecycle in mule based on RAML?

APIKit router plays a key role in mapping the resources(RAML) and Mule flows. Any Request entering into mule app first hits APIKit router and the request will be validated according to the RAML description.

Success Requests will be mapped to their corresponding flow and Bad requests will be mapped to their corresponding error flows and respond back with appropriate HTTP Status code.

4. What are the parameters defined for methods in RAML?

URI parameters and Query parameters can be defined in RAML.

URI Parameters: Unique Resource Identifier as the name suggests, it should get a unique resource. It is sent as a part of the request with a unique id each time. They are defined by using keyword "uriParameters".

Query Parameters: The question mark, the parameter, and its real value make what is referred to as the query string. It has two distinguishing features from the hierarchy parameters: They are optional. They are non-unique, in the sense that you can specify any one parameter multiple times. They are defined by using keyword "queryParameters".

5. Difference between POST and PUT?

POST is used while Inserting or creating a new record in the Database or System of record whereas PUT is used while updating an existing record. Both methods will support URI and Query parameters.

What is RAML and why we use it?

RAML – *RESTful API Modeling Language*

RAML is similar to WSDL, it contains endpoint URL, request/response schema, HTTP methods and query and URI parameters.

RAML helps client (a consumer of the service) know, what the service is and what/how all operations can be invoked.

RAML helps the developer in creating the initial structure of this API. RAML can also be used for documentation purpose.

1. What is MuleSoft?

MuleSoft is a cloud-based solution provider that integrates with Data, Applications, and APIs on-premises over the cloud platform. It works on "AnyPoint Connectivity Model" which connects existing SaaS-based applications or set of APIs through one single API interface. It is flexible in accessing all the required applications with this integration support that is based on service-oriented architecture. MuleSoft runs the programmable web portal for building web, mobile and other user applications.

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2. What is ESB?

The Mule ESB is an integration platform based on ESB Java that enables the developers to quickly connect the applications and exchange the data. It integrates with the existing system irrespective of the technologies which the applications use such as Web Services, HTTP, JDBC, JMS etc

3. What are all the Primitives used in Mediation?

There are many different types of primitives in mediation.

- Message Filter
- Type Filter
- Endpoint Lookup
- Service Invoke
- Fan-out
- Fan-in
- XSLT
- BO Map
- Message Element Setter

- DB lookup
- Data Handler
- Custom Mediation
- Header Setters
- Message Logger
- Even Emitter
- Stop
- Fail
- Sub Flow

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4. Name the types of Exception Handling in MuleSoft?

There are five types of Exception Handling in MuleSoft.

- Global Exception Handling.
- Catch Exception Handling.
- Choice Exception Handling.
- Default Exception Handling.
- Rollback Exception Handling.

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5. What is Transient Context?

It is used in passing the required values within the existing flow. The flow can be either a requesting flow or a responding flow. Transient flow is not used across since it will not make the link requests or responses together. It will not be used in saving an input message before service gets invoked into request or response flow. In general the transient acts as temporary storage of messages. After service invokes a call, the next primitive creates another message by combining the invoking response and the original message that is stored in the Transient Context.

6. How can you develop and consume SOAP services in Mule?

SOAP services are created in a similar fashion as that of creating a Mule project by using RAML. The difference is that Concert WSDL is imported instead of RAML and SOAP services are consumed by using Web Service consumer or Mule flow CXF components.

7. Difference between Callout and Service Invoke?

Service Invoke: This primitive is used in making the service request either a request or response of a mediation flow. The service can be a request/response or a one-way. A series of service invocations are performed after permitting the multiple instances of a service invoke primitives in a flow.

Callout: The callout receives the message and calls the requested service and operation. For each connected target operation, there will be a callout node in the median flow.

- When the call is successful, the callout response node in the response flow receives a response message
- When the call is unsuccessful, the callout will be set to retry service invocations depending on the type of fault received.

8. What is the advantage of using Mule ESB?

Mule ESB is a lightweight and highly scalable integration framework which enables the developers to start and connect various applications. The Mule manages the exchange between the components, applications transparency and ESB is taken care of by various applications. Mule can easily integrate third-party applications.

9. What is the functionality of Fan-in and Fan-out?

Fan-out: The Fan-Out primitive is used to fire the output terminal once or multiple times. Fan-Out can be used in isolation or as a combination of Fan-Out and Fan-In.

Fan-in: The Fan-In acts as a decision point to continue the flow execution. Until a decision point is reached it receives a number of messages, at which point the last message to be received is propagated to the output terminal. The Fan In primitive can be used in combination with Fan-Out.

10. Name the features in Mule ESB?

An ESB follows a service-oriented approach and is used in the purpose of integration. The features include.

- Message Routing Service.
- Message Transformation Service.
- Set of Service Container.
- Web Service Security.

11. What are all the configurations required for JDBC Adapter implementation?

The configuration of JDBC adapter is not a complex task, it just needs a data source to connect and configure with a database. A security authentication program must be created if the DB has secure access.

12. What is the Difference between SDO and SMO?

SDO: It is a Service Data Object which represents a variable or an object.

SMO: It is a model pattern that is used for SDO [Data Objects](#) to represent messages.

13. What is the Difference between Stop and fail?

It is used to stop the particular path inflow without generating any exception.

Fail: Generates a failure in the flow.

14. What are the different types of variables in Mule?

There are three different types of mule variables.

- Record variable.
- Session variable.
- Flow variable.

15. Explain the core principles in ESB Integration.

ESB Integration has four core principles.

- **Transportation** - Transport protocol negotiation between different formats including JMS, JDBC, HTTP, etc.
- **Transformation** - Data transformation between the data formats as required by every individual ESB connector.
- **Non-functional consistency** - Applying the consistent implementation of monitoring and security policies.
- **Mediation** - As the name suggests which involves offering different interfaces.
- Enabling different channels for the same underlying implementation of components.
- Support for backwards compatibility with different service versions.

16. What are various parts of composing a message in Mule?

Properties - It contains the header or meta-information or header similar to SOAP.

Payload - It is the main data context carried by a particular message.

Multiple name attachments - Provides the support for multiple messages or payload during event processing.

17. What are the different types of messages in MuleSoft?

Echo and Log message: These are log messages and it moves from inbound to outbound routers.

Bride Message: It is a passed message from inbound to outbound routers.

Build Message: These are messages which are created from fixed or dynamic values.

18. What is the difference between ESB and JMS?

ESB provides the middleware and interfaces services which allow the business enterprises to connect their applications without writing any code.

JMS provides the communication facility and messaging capability between the modules of an application.

19. Explain the parameters that are used in configuring a scheduler.

The related parameters which are used to configure a scheduler are:

Frequency - These are frequencies used by scheduler for triggering flows.

Start Delay - It is the waiting time that is used before triggering any flow.

Time Unit - The time unit for frequency and start delay.

```
implementation="org.mule.components.simple.BridgeComponent">
```

```
  className="org.mule.providers.file.filters.FilenameWildcardFilter"/>
```

20. What is Mule Data Integrator?

A mule data integrator is a tool that maps data by visualizing it. It offers a drag and drop feature interface which makes the developers code easily.

21. What are the advantages of SOAP?

- A perfect medium for web services to talk with client applications.
- Being a lightweight protocol it can be used for data exchange between applications.
- The SOAP protocol works on any programming languages based on Windows and Linux platforms.
- Customization is not required to run the web services built on the SOAP protocol for working on WWW.

22. Name the configuration patterns provided by the MuleSoft?

The configuration patterns of MuleSoft are:

- Bridge
- Validator
- WS proxy
- Simple service pattern
- HTTP proxy

23. What are the characteristics of a global endpoint?

The characteristics are:

- The global endpoint is not typified or outbound routing.
- It is used in configuration files at different locations.
- The global service name is applied for referring to the endpoint.
- It helps in clarifying the usage of particular destinations.

24. What is the use of Outbound Endpoint in MuleSoft?

The Outbound Endpoint performs the following things:

- Sending the SOAP messages.
- Writing to file streams.
- Sending email messages.

25. What Are Available Esbs Apart From Mule?

There are different ESBs available in the market which is both licensed and open source. They are:

- JBoss Fuse.
- ESBMule.
- ESBTalend.

26. What are the various categories of Mule Processors?

There are six categories in Mule Processors.

1. Components.
2. Exception strategies.
3. Business events.
4. Routers.
5. Connectors.
6. Transformers.

27. Name the languages supported by MuleSoft?

The supported languages of MuleSoft are:

1. Ruby.
2. Python.
3. Groovy.
4. Javascript.

28. What are the benefits of the logger component?

The benefits are:

- This core component can be added by users anywhere in the workflow.
- It can be configured for any combination of strings and expressions.

29. What are the advantages of using ESB?

The advantages are:

- It provides a high-level control facility from the portal based on the web.
- By using SaaS-based applications, ESB provides numerous connectivity options.
- ESB provides API and Analytical management.
- ESB is equipped with numerous facilities which can fix the bugs and automates testing.
- Integration of EDI (Electronic Data Interchange)/B2B (business to business).
- Supports batch integration by applying real-time integration methods.

30. Name the flow processing strategies in MuleSoft.

The strategy types in flow processing are:

- Non-blocking flow processing
- A queued asynchronous flow processing
- Asynchronous flow processing
- Custom flow processing
- Tread per processing
- Synchronous flow processing
- Queued flow processing

31. Explain about configuration builder in MuleSoft.

The configuration builder in mule helps in transferring the human organized configuration file into complex graph objects that substitute a running node in ESB. There are two types of configuration builder.

1. Spring Driven Builder - works with XML files.
2. Script Builder - accepts scripting languages files.

32. How do you know about the Mule's shared resources?

To make the connectors as reusable components, it must be first defined as a common resource and then to expose all the applications that have been deployed under the same domain. These are called shared resources. It must be defined inside the Mule Domain Project and has to be referred to every project that is intended to incorporate the elements in it.

33. Explain the concept of Auto-Delete feature in file connector?

By default, the auto-delete feature value is zero which means an inbound endpoint deletes the file automatically from the source directory. If the source directly files doesn't need any auto-delete then the value must be set to false.

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34. Define Batch Jobs in Mule ESB.

A batch job is an element in a mule which splits large messages into records and processes asynchronously in a batch job.

35. What is a Scatter-Gather Router?

Scatter-Gather Router is a routing event processor which can send the request messages for more than one target concurrency. The Scatter-Gather Router will then collect responses from all routes and aggregate back into one response.

36. What is a Choice Router?

Choice Router is a dynamic router which routes the messages inflow. The message content is evaluated based on a set of DataWeave expressions.

37. What is VM transport in MuleSoft?

VM transport is a special type of transport which sends the messages via memory. When a Mule instance runs then messages will never leave JVM.

38. What are message sources in Mule ESB?

Message sources are Anypoint connectors, connectivity elements to a specific external source via standard protocols such as HTTP, FTP, SMTP or a third-party API such as Salesforce.com, Twitter or MongoDB.

39. Define Payload in a MuleSoft?

The payload is a runtime variable in MuleSoft which stores arrays or objects. It is wrapped under the "mule.api.MuleMessage" library which helps in getting different means of accessing payloads under different forms. Mule messages are similar to SOAP, JMS messages which have containers properties, header and multiple names attached to it. The main content of the message is called Payload.

40. Explain Mule Transformer?

Mule Transformer is an event instance of "org.mule.api.MuleEvent" library. This object carries the actual context of the message with the context of the event. The main purpose is to translate a message from one form to another. It can also create a chain of transformers. The message transit occurs between one medium to another medium while staging into different services.

41. What is the Transport Service Descriptor in Mule?

It is a technical hidden configuration that is used in each instance of a connector. It provides the definitions to parameters such as the use of particular parameters, classes which are required for a particular message receiver, dispatchers and requests. These definitions are the default transformations which are used inbound or outbound and utilizing the response of the router.

42. Explain the concept of the endpoint.

An endpoint is a destination shared by many routers in a group which also helps in creating a global endpoint. The global endpoint is useful in different places in the configuration file but it is not similar for inbound and outbound routing services. The entire endpoint names must be specified during the services. These names identify the global endpoint in the group of routers. The global endpoint offers the usage clarification for a specific destination.

43. What is a Router in a MuleSoft?

The router is a critical service in MuleSoft. It finalises and assigns the running territory for the messages to move from one service to another. The routing is a transitory control processing decided by the router which transits the message from one source to another. It is also called as a gatekeeper of endpoint services. It tracks the targeted successions to ensure message delivery on the right intended destination. Routers act as a bundle of classified tasks such as split, sort, group or regroup, messages based on specified conditions or certain mappings.

44. What is the use of Filter in a MuleSoft?

Filters are the powerful capabilities for routers that help in making the smart decision on message delivery or request and response environment. It also provides insights to the router for deciding what to do with the messages in the transit stage. Few filters analyse the messages deeply to find the actual value for desired outputs.

45. Define the purpose of Endpoint in MuleSoft?

Endpoint defines the specific usage of a transport protocol in reading the message, writing, listening or polling to the target destination. The endpoint control entity ensures the usability of connectors. The target destination will be defined as a URI that depends on connectors which treat the destination as URI, URL or JMS.

46. Define Fileage properties in file connector in MuleSoft?

Fileage defines the duration of waiting time of an endpoint that starts before reading the file again. For example, file age of 60000 indicates an endpoint that has to wait for 1 minute before the start for the next processing.

47. What is streaming property in the file connector in MuleSoft?

Streaming properties represent the format of true and false values in connectors. You will be working with the streaming data on the connectors if the streaming value is true else you will work on the file system if it's false.

48. What is the context of MuleSoft?

In general, the context of a message defines the overall purpose of the message but the context of mule defines the temporary area that is created along with SMO (Server Message Object) in median flow while message transit. SMO contains shared context in the message flow that is used at the time of aggregation. The context of aggregation maintains the data between FanOut and FanIn primitives. The context of data which is present in the request flow is not persistent in the throughout request and responses flow as it belongs only to a request BO.

49. How can you judge the need for ESB in a project?

The application needs of a project must be analysed carefully for avoiding unnecessary arrangements. ESB benefits the project needs in several ways by operating it on a huge setup of multifunctional application support. The analysis depends on many factors such as.

- When a project needs a group of actions for additional applications and servers where the interaction happens between two or more complex resources.
- When a project requires communication for multiple applications over a single setup during a broad scale-up for the future scope.
- When the project requires to route the messages from one end to other such as forking and aggregating the routing messages.
- When a project needs application support that requires huge investment.

50. What is polling frequency in the file connector in MuleSoft?

When you need a new file inbound endpoints to poll direction for reading the new content then you must set the polling frequency value to few milliseconds to achieve this. Here polling frequency defines the value of the poll.

51. What is Mule ?

Mule ESB (a.k.a. Mule) is a lightweight Java-based enterprise service bus (ESB) and integration platform that allows developers to connect applications quickly and easily, enabling them to exchange data. Mule ESB enables easy integration of existing systems, regardless of the different technologies that the applications use, including JMS, Web Services, JDBC, HTTP, and more.

52. What is Shared Context?

Shared Context: Context is a temporary area which is created along with Service Message Object (SMO) in the Mediation Flows. Shared Context is a type of context which is present in the SMO. Shared Context is mainly used when we are using Aggregation process where we need to iterate the BO for Certain times. Shared Context maintains Aggregation data between Aggregation primitives. The Content (data) which is present in the shared context BO does not persist across Request and Response flows i.e. The Data in the Shared Context which is used in Request flow cannot be used again in Response flow.

53. What is Correlation Context?

Correlation Context: Used when Mediation primitives want to pass values from the request flow to the response flow.

Used to pass values from the request message onto the response.

54. How can you implement a loop in mediation?

By using Fan-in and Fan-out primitive.

55. How can you change the runtime changes using mediation primitive?

We have feature called Promotable properties in ESB. We can configure this feature while development. Then we can make it changed at runtime without restarting the server it can be published.

56. Why am I getting 'Unable to get resource from repository' while building the Mule examples?

If you are getting this error message while building the Mule examples:

```
[WARNING] Unable to get resource from repository atlassian-m2-repository
(http://repository.atlassian.com/maven2)
```

```
Downloading: http://repository.atlassian.com/org.apache.maven.plugins/poms/maven-dependency-plugin-2.0-
20060907.213033-3.pom
```

```
[WARNING] Unable to get resource from repository atlassian-m1-repository (http://repository.atlassian.com)
```

```
Downloading: http://repository.codehaus.org//org/apache/maven/plugins/maven-dependency-plugin/2.0-
SNAPSHOT/maven-dependency-plugin-2.0-20060907.213033-3.pom
```

```
[WARNING] Unable to get resource from repository Codehaus (http://repository.codehaus.org/)
```

```
Downloading: http://snapshots.repository.codehaus.org/org/apache/maven/plugins/maven-dependency-
plugin/2.0-SNAPSHOT/maven-dependency-plugin-2.0-20060907.213033-3.pom
```

```
[WARNING] Unable to get resource from repository codehaus-snapshots
(http://snapshots.repository.codehaus.org)
```

```
[INFO] -----
```

```
[ERROR] BUILD ERROR
```

```
[INFO] -----
```

```
[INFO] Failed to resolve artifact.
```

You may need to change the version value of the dependency in the pom.xml, for instance,

```
org.apache.maven.plugins
```

```
maven-dependency-plugin
```

```
2.0-alpha-1
```

The issue regarding dependencies review is tracked at MULE-1446.

57. How to use Mule HQ with Mule?

Mule Configuration

Add the following code snippet to your Mule configuration:

```
className="org.mule.management.agents.RmiRegistryAgent"/>

className="org.mule.management.agents.JmxAgent">

    value="service:jmx:rmi:///jndi/rmi://localhost:1099/server"/>

    value="true"/>
```

Start your Mule instance.

Ensure the HQ agent is running on the server the Mule instance is configured on and is pointing to the desired HQ server.

Check the Mule HQ server page to see if information about the Mule instance is being received.

58. What is a Mule UMO?

A Mule UMO is a Universal Message Object

UMO is now a legacy term. What was once referred to as UMO Components are now referred to as Service Components?

59. Mule won't start? Check your Environment?

The java and mule environment variables must be setup correctly for mule to start. If you are experiencing problems check the following variables:

MULE_HOME – should be the location of the mule install

JAVA_HOME – should be the location of the JDK

PATH – should have both JAVA_HOMEbin and MULE_HOMEbin in the path.

Check all of the above carefully. Some systems with multiple JDK's installed can end up with incorrect mappings between the PATH and the JAVA_HOME, which will stop mule from loading.

60. How to add classes/jars to the Mule class path?

Use the MULE_LIB variable (generally set in the run script)

To include JAR file(s) in a mule class path, declare each dependent jar file in the MULE_LIB entry.

For spring resource, if the XML bean declaration is placed within a project, include the project JAR file in the class path too (i.e., if not included, Mule will throw a **.xml not found on class path)

61. How to configure a simple FTP handler in Mule?

ftp_handler-config.xml

```
"http://mule.mulesource.org/dtds/mule-configuration.dtd">
```

Ftp get to a remote server and place into a local directory on the MULE server

62. Where to find the class Abstract Mule Test Case?

This is in 1.4/1.4.1 distributions but was missing from the 1.3.3 distribution – the class is defined in /lib/mule/mule-core-jar.

Our design of course tutorials and interview questions is practical and informative. At TekSlate, we offer resources to help you learn various IT courses. We avail both written material and demo video tutorials. For in-depth knowledge and practical experience explore online mule ESB Training.

Messaging framework to an enterprise-wide highly distributable object broker.

63. What Difficulties Mule Does Encompass?

Transport: applications can accept input from a variety of means, from the file system to the network.

Data format: speaking the right protocol is only part of the solution, as applications can use almost any form of representation for the data they exchange.

Invocation styles: synchronous, asynchronous, or batch call semantics entail very different integration strategies.

Lifecycles: applications of different origins that serve varied purposes tend to have disparate development, maintenance, and operational lifecycles.

64. Why was Mule Designed?

Mule's core was designed as an event-driven framework combined with a unified representation of messages, expandable with pluggable modules. These modules would provide support for a wide range of transports or add extra features, such as distributed transactions, security, or management. Mule was also designed as a programmatic framework offering programmers the means to graft additional behavior such as specific message processing or custom data transformation.

65. Why the Name Mule?

There is a lot of infrastructure work to be done before we can really start thinking about implementing any logic. So this infrastructure work is regarded as "donkey work" as it needs doing for every project. A Mule is also commonly referred to as a carrier of load, moving it from one place to another. The load it specializes in moving is our enterprise information.

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66. What Are Differences Between Mule And Other Commercial Esbs ?

Prescriptive deployment model, whereas Mule supports a wide variety of deployment strategies. Prescriptive SOA methodology, whereas Mule can embrace the architectural style and SOA practices in place where it is deployed. Mainly focused on higher-level concerns, whereas Mule deals extensively with all the details of integration. Strict full-stack web service orientation, whereas Mule's capacities as an integration framework open

it to all sorts of other protocols. Comprehensive documentation, a subject on which MuleSource has made huge progress recently

67. What Is Model Layer In Mule?

The first logical layer is the model layer. A Mule model represents the runtime environment that hosts services. It defines the behavior of Mule when processing requests handled by services. The model provides services with supporting features, such as exception strategies. It also provides services with default values that simplify their configuration.

68. What Is Service Layer In Mule?

A Mule service is composed of all the Mule entities involved in processing particular requests in predefined manners. A service is defined by a specific configuration. This configuration determines the different elements, from the different layers of responsibility that will be mobilized to process the requests that it will be open to receive. Depending on the type of input channel it uses, a service may or may not be publicly accessible outside of the ESB.

69. What Is Transport Layer In Mule?

The transport layer is in charge of receiving or sending messages. This is why it is involved with both inbound and outbound communications. A transport manifests itself in the configuration by the following elements: connectors, endpoints and transformers.

A transport also defines one message adapter. A message adapter is responsible for extracting all the information available in a particular request (data, meta information, attachments, and so on) and storing them in transport-agnostic fashion in a Mule message.

70. What Is Connector In Mule?

A connector is in charge of controlling the usage of a particular protocol. It is configured with parameters that are specific to this protocol and holds any state that can be shared with the underlying entities in charge of the actual communications.

For example: a JMS connector is configured with a Connection, which is shared by the different entities in charge of the actual communication.

Dataweave Interview questions for Mulesoft

Differences between DW1.0 & DW2.0

Few of the changes from Dataweave 1.0 of Mule 3 to Dataweave 2.0 of Mule 4

- Header Changes,
- Conditional Logic
- Operators are functions
- Type Names
- Comments

What is p function in Dataweave?

p function is used for reading data from properties file in transform message generally we use “p” function.

What is lookup function in Dataweave?

lookup is used to invoke flows from [Dataweave code](#)

What is causedBy function in Dataweave?

The *causedBy* function matches an error by its type, like an error handler does

What is the difference between map, mapObject and pluck?

- [map](#) is used to iterate over array
- mapObject is used to iterate over object and return object
- pluck iterates over an object and returns an array of keys, values, or indices in that object.

Difference between Map, flatMap and MapObject?

Map: Iterates over items in an array and outputs the results into a new array.

flatMap: Iterates over each item in an array and flattens the results.

MapObject: Iterates over an object using a mapper that acts on keys, values, or indices of that object.

How to convert an Object into Array using Data weave?

Pluck – Iterates over an object and returns an array of keys, values, or indices from the object

How to access secure property inside dataweave?

By using **p(secure::<property name>)** you can access the secure property defined as part of secure property file in mule application

I have array within it I have nested array of data, how can I make simple array?

Flatten – Turns a nested array into a simple array

How will you combine 2 arrays into single array in MuleSoft?

You can use flatten Dataweave function to combine two arrays in to single array

Difference between flatMap and flatten?

[flatMap](#): Iterates over each item in an array and flattens the results, can act on values and indices of items in the array.

Flatten – Turns a nested array into a simple array. only acts on the values of the arrays

How to call Flow from inside a Dataweave?

You can use Dataweave lookup function to call flow from inside a Dataweave It works in Mule apps that are running on Mule Runtime version 4.1.4 and later It takes the flow's name and an input payload as parameters. For example, lookup("flowname", payload)

Mulesoft Anypoint Studio interview questions

What connectors have you used?

A. Salesforce, Sharepoint, JMS, HTTP, HTTPS, FTP, SFTP, Database, File...etc.

Difference between VM queue & JMS

A. VM Transport is for intra JVM communication between Mule flows. We can use VM transport to communicate with different flows of the same application whereas JMS is used for communicating with the external applications also.

How to filter duplicate records?

A. Using Idempotent filter to discard duplicate records

What are the core components of Mule ESB?

A. Scheduler, For Each, Logger, Batch (Batch Aggregator, Batch Job, Batch Step), Flow reference, Transform message, Error handling Routers, Scopes and Transformers. Refer [Core components of Mule](#)

How to read a file as input other than FTP connector

Scenario: How to fetch and process 5M records(few duplicates) into DB

Different types of flows, flow processing strategies

Difference between sub-flow, private flow

A. **Subflow** processes messages synchronously but inherits processing strategy and exception handling strategy from the calling flow (parent flow). It can be used to split common logic and be reused by other flows.

Private flow does not use a source. It can be synchronous or asynchronous based on the processing strategy selected. Also, has its own exception handling strategy. Allows you to define different threading profile.

How can we create a custom policy in Anypoint Studio?

A. Navigate to File > New > API custom policy Project(Beta) in Anypoint Studio

Difference between async flow & async scope

Async Flow: Async Flow is a transactional flow in which the requestor does not wait for the response from the target service and processes the flow

Async scope: The Async scope is a branch processing block that executes simultaneously with the main flow. The main flow continues to execute while it initiates and processes the Async scope

Error handling in Mule 3.9

Refer here for the [Error handling mechanism](#) in Mule 3.9

How do you consume a SOAP-based web service in Studio

A. Using the Web Service Consumer component. Configure the connector with WSDL or SOAP web service configuration details like Service name, URL, Binding port, etc.

What are the steps involved in MUnit creation?

How do you set environment-specific values in Mule application?

At first, Create a properties file for each development environment in your application. Configure a properties placeholder in your app to look for the environment upon launch. Set an environment variable to point to a specific development environment during application deployment.

Explain Collection aggregator, splitter, object store, cache

What are the variables available in Mule 3?

A. Flow variables, Record variables, Session variables

How do you import a RAML and use in Anypoint Studio?

What are various types of Exception Handling?

- Choice Exception Handling.
- Catch Exception Handling.
- Rollback Exception Handling.
- Global Exception Handling.
- Default Exception Handling.

What is Shared Resource in Mule and how they have been used?

We can make connectors as a reusable component by defining them as common resources and expose them to all applications deployed under the same domain, these resources are known as shared resources. These shared resource needs to be defined inside Mule Domain Project and then referred to each of the projects that are meant to use the elements in it.

What is the definition of Web Services?

Web service is a function or program in any language that can be accessed over HTTP. Message format can be XML or JSON or any other program as long as the other programs can understand and communicate. Any web service has a server-client relationship. Web services can be synchronous or asynchronous. Any web service can have multiple clients.

What are the different types of Flow Processing Strategies?

There are six different types of Flow Processing Strategies. They are

- Asynchronous Flow Processing Strategy.
- Custom Processing Strategy.
- Thread Per Processing Strategy.
- Queued Asynchronous Flow Processing Strategy.
- Synchronous Flow Processing Strategy.

- Non-blocking Flow Processing Strategy.
- Queued Flow Processing Strategy.

How to create and consume SOAP service in Mule?

Creating SOAP Service – We can create a SOAP service same as we create Mule Project With RAML, the only change is instead of RAML we need to import Concert WSDL.

Consuming SOAP Service – We can use Web Service Consumer or CXF component in our mule flow to access/consume SOAP service.

How a message in Mule is composed ?

A Mule message is composed of different parts: The payload, which is the main data content carried by the message.

The properties, which contain the meta information much like the header of a SOAP envelope or the properties of a JMS message. Optionally, multiple named attachments, to support the notion of multipart messages.

Optionally, an exception payload, which holds any error that occurred during the processing of the event.

[RAML Mule Interview questions and answers](#)

What are Inbound endpoints and Outbound endpoints in Mule ESB ?

Inbound endpoints are message sources, which as the name suggests is where message is generated and triggered. They can be created based on external events (like an incoming HTTP request or JMS message) or by a file poller.

Outbound endpoints and anything else you see in a flow (except exception strategies) are message processors which means they do something with the message in the flow

How to implement Custom Connectors in Mule ESB?

Creating Connector Project in Anypoint studio,

- Write Connector Code,
- Write Connector unit Tests (optional but recommended),
- Document the Connector for reference,
- Package the custom Connector.

How to achieve the parallel processing in MuleSoft?

using scatter gather we can implement parallel processing

How can you raise error explicitly in Mule flow/subflow?

By using the Raise Error component you can explicitly raise the error in between the flow

How can you handle errors in middle of the processing in MuleSoft?

You can use the Try block to handle the errors in between the flows, try block also support the transaction. By using try block you can use the on-error continue or on-error propagate to handle the error

What are target variable in MuleSoft

When creating a flow in mule app, you can store data in variable so that any component in a flow can use it. In Mule 4 Target Variable replaces the Message Enricher of Mule 3, target variable is available with most of the connectors

[Mulesoft Dataweave Interview questions and answers](#)

How many phases are there for batch job in Mule 4

Each batch job contains 3 different steps

- Load and Dispatch.
- Process
- On Complete.

When we should use batch processing

You can use batch processing when:

- Synchronizing data sets between business applications, such as syncing contacts between two different ERPs
- Extracting, transforming and loading (ETL) information into a target system, such as uploading data from a flat file (CSV) to Hadoop.
- Handling large quantities of incoming data from an API into a different systems

Why we use Batch Aggregator scope

Batch Aggregator scope is used to accumulate a subset of records from a batch step and process them to external source or service for example rather than processing single record to target system you can use batch aggregate to process all the records at once. You can also configure batch aggregator scope to process fixed size groups of records inside a batch aggregator scope

What is the difference between For each and Batch Process

For each do the processing in single thread while Batch Process performs multi-threaded processing

How to manage the Transaction in MuleSoft

MuleSoft supports two types of transactions

- Single resource(Local)
- XA transactions (extended architecture)

You can configure the transaction depends on the requirement

You can start a transaction from a message source. In this case, the entire flow takes part in transaction This is useful when working with messaging connectors to prevent the consumption of the message if a problem occurs when processing it, allowing you to retry later (because of the rollback).

You can also use the Try block which support the transaction when you want to handle the transaction in middle of the flow

There are below transaction Action define the type of action that operations take regarding transactions

- Always_Begin
- Always_Join
- Begin-or-join
- Join-if-possible
- Indifferent
- None
- Not Supported

What are various types of error handling in Mule 4?

Mule 4 has redesigned error handling by introducing the error-handler component, which can contain any number of internal handlers and can route an error to the first one matching it. Such handlers are on-error-continue and on-error-propagate.

How can you process message asynchronously in MuleSoft?

You can use the Async Scope to process the message asynchronously.

Async can be used to do the processing in parallel with the main flow, it can be used to process the time consuming operations that does not expect a response back

How can you achieve the Reliability in MuleSoft?

If your application uses a transactional transport such as JMS, VM, DB etc. Reliable messaging is ensured by the built in support of transactions in the transport. Reliable messaging pattern is important when dealing with non-transactional endpoints

How can you improve performance of Mule Application in MuleSoft?

There are multiple ways to improve the performance

- Try to save the result and reuse them later
- If processing is expansive do it only once and save result and use them later
- Try to put the validation at start of the flow
- Use Streaming to process data
- Try to process the data asynchronously wherever is possible

[Mulesoft Anypoint Platform Interview questions and answers](#)

How can you optimize memory efficiency of the Code in MuleSoft?

- Do not save the Payload in to the flow Variable as it is more memory consuming element
- Do not load unnecessary part of the document
- Use better Database polling for highly concurrent scenarios

What is distributed file polling?

In mule 4 distributed file polling is used to poll files in cluster nodes

There are few connectors where this feature is enabled by default

- File Connector
- FTP Connector
- SFTP Connector

What happens if async scope is inside until successful

The flow will stop. As async scope will not return anything and Until successful will expect some response to fulfil until-successful condition.

How will you implement caching in MuleSoft?

There are 2 ways to implement the caching in MuleSoft

1.
 1. Use Object Store
 2. Use Cache scope to cache the Response

What is API Route Kit how it works? On what basis routing is going to happen?

APIkit Route Kit is a tool for building Mule REST API flows. By using this we can route to appropriate flow based on method name and resource.

Based on Method name and Resource name its going to route the request to particular flow.

[Mulesoft Top Interview questions and answers](#)

What is SOAP Route Kit? How the routing works in SOAP Route Kit?

SOAP Route Kit is a tool for building Mule SOAPAPI flows.

Based on Method name it's going to route the request to particular flow.

What is the security mechanism your using to integrate Salesforce? How you got the token?

OAuth 2.0 or explain how its implemented in your project.

Difference between Object Store and VM connector and when to use what?

Object Store: An Object Store is a Mule component that allows for simple key-value storage. Although it can serve a wide variety of use cases, it is mainly design for:

- Storing synchronisation information, such as watermarks.
- Storing temporal information such as access tokens.
- Storing user information.

Virtual Machine: Virtual Machine (VM) (VM Connector) handles intra-app and inter-app communication through asynchronous queues that can be transient or persistent.

- Transient queues are faster than persistent queues, but they are not reliable in the case of a system crash.
- Persistent queues are slower but reliable.

If two services are invoked in parallel using Scatter Gather. One message got success response and one got failed with connectivity error. What will happen in scatter-gather?

The complete flow gets failed and will raise error. Based on error handling implementation it will go to error flow.

What Is Filter In Mule ?

Filters are a powerful complement to the routers. Filters provide the brains routers need to make smart decisions about

what to do with messages in transit.

- Filters are used to decide on what message continues processing through a flow.
- Filtering can be done based on the message type, payload or headers.
- Payload-type-filter
 - Message passes through this filter only if the payload is of a required type
- Expression Filter
 - Is used to filter based on the result of a MEL expression
- Wildcard Filter
 - Messages that match a shell-like wildcard will be allowed to pass.
 - Is generally used for filtering text messages.
- Regular Expression Filter
 - Filtering is done based on a regular expression.
 - Is very useful for filtering text messages.

What is object-store

An object store is a facility for storing objects in or across Mule applications. Mule uses object stores to persist data for

eventual retrieval. Mule uses object stores in various filters, routers, and other message processors that need to store state between messages. In most cases, Mule creates and manages object stores automatically.

Why we use VM transport?

You can use the Java Virtual Machine (VM) transport for intra-JVM communication between Mule flows. This transport

by default uses in-memory queues but can optionally be configured to use persistent queues (persistent queues are used to help prevent data loss). Note: VM file persistency does not work on clusters. The VM transport has often been used to implement complex integrations made up of multiple applications.

VM is still useful in certain situations. Suppose, for instance, that most of the parts of your solution are local, but some

need to be decoupled for testing, or because some need to be made remote. You can use Flow References to directly reference one flow from another without a transport in the middle. Use VM

transports if you need redelivery of messages, this won't be possible in private flow and flow ref because for rollback exception strategy even if we define that no internal queue exists.

SEDA is staged event driven architecture. It breaks the complex event driven application into number of stages which are connected by queues.

It decouples receiving, processing and dispatching phases.

A Stage is analogous to an "Event". To simplify the idea, think of SEDA as a series of events sending messages between them.

Example: In a restaurant:

if waiter A has work of taking in the customers and allot a seat.

Waiter B has work of taking and processing order.

Waiter C has work of giving/taking bill and take out the customer.

Advantages:

1. we create a new thread for each request
2. it is easy to program
3. We can achieve higher parallelism in SEDA.
4. less time requires
5. operation become robust.
6. Performance increases.

Disadvantages:

1. Caching can be a problem.
2. Scheduling can be problem
3. Locking can be a problem

2. How SEDA can be implemented in mule?

Mule is built up on SEDA architecture.

In Mule, each component is treated as a stage with its own thread pool and work queue.

If two flows are connected by queue let's say by JMS component or by VM.

3. What is RAML?

RAML stands for Restful API Modelling Language. RAML can be used for:

1. API specification
2. We can mock API by RAML and API console.
3. We can generate interface by RAML.
4. We can use RAML for API documentation.
5. Its a contract between APIs.

RAML lets you see what your API looks like as you design it, using easy to read plain text. RAML makes it easy to manage the API lifecycle from design to deployment to sharing. RAML is used for API modelling. RAML is both machine and human readable.

4. What are overlays in RAML?

Overlays is used to extend non-behavioral aspects of an API, such as descriptions, usage directions, and user documentation items.

Example: RAML 1.0 Overlay

Example: in api documentation if documentation is needed in multiple languages then overlays can be used.

5. What are Extensions in RAML?

Extension is used to extend or override behavioral aspects of the API.

Example: RAML 1.0 Extension

As you may infer from the name, extensions are used to extend an API by adding new behaviors and/or modifying existing behaviors of an API. An analogy from the object-oriented programming world would be a subclass extending a super class, where the subclass can add new methods and/or override existing methods. An extension may also extend an API's non-functional aspects.

An extension might be used, for example, to define additional resources that are exposed only to a select set of users, such as administrators or users having been assigned a particular role. An extension could also be used to add features for a newer version of an API.

6. What is transport barrier?

A "transport barrier" is when your Mule message goes through an endpoint that makes use of a transport (say for example, HTTP endpoints, JMS, TCP, VM, etc).

There are two ways of calling flows in Mule:

1. Via an endpoint.
2. Via the Flow-Ref component.

The first case makes use of transports, therefore, your message goes through the whole serialization and deserialization processes. At the end of the day, you will still end up with the same payload BUT it will be a different Mule message. It's just a fancy way of saying we are sending a message across a transport.

7. What are inbound and outbound properties in mule?

Mule Inbound Properties

- Inbound properties cannot be set by you.
- Message sources (such as inbound endpoints) set them for you when they receive a message.
- Inbound properties are lost when crossing a "transport barrier."

Mule Outbound Properties

- Outbound properties can be set by you.
- When crossing a "transport barrier", outbound properties are automatically turned into inbound properties, and no longer exist as outbound properties.

8. What are Invocation and Session properties in mule?

Mule Invocation Properties

- Invocation properties can be set by you.
- Invocation properties are lost when crossing a "transport barrier".

Mule Session Properties

- Session Properties can be set by you.
- Session properties are preserved when crossing a "transport barrier".

9. What are Invocation and Session properties in mule?

Mule Invocation Properties

- Invocation properties can be set by you.

- Invocation properties are lost when crossing a “transport barrier”.

Mule Session Properties

- Session Properties can be set by you.
- Session properties are preserved when crossing a “transport barrier”.

10. When to use VM and when to flow ref?

VM endpoints enable message redelivery strategies to be configured in your exception handling blocks – this is not possible with flow-refs. VMs can do this because they internally use a queue to hold messages while flow-refs are like simple method calls.

Contrast that to a private flow and chaining with flow-refs, if an exception occurs in the called flow even though we have a rollback strategy configured it will NOT execute because there is no internal queue involved.

11. What is Anypoint platform?

Anypoint Platform is a complete solution for API-led connectivity that helps companies build application networks of apps, data, and devices, both on-premises and in the cloud.

This hybrid integration platform includes iPaaS, ESB, and a unified solution for API management, design, and publishing.

Anypoint platform gives us:

API Manager

Runtime manager

Design centre

Access management

Anypoint exchange

Anypoint visualizer: Anypoint Visualizer displays views of different aspects of application network graph. It provides a real-time graphical representation of the APIs and the Mule applications that are running and discoverable.

It collects the data from the APIs, Mule applications, and proxies deployed to CloudHub or standalone mule runtime engine to discover all incoming and outgoing connections through an embedded plugin.

Anypoint Monitoring: Anypoint Monitoring provides visibility into integration across your Application Network graph. It provides feedback from the Mule flows and components in your application network graph.

Secret Manager: Secret Manager uses secure vault technology to store and control access to private keys, passwords, certificates, and other secrets.

Use Secret Manager to write, read, manage your secret, keys, Transport Layer Security Certificate (TLS), and many other documents.

12. What is API Manager?

API Manager is the component of Anypoint Platform that helps in managing the APIs for an organization.

From API manager we can:

- Apply policies
- Alerts by email if someone violates policy or response time is more, etc
- SLA tiers
- Autodiscovery

13. How many policies you have applied?

Give details of policies applied by you.

14. OAuth policy, how we can apply?

<https://www.youtube.com/watch?v=xRTKx4omiRE>

First of all we have to generate token via third party say okta.

for that we have to register in okta. It gives server URL and client URL.
Server URL is applied in Anypoint platform under oAuth policy section.
Client URL is used to generate token via client credentials like client id and client secret of organization.
Once token is generated, we can use that token in our API to access it.

15. What are exception strategies?

1. Default exception strategy
2. Catch exception strategy
3. Rollback exception strategy
4. choice exception strategy
5. reference exception strategy.

16. What are processing strategies?

1. Synchronous processing strategy
2. Queued asynchronous strategy
3. Non-blocking processing strategy.

17. What is Jenkins? How to create job in Jenkins?

Jenkins is used for CI/CD (continuous integration/continuous deployment).

We can deployment our project to any server or cloud via Jenkins. If we make any change in code Jenkins will take the change automatically and deploy that code to the server.

We need to login to Jenkins portal to create job. Jenkins job is used to fetch the code form our repository and deploy the code to our server.

18. What is mq and why we should use it?

MQ or message queue is used to connect applications from each other to send data or messages.

Let's say if we want to connect mule application with Mainframe, we can use MQ to connect it and exchange data.

We can also use web services to communicate between applications but MQ is preferred because:

1. If connection is lost then MQ persist the message and resumes from the same point from where connection get lost.
2. We can configure MQ to process asynchronously.

19. What is Anypoint MQ?

Anypoint MQ is the queueing facility given by Mulesoft in Anypoint platform. It comes only in paid version. We can create queues and call that queues from our mule flows via client id and client secret from of our organization and queue name.

20. How we can deploy our API in mule cloud?

By runtime manager of Anypoint platform we can deploy mule applications to mule cloudhub.

In mule 3 cloudhub deployment, we need .zip project file.

In mule 4 cloudhub deployment, we need .jar project file.

21. What is API kit?

API kit is required to convert RAML into Mule flows.

22. What is API kit router?

API kit is used to route the request to various flows via http method and resource name.

23. What is dataweave? Name 10 dataweave inbuilt functions?

Dataweave is a language which is used to transform data from one form to another. For example: json to xml. There are many facilities given in dataweave language to do modifications in our data.

24. What is message enricher? How it works?

Message enricher is used to enrich our payload with some other data.

We can modify our payload with some other data.

There are two main attributes in message enricher: source and target.

Source is our existing payload.

Target is message or data which is used to modify our existing payload.

25. What is SAML? How we can apply SAML in our application?

SAML or security assertion mark-up language is used to do single sign on.

So users do not need to manually sign on each time and SAML is secured too.

We need to sign on third party for generation of token.

In Anypoint platform go to access management -> external identity and select SAML 2.0. and feed server URL given by third party in SAML policy section in Anypoint platform.

Access token is generated by client URL provided by client id and client secret of organization.

Once token is generated, we can access our API by applying access token in request.

26. How to convert HTTP to HTTPS in mule?

Ans: For conversion of HTTP to HTTPS we need to do TLS configuration in these we need to generate security certificates like key store and trust store

Key Store: It is a server-side asset which contains data in encrypted and decrypted form.

Trust Store: It is a client-side asset which contains data in encrypted and decrypted form. We need to give location of both key store and trust store in TLS configuration to convert HTTP to HTTPS

27. What is aggregator and splitter in mule?

Ans: Aggregator: Is used to aggregate or some of the response data. Ex: collection aggregator

Splitter: It is used to split or bifurcate the data. Ex: Collection splitter

28. What are core principles of ESB integration?

Ans: 1) Data orchestration

2) Data transportation

3) Data transformation

4) Data mediation

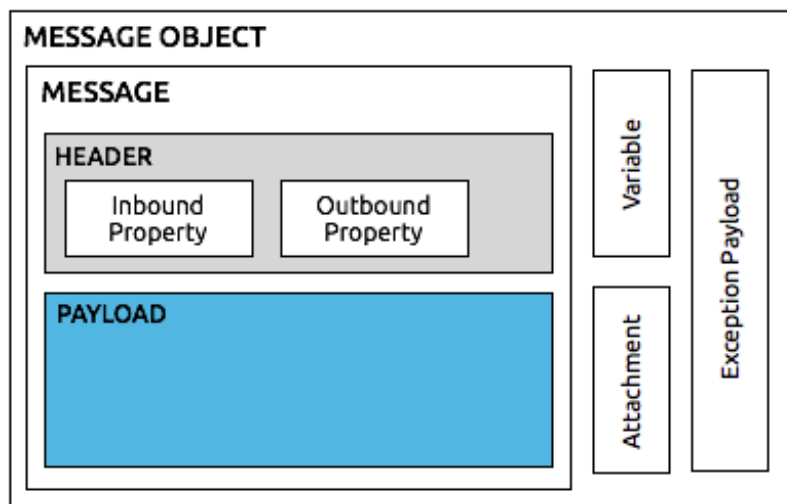
5) Non- functional consistency

29. Integration design patterns?

Ans: 1) Canonical design pattern

2) Facade design pattern

30. Mule object structure



31. Name few ESBs

Ans: 1) Mule ESB

2) Tibco ESB

3) Dell-boomi

- 4) Apache Camel
- 5) Fuse Jboss ESB

32. What is salesforce? Have you worked on salesforce connector?

Ans: Salesforce is a CRM application and used for storing customer related data.

33. Dynamic and parameterized query in DB connector, how to execute and how to execute stored procedure in mule?

Ans: We used parameterized query in DB operations to avoid SQL injection. Dynamic queries lead to SQL injections. SQL injections is vulnerable thing and used to attack data driven applications and to hack database application. We can call store procedure from database connector in Mule

34. Types of variable in mule?

Ans: There are three types of variable in mule:

- 1) Flow variable
- 2) Session variable
- 3) Record variable

Flow variable: We can store data or values in Flow variable and can access in the flow. A flow variable cannot cross the transport barrier or end points.

Session variable: We can access Session variable anywhere in the flow and Session variable can cross transport barrier in some cases for ex. VM

Record variable: Record variable are used in batch processing.

35. How to perform batch processing in mule

Ans: Batch processing is a technique to process data in batches. In mule there are four phases in batch processing.

- 1) Input – In these stage data is gathering and transform into desired input.
- 2) Load And Dispatch – In these phase batches are created for processing.
- 3) Process – In these phase batch steps are processed one by one and output is forwarded to final stage.
- 4) On Complete – These is the optional stage which is used to show the report of the records process.

36. How to consume soap web service in mule

Ans: By using web service consumer. We need to give WSDL path in web service consumer component to consume soap web service.

37. how to create soap web service in mule

Ans: By using apache CXF component in mule.

38. How to consume rest web service in mule?

Ans: By using HTTP connector.

39. What are advantages of ESB?

- Ans:
- 1) Defining API specifications
 - 2) Connection of applications by que and web services
 - 3) Routing of data
 - 4) Deployment of application to server or cloud
 - 5) Manage applications
 - 6) Data format conversion

40. What is flow, sub flow and private flow?

Ans: Flow – Flow contains message source and message processors which are used to perform a certain operation.

Sub Flow – Sub Flow is used to break the main flow into smaller flows to increase the readability. In sub flow main flow's encryption strategy is used. Sub Flow are synonymous. We can connect sub flow from the main flow way flow reference components.

Private Flow – Flow without message source is known as private flow it has its own exception strategy.

41. Have you worked on mule 4?

Ans: Yes

42. Difference between mule 3 and mule 4

Ans:

Sr.no	Mule 4	Mule 3
1	In Mule 4 the project is by default mavenized, so we do not need to convert into maven project manually.	We need to convert project into maven project manually.
2	In cloud hub deployment we need to deploy .jar project file from runtime manager	In mule 3 cloud hub deployment we need to deploy .zip project file from runtime manager
3	In mule 4 exception handling is changed, we use on error continue and on error propagate to handles error	We use mule exception strategy like catch, rollback, choice etc.
4	Mule expression language is eliminated in mule 4	Mule expression language is present in mule 3
5	Data weave version is changed to 2.0	Data weave version is 1.0
6	Transformer are eliminated in mule 4	Transformer are present in mule 3
7	In mule 4 there is no need to manually tune the runtime it has self-tuning engine	In mule 3 we need to manually tune the runtime

43. What is invoke component in mule?

Ans: It is used for JAVA support in mule. We can execute JAVA code from invoke component.

44. What is API LED connectivity in mule

Ans: API LED connectivity is used to propagate message from an application to another via different layers.

1) System API – It is used to expose all data from backend and convert into another desirable form.

2) Process API – In process API we can filter desired data and perform business operation in it.

3) Experience API – It is used to connect different UI devices. It facilitates UI to capture the same API in different platforms.

45. What is API lifecycle in mule?

Ans: 1) Design

2) Simulate

3) Feedback

4) Validate

46. How to read flat file in mule and how to read fix width format in mule?

Ans: Both flat file and fix width format functionality support is present in transform message component in mule.

47. In how many connectors you worked? Explain in detail

Ans: List the connectors you have worked on:

48. What is Munit? How to perform it?

Ans: Munit is the unit testing framework in mule. It is like Junit in JAVA. We need to generate Munit flows from mule flows and performed Munit testing. Green bar indicates successful testing and Red bar indicates error.

49. What is VM connector? Diff between flow reference and VM?

Ans: In VM:

1) VM is used for asynchronous processing.

2) It is used to connect different mule applications.

3) It is used to connect applications via queues.

4) When we distribute our work across cluster.

In Flow reference:

- 1) We use flow reference to connect sub flow from main flow.
- 2) Process is synchronous.

50. What is Synchronous & Asynchronous flow?

Ans: Synchronous flow:

- 1) In this same thread is responsible for execution of whole process.
- 2) So, more time is elapsed in it.

Asynchronous flow:

- 1) More than one thread is required to execute a process.
- 2) There is separate thread for message receiving, message processing and message dispatching.

51. What is API proxy and API gateway.

Ans: API proxy: API proxy is proxy URL which is used to secure API by avoid accessing original URL. Proxy URL is generated from API manager in Anypoint platform.

API gateway: API gateway restricts the unwanted malicious attacks from the hackers and the security policies applies on API gateway.

52. How to iterate collections in mule?

- Ans: 1) By using map operator.
2) By using for each scope.

53. What are scopes. How many scopes you have used?

Ans: Scopes: Scopes contains multiple message processors for performing certain operations.
I have used Scopes like for each, async, transactional are used.

54. How to handle transactions in mule?

Ans: By using transactional scope.

55. What is MEL? How to consume headers in mule?

Ans: MEL or Mule expression language is used to access in bound properties in mule like query parameters, URI parameters and headers.

We can consume headers in mule by: `message.inboundproperties.headers`

56. How to apply basic authentication in mule?

Ans: Basic authentication is used to secure API by username and password, so we need to go to policy section in API manager and select basic authentication. Before it we need to enable security manager and choose username and password.

57. What is Mongo DB connector? Have you used it?

Ans: Mongo DB connector is used to connect Mongo DB. Mongo DB credentials are applied to Mongo DB connector to access it.

58. What is cache scope and until successful & first successful scope in mule?

Ans: Cache Scope: Cache Scope is used in mule to store the data which is used again and again. There are two types of object store we used in caching.

- 1) In memory object store
- 2) Persistent store

Until Successful Scope: Until Successful scope is used for a resource to execute successfully. In this scope we give retries to connect to the resource if the resource connects successfully within given retries then scope gives successful response.

First Successful in Mule: First successful components return payload of the branch which gives successful response first.

59. What is poll scope and watermarking in mule?

Ans: Poll scope: Poll scope is used for polling. If you want to execute the operation within a gap of certain time then we can use poll scope.

Watermarking: Adding new data without replacing old data is known as watermarking. If 10 records are read from database and return in file and new 2 records are added in database so that 2 new records will be added in file without replacing all the records.

60. What is CORS?

Ans: CORS or Cross Origin Resource Sharing is a technique by which we can access our API in JAVA script and other JAVA script supported languages for this we need to enable CORS policy in Anypoint platform.

61. How to validate schema in mule?

Ans: By schema validation filter.

62. How we can send email in mule?

Ans: By using SMTP and attachment transformer in mule.

63. How to use property file in mule?

Ans: For this we need to configure property place holder in global elements. We can access value by \${keyname} anywhere in the mule components.

64. What is filter in mule? Explain At least 5 types of filters?

Ans: Filter is used for filtering task if the given condition is matched then it allows to pass the payload otherwise restrict it.

5 Types of filters are as follows:

- 1) Message filter
- 2) Schema validation filter
- 3) Or filter
- 4) And filter
- 5) Not filter

65. What is scatter gather in mule? How it works in detail.

Ans: Scatter gather is a routing component which routes the payload via different branches, and we get output in the form of array in output node.

By default, if any branch give error than whole operation will not stop, and it return output of the successful branches. If any branch fails, then it gives composite routes exception and we can track the exception in particular branch by sequential ID.

66. What is choice in mule?

Ans: Choice is a routing component in mule which is used to route the message on the basis of given conditions.

67. What is Anypoint exchange?

Ans: Anypoint exchange is an online portal given by mulesoft to store and access assets publicly and private. We can store connectors, components etc in Anypoint exchange.

68. What is object store?

Ans: Object Store is used to store object. There are 2 types of object store mainly we have-

- 1) In memory object store
- 2) Persistent store

69. How can we apply encryption and decryption in mule?

Ans: Encryption and decryption is applied via encryption component in mule. Encryption technique like:

- 1) PGP Encryption
 - 2) JCE Encryption
 - 3) XML Encryption
- can be applied

70. LDAP connector in mule?

Ans: LDAP or Light Weight Directory Access Protocol is used to access network directories. LDAP connector is used in mule to access it. Operations like search, create, update, delete can be used.

71. Diff between map object and map operator in mule?

Ans:

Sr.no	Map Operator	Map Object
1	Output of map operator is array	Output of map object is object
2	Transformation function lambda invokes 2 parameter index and value. Index refers to the positions of key and value.	Transformation function lambda invokes 2 parameter key and value.
3	Index is denoted by \$\$ sign and value by \$	Key is denoted by \$\$ and value by \$

72. Name 10 dataweave operators.

Ans: 1) Split space by

2) Pluck

3) Flatten

4) Now

5) Seil

6) ABS

7) Min

8) Max

9) SQRT

10) Random

73. Thread pools in mule

Ans: There are 3 thread pools in mule.

1) Receiving thread pool – It is used for receiving message in message source.

2) Processing thread pool – It is used for processing the message

3) Dispatching thread pool – It is used to dispatch the message out of the flow.

74. minimum components required for valid flow in mule

Ans: At least one message processor.

75. What is autodiscovery in mule

Ans: Autodiscovery is a technique for API management. It is used for API pairing or API tracking.

76. How can we use if-else condition in mule?

Ans: By using when-otherwise in mule.

77. What is C4E in Mulesoft?

Ans: <https://www.goformule.com/2020/06/c4e-mulesoft.html>

78. What are API best practices?

Ans: <https://www.goformule.com/2020/06/best-practices-for-designing-restful-api.html>

1. What are Web Services?

It's a method of communication between two devices over network, it's accessible via HTTP.

Two machines which are working in different languages eg one is Java another is .net can talk with the help of Web Service using XML/JSON. (it's Language independent way of communication)

1. SOAP web services.

2. RESTful web services.

Any web service has server-client relationship. Any web service can have multiple clients. Eg: a travel portal like Make My Trip is selling tickets of an airliner, So in this, this Portal is client and the Airline is the server where the service is running.

2. What is REST?

REST stands for Representational State Transfer or we say it's RESTful web service. REST is a client-server architecture, in this each unique URL is representation of some resource.

REST API uses HTTP methods.

For example

To create a resource on the server, use POST.

To retrieve a resource, use GET.

To change the state of a resource or to update it, use PUT.

To remove or delete a resource, use DELETE.

Example: If we want to create a REST service that fetches the record of a customer then our URI will be:

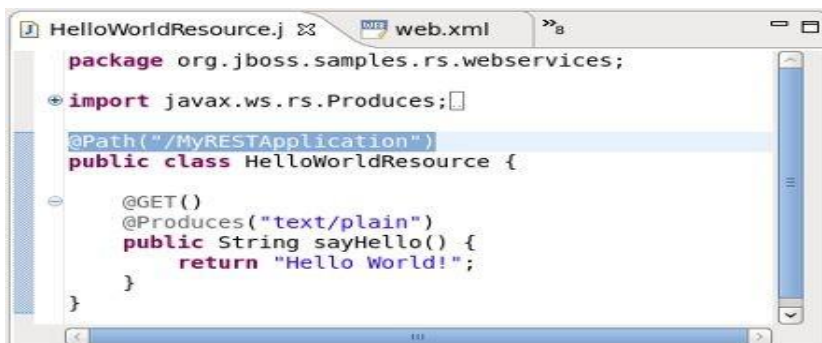
– GET `http://tutorialsAtoZ.com/customer/{customerID}`

where HTTP method is GET,

Resource is customer

And URI parameter is customerID which depicts

for which resource we want to fetch the records.



3. Which HTTP method is safest?

Ans Safe methods are methods that can be cached, prefetched without any impact on the resource.

An idempotent HTTP method is a HTTP method that can be called many times without different outcomes

HTTP Idempotent Safe

OPTIONS	yes	yes
GET	yes	yes
HEAD	yes	yes
PUT	yes	no
POST	no	no
DELETE	yes	no
PATCH	no	no

4. What is a web service API?

An API (Application Programming Interface) is the means by which third parties can write code that interfaces with other code. A Web Service is a type of API, one that almost always operates over HTTP (**though some, like SOAP, can use alternate transports, like SMTP**)

a Web service API almost always uses HTTP (hence the Web part), and definitely involves communication over a network.

5. What's the difference between SOAP and REST?

SOAP is a protocol	REST is architectural style
SOAP means Single Object Access Protocol	REST means Representational state transfer
SOAP cannot use REST	REST can use SOAP as a protocol
P uses service Interfaces to expose the business logic	REST uses the URI to expose the business logic

JAX-WS is java api for soap	JAX-RS is java api for REST
It permits XML only	It permits XML,JSON,HTML and others
Consumes more bandwidth	Consumes less bandwidth

6. What is RAML and why we use it?

RAML – RESTful API Modeling Language

it's similar to WSDL(Web Services Description Language), it contains endpoint URL, request/response schema, HTTP methods and query and URI parameter.

it helps client (I mean a consumer of the service) in knowing, what the service is and what are all operations and how can be they be invoked.

it helps the us in creating the initial structure of this API.

And can also be used for documentation purpose.

7. What Is Mule ?

Mule ESB is a Java-based enterprise service bus (ESB) and integration platform, developer can connect their application with ESB. Mule use service oriented architecture. Apart from of the different technologies the applications use, including JMS, Web Services, SMTP, HTTP. The advantage of ESB, it allows different applications to communicate with each other. Messages can be of any format XML to JSON. Mule ESB Development provide messaging framework that enable exchange of data among application.

8. Why we use ESB? What is the basic responsibility of ESB?

Normally when we connect two application for ex using webservices it would be

Point-to-Point Integration Challenges

- Applications connect to other applications in a point to point manner
 - Approach works for small scale integration projects
 - Complexity increases as
 - integration points grow
 - Data received over multiple protocols
 - Need of such data format which could be understood by all
 - applications become mission critical
 - Centralization of concerns like security, auditing, routing
 - Appropriate error handling
- An ESB should provide means to handle all these challenges properly.
- Mule provides solutions for the following technical integration problems
 - Connectors for integrating with different systems
 - Different protocols for input – REST, File etc
 - Different data Format for exchange – JSON, XML, Java Object
 - Support for multiple Invocation Styles – Synchronous, Asynchronous, RPC, Messaging, Batch call
 - Life Cycle and Management of mule elements
 - Error Handling

9. Why Mule Esb?

Mule ESB is lightweight but highly scalable, allowing us to start small and connect more applications over time. Mule manages all the interactions between applications and components transparently, regardless of whether they exist in the same virtual machine or over the Internet, and regardless of the what is transport protocol used.

There are currently several commercial ESB implementations on the market. However, many of these provide limited functionality or are built on top of an existing application server or messaging server, locking you into that specific vendor. Mule is vendor-neutral, so different vendor implementations can plug in to it. You are never

locked in to a specific vendor when you use Mule.

10. What Difficulties Mule Does Encompass?

Transport: applications can accept input from a variety of means, from the file system to the network.

Data format: speaking the right protocol is only part of the solution, as applications can use almost any form of representation for the data they exchange.

Invocation styles: synchronous, asynchronous, or batch call semantics entail very different integration strategies.

Lifecycles: applications of different origins that serve varied purposes tend to have disparate development, maintenance, and operational lifecycles.

11. Why The Name Mule ?

There is a lot of infrastructure work to be done before we can really start thinking about implementing any logic. So this infrastructure work is regarded as “donkey work” as it needs doing for every project. A Mule is also commonly referred to as a carrier of load, moving it from one place to another. The load it specializes in moving is our enterprise information.

12. What Are Available ESbs Apart From Mule ?

All major JEE vendors (BEA, IBM, Oracle, Sun) have an ESB in their catalog. It is unremarkably based on their middleware technologies and is usually at the core of a much broader SOA product suite. There are also some commercial ESBs that have been built by vendors not in the field of JEE application servers, like the ones from Progress Software, IONA Technologies, and Software AG.

13. What Are Differences Between Mule And Other Commercial Esbs ?

Prescriptive deployment model, whereas Mule supports a wide variety of deployment strategies. Prescriptive SOA methodology, whereas Mule can embrace the architectural style and SOA practices in place where it is deployed. Mainly focused on higher-level concerns, whereas Mule deals extensively with all the details of integration. Strict full-stack web service orientation, whereas Mule’s capacities as an integration framework open it to all sorts of other protocols. Comprehensive documentation, a subject on which MuleSource has made huge progress recently.

14. What Is Model Layer In Mule ?

The first logical layer is the model layer. A Mule model represents the runtime environment that hosts services. It defines the behavior of Mule when processing requests handled by services. The model provides services with supporting features, such as exception strategies. It also provides services with default values that simplify their configuration.

15. What Is Service Layer In Mule ?

A Mule service is composed of all the Mule entities involved in processing particular requests in predefined manners. A service is defined by a specific configuration. This configuration determines the different elements, from the different layers of responsibility, that will be mobilized to process the requests that it will be open to receive. Depending on the type of input channel it uses, a service may or may not be publicly accessible outside of the ESB.

16. What Is Transport Layer In Mule ?

The transport layer is in charge of receiving or sending messages. This is why it is involved with both inbound and outbound communications. A transport manifests itself in the configuration in connectors, endpoints and transformers.

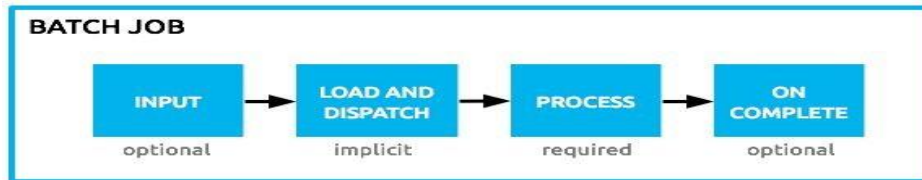
A transport also defines one message adapter. A message adapter is responsible for extracting all the information available in a particular request (data, meta information, attachments, and so on) and storing them in transport-agnostic fashion in a Mule message.

17. Batch processing

Mule possesses the ability to process messages in batches. A batch job is a block of code that splits messages into individual records and perform operations on each record.

A batch job executes when triggered by either a batch executor in a Mule flow or

a message source in a batch-accepting input; when triggered, Mule creates a new batch job instance.



Batch Processing in Mule has four phases.

➤ Input Phase

- Input Phase is an optional part in a batch job. You can trigger a batch job, by placing an inbound connector (a message source) in the input Phase of batch job.
- If using batch executor (in Mule flow) to trigger the batch job, then we can accommodate any transformations or adjustments to a message payload before Mule begins processing it as a batch, as batch Step accepts only payloads of type interface java.util.Collection, java.lang.Iterable..

➤ Load and Dispatch

- This is implicit and is running behind the scenes. During this phase, Mule creates a batch- job-instance for a message received.
- Mule creates a persistent queue for a batch-job-instance and stores all the records of the payload. Then each record is sent to the first BatchStep for processing.

➤ Process Phase

- We can use multiple batch steps here to process the records.
- Each batch Step has the following configuration
 - Accept Expression: BatchStep processes a particular record only when this expression is evaluated to true (corresponding to that record). If false the corresponding record will be skipped, and batchStep tries to execute next record in the queue.
 - AcceptPolicy: There are three categories under this. They are whether to accept all the records (ALL), Only FailedRecords (ONLY_FAILURES), Only SuccessfulRecords (NO_FAILURES). The default Accept policy is NO_FAILURES.

➤ OnComplete Phase

- This is the fourth phase in the batch job. After mule has executed entire batch, the output is an object BatchJobResult.
- This can be used to create report of the no. of Records processed, Count of successful and failed records. So, that we can take appropriate action on what type of input data records are getting failed.

Batch Commit

- We can use a batch Commit in a batch Step to accumulate a subset of records within a batch for bulk upsert to an external source or service.
- We can set the chunk size or the number of records to be taken as subset using CommitSize attribute.
- However, if we want all the records to be written in one shot, we can use the Streaming attribute of BatchCommit. This will wait until all the records reach batchCommit and write all records in one go with streaming enabled.
- BatchCommit will be helpful particularly when writing to Database, Salesforce connectors, as records number is high we cannot write each record everytime to them.

18. What are different elements of a Mule Application

Elements of a Mule Application

- a. Message Sources
- b. Messages
- c. Flows, Sub-flows and Batch Processing
- d. Message Processors
- e. Exception Strategies

19. Diff between async scope and asynchronous flow

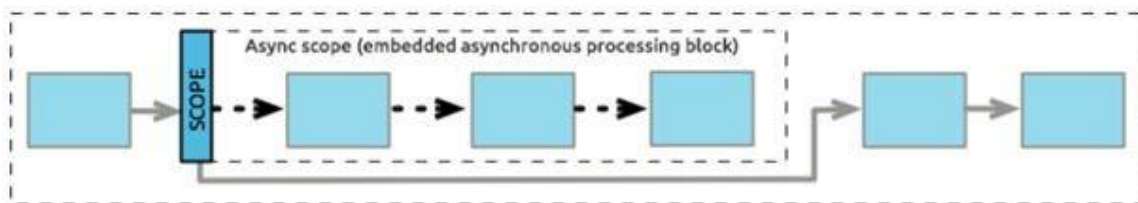
Async Scope: An **async scope** is a branch processing block that executes simultaneously with the parent message flow, Async scope invokes the contained message processors in a separate thread

- Cannot be reused
- **Source cannot be added**
- Cannot have its own exception strategy

Asynchronous Flow: contents of the flow will be processed asynchronously with the triggering flow, it can perform **time-consuming tasks**, without stalling the flow that triggered its execution. In clusters of Mule servers, messages can migrate between nodes when sent to an asynchronous flow. This allows for load balancing between nodes and higher performance of application. This type of flow does not inherit processing or exception strategies from the triggering flow instead they have their own.

20. What is async scope

An **async scope** is a branch processing block that executes simultaneously with the parent message flow. This type of processing block can prove useful for executing time-consuming operations (such as printing a file or connecting to a mail server) — as long as those operations do not require sending a response back to the initiating flow. In other words, the main flow can continue execution while it initiates and processes the asynchronous scope; it does not have to pause until the last message processor embedded in the asynchronous flow has completed its task. To facilitate this simultaneous branch processing, the async scope sends one copy of the message it has received to the first embedded message processor in its own processing block; at the same time it sends another copy of the message to the next message processor in the main flow (see below).



21. What happens if async scope is inside until successful

The flow will stuck.

As async scope will not return anything and Until successful will expect some response to fulfill until-successful condition.

22. What are the different ways to connect a database with Mule DB connector?

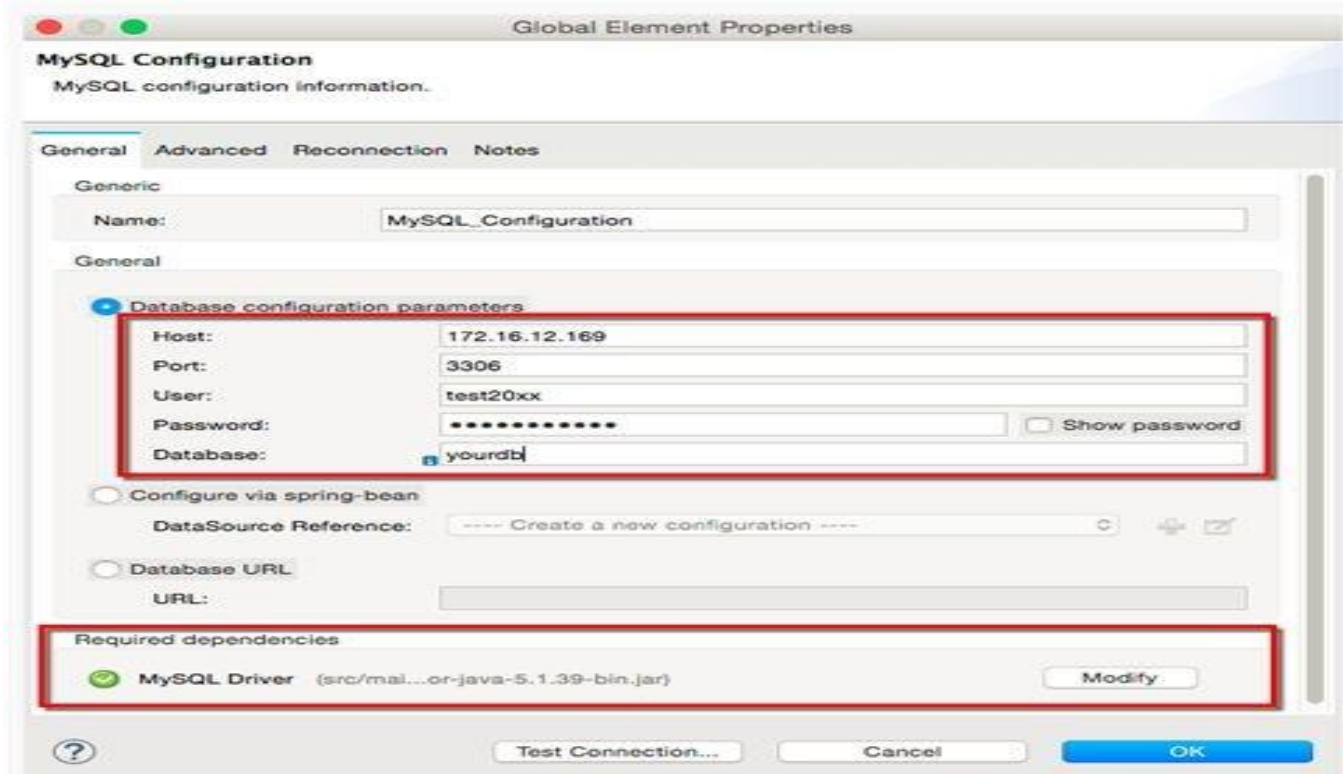
The Database connector allows you to connect with database with almost any Java Database Connectivity (JDBC) relational database.

Database connector and its required fields

Required fields:

- Host
- Port
- Username
- Password
- Database

Adding jar is very important. (In required dependencies)



23. What are Shared Resources in Mule and how are they been used?

We can make connectors as a reusable component by defining them as common resources and expose them to all applications deployed under a same domain, these resources are known as shared resources. These shared resource needs to be defined inside Mule Domain Project and then referred to each of the projects that are meant to use the elements in it.

Mule project is normal project, Normal project will have the all dependencies within the project scope itself, if you Created Mule Domain Project where we can keep all the common functionality, so that we can access from the multiple projects.

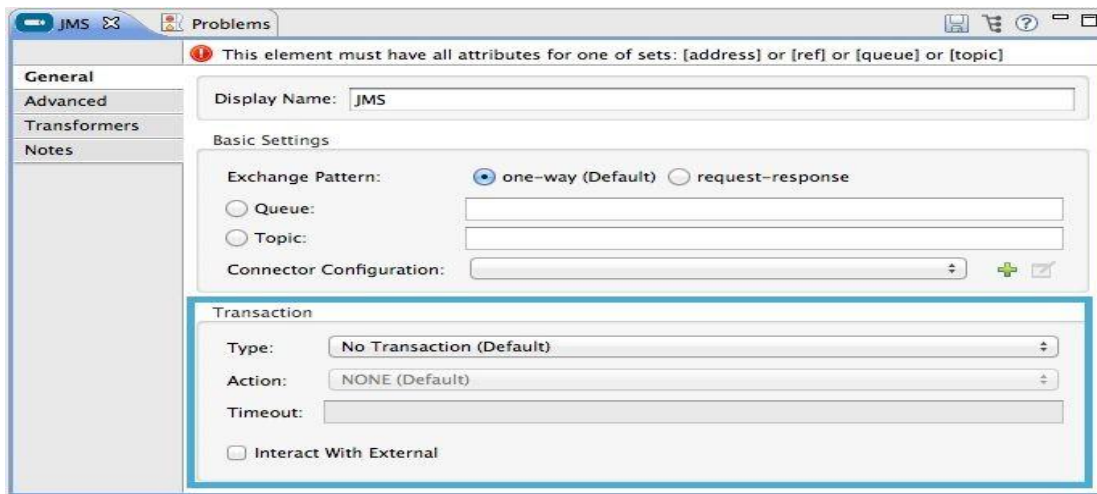
*to get this, we need to create domain project, to access in Mule Project we need to add the the domain project name in the **mule-deploy.properties***

24. What and how to use transaction management in Mule?

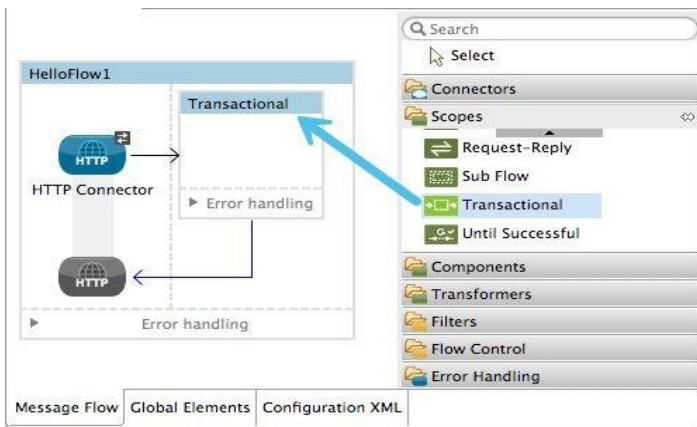
Transactions in mule

where a series of steps in a flow must succeed or fail as one unit, Mule uses a transaction to denote such a unit. For example, you might use a transaction to encapsulate several steps in a flow for which the end result involves committing information to a database. In this type of scenario, the commit is either entirely complete and succeeds, or is incomplete and it fails. Even if partially complete, the commit – or transaction – fails. Where a transaction fails, Mule rolls back the operations within the transaction so that no one part results in partial completion.

Transaction to a Connector

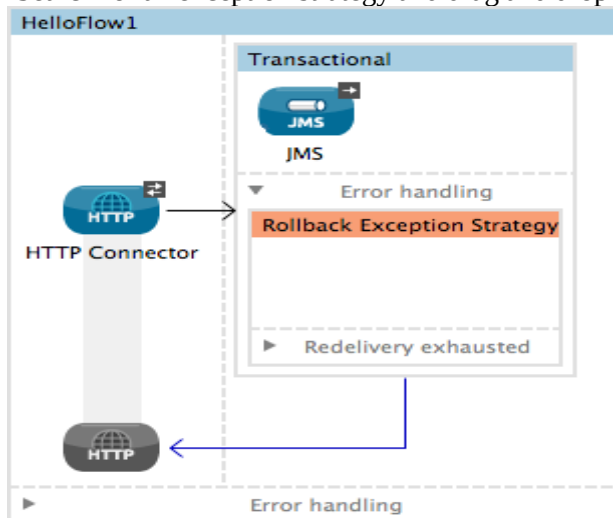


Transaction as a Scope



Exception strategy in Mule

Add a **Transactional** scope to your flow , then add building blocks within the scope to build a transaction. Search for an exception strategy and drag and drop it into the exception strategy section at the bottom of the scope.



Configure the exception strategy as needed, keeping in mind Mule uses this exception strategy to handle any exceptions thrown while processing the transaction.

25. what is Mule Flow?

Mule applications are built around one or more flows, flow of message from message source to one or more message

processors. Typically, a Mule application begins processing a message it receives at an inbound endpoint in a flow.

This flow can then either:

- a. implement all processing stages, or
- b. route the message to other flows or subflows to perform specific tasks.

26. What are the different types of flow

Subflow: A subflow processes messages synchronously (relative to the flow that triggered its execution) and always inherits both the processing strategy and exception strategy employed by the triggering flow. While a subflow is running, processing on the triggering flow pauses, then resumes only after the subflow completes its processing and hands the message back to the triggering flow.

Synchronous Flow: A synchronous flow, like a subflow, processes messages synchronously (relative to the flow that triggered its execution). While a synchronous flow is running, processing on the triggering flow pauses, then resumes only after the synchronous flow completes its processing and hands the message back to the triggering flow. However, unlike a subflow, this type of flow does not inherit processing or exception strategies from the triggering flow.

It's like single thread processing. Can be used in transactional processing

Asynchronous Flow: An asynchronous flow simultaneously and asynchronously processes messages in parallel to the flow that triggered its execution. When a flow passes a message to an asynchronous flow, thus triggering its execution, it simultaneously passes a copy of the message to the next message processor in its own flow. Thus, the two flows – triggering and triggered – execute simultaneously and independently, each finishing on its own. This type of flow does not inherit processing or exception strategies from the triggering flow instead they have their own.

This type of flow processes messages along multiple threads.

Advantages of Using Multiple Flows in an Application

- Asynchronous Flow B can perform time-consuming tasks, such as writing data to an external database or emailing a message, without stalling Flow A, the flow that triggered its execution.
- Flow A and Flow B can respond differently to errors.
- Breaking up complex operations into a series of smaller flows or subflows makes applications – whether in a GUI or in XML code – easier to read.
- The processing actions in a flows or subflows can be called and used by multiple flows in an application.

27. Difference between flow and sub-flow.

Flow is a message processing block that has its own processing strategy and exception handling strategy. Used in integration tasks, data processing, connecting applications, event processing, etc.

Subflow always processes messages synchronously but inherits processing strategy and exception handling strategy from the calling flow. It can be used to split common logic and be reused by other flows.

28. Difference between private flow and sub-flow.

Private flow has Processor and own Exception Strategy. **Private flow** does not have source define. It can be synchronous or asynchronous based on the processing strategy selected.

Sub-flow consists only of Processor. **Subflow** always processes messages synchronously but inherits processing strategy and exception handling strategy from the calling flow. It can be used to split common logic and be reused by other flows.

29. What are Flow Processing Strategies?

Flow processing strategies A flow processing strategy determines how Mule implements message processing for a given flow. Mule automatically determines best processing strategy Implicitly, set to either synchronous or queued-asynchronous.

A flow is synchronous when – (single thread)

1. The flow's exchange pattern is request-response –
2. The flow partakes in a transaction –
3. The flow is really a sub-flow

A flow is asynchronous in all other cases (multi thread) : flows with one-way exchange patterns and which are not transactional

(Unless overridden by manually setting a processing strategy)

Synchronous vs asynchronous flow difference

Synchronous – (You cannot force a flow with a request-response exchange pattern and/or transactionality to be asynchronous)

1. Same thread is used across the flow
2. If outbound endpoint is one way (JMS), then the thread will be used to continue the message processing
3. If outbound endpoint is request-response, then the thread will wait for the response from the endpoint and continue the message processing after receiving the response

Asynchronous-(You can override an implicitly selected queued-asynchronous processing strategy by explicitly declaring a synchronous processing strategy (or, in rare use cases, a different kind of asynchronous processing strategy) instead.)

1. Receiver thread receives the message and places it in Staged Event-Driven Architecture (SEDA) Queue
2. Receiver thread will be released for receiving message after placing in queue
3. Message will be placed in a dispatcher queue for sending out

In Async Mule uses a queue to decouple the receiver thread from the rest of the flow.

30. Default processing strategy of flow

Exchange Pattern	Transactional?	Flow Processing Strategy
Request-Response	Yes	Synchronous
Request-Response	Yes	Synchronous
One-way	Yes	Synchronous
One-way	No	Queued-Asynchronous

31. What is a Mule Message Processor?

Message Processors operate on the messages in a flow.

- [Connectors](#)
- **Transformers**
- [Routers](#)
- **Filters**
- [Components](#)
- **Scopes**
- [Exception Strategies](#)

It is possible to create custom message processors by implementing the `org.mule.api.processor.MessageProcessor` in a **custom** class

32. What Is Connector In Mule ?

Connectors are message processor in Mule which are used for:

Sending Data to a flow
Receiving Data from a flow
Interacting with APIs

Types of Connectors

Endpoint based Connectors

Used for sending and receiving data over a protocol like Http, JMS
Follow either a one-way or request-response exchange pattern

Operation Based Connectors

Used for interacting with APIs like Twitter, Salesforce, MongoDB
Operation-based connectors follow an information exchange pattern based on the operation selected

EndPoint Based connectors

- Endpoint based connectors are configured as either inbound or outbound endpoints in a flow.
- Inbound endpoints serve as a message source for a flow.
- Outbound endpoints can occur mid-flow or at the end of flows, and send information to external systems. Eg File,Http,JMS,Email,FTP,JDBC,VM

Operation Based connector

- **Operation-based connectors** define a specific operation for the connector to perform.

Global Connector Configuration

- Connectors can be configured as a global element rather than at the level of the message processor within the flow.
- The global connector can be referenced by connectors of the same type in an application.
- For operation-based connectors, the global connector configuration is mandatory.
- For endpoint-based connectors global connector configuration it is optional.
- Eg:

<connectorName>:config for operation-based connectors

<connectorName>:connector for endpoint-based connectors.

33. What is the difference between Endpoints and connectors?

Endpoints are a type of connector. Connectors in Mule are either endpoint-based or operation-based. Endpoint-based connectors follow either a one-way or request-response exchange pattern and are often (but not always) named and based around a standard data communication protocol, such as FTP, JMS, and SMTP. Operation-based connectors follow an information exchange pattern based on the operation that you select and are often (but not always) named and based around one or more specific third-party APIs.

34. Which connectors you have used.

HTTP, HTTPS, FTP, SFTP, Database, File.... Etc.

35. What are required fields of connectors you have used

Remember the fields of connectors which u have used. For example: What if file Age and polling frequency in FILE connector.

36. When u use FTP and SFTP connector

FTP: The FTP Connector implements a file transport channel so that your Mule application can exchange files with an external FTP server. You can configure FTP as an inbound endpoint (which receives files) or outbound endpoint (which writes files to the FTP server).

SFTP: The SFTP Connector implements a secure file transport channel so that your Mule application can exchange files with external resources. SFTP uses the SSH security protocol to transfer messages. You can implement the SFTP endpoint as an inbound endpoint with a one-way exchange pattern, or as an outbound endpoint configured for either a one-way or request-response exchange pattern

37. Diff between HTTP and HTTPS

HTTPS is secure HTTP, HTTP with SSL(secure socket layer)

It uses .jks file for handshake, which is present in src/main/resource folder.

<https://docs.mulesoft.com/mule-user-guide/v/3.8/tls-configuration>

TLS (Transport layer security its a successor of SSL) configuration Element is used to provide security with the help of trust store and keystore

```
<tls:context name="customContext">
```

```
  <tls:trust-store path="trustStore" password="mulepassword"/>
```

```
  <tls:key-store path="clientKeystore" keyPassword="mulepassword"
```

```
password="mulepassword"/>
```

```
</tls:context>
```

The tls:context element defines a configuration for TLS, which can be used from both the client and server sides. It can be referenced by other configuration objects of other modules (or defined as a nested element of one of them). Inside it, you can include two nested elements: key-store and trust-store. You don't need to include both, but at least one of the two must be present:

- From the server side: the trust store contains certificates of the trusted clients, the key store contains the private and public key of the server.
- From the client side: the trust store contains certificates of the trusted servers, the key store contains the private and public key of the client.

Adding a trust store or a key store to a TLS configuration implicitly implements the corresponding kind of authentication. Adding both a keystore and a trust store to one same config (as in the code example above) implicitly implements **two way TLS authentication**, also known as **mutual authentication**.

38. How to build a Custom connector

<https://docs.mulesoft.com/anypoint-connector-devkit/v/3.8/>

Annotations:

@Connector: For java class, for giving name to connector

@processor: java methods

@Configurable: For data members

```
@Connector(name="connpom", friendlyName="Connpom")
```

```
public class ConnpomConnector {
```

```
    @Config
```

```
    ConnectorConfig config;
```

```
    * @param friend Name to be used to generate a greeting message.
```

```
    * @return A greeting message
```

```
    @Processor
```

```

    public String greet(String friend) {

        return config.getGreeting() + " " + friend + ". " +
config.getReply();

    }

    public ConnectorConfig getConfig() {

        return config;}

    public void setConfig(ConnectorConfig config) {

        this.config = config;}}

```

ConnectorConfig Class:

```

@Configuration(friendlyName =
"Configuration")

```

```

public class ConnectorConfig {

    @Configurable

    @Default("Hello")

    private String greeting;

    @Configurable

    @Default("How are you?")

    private String reply;

    public void setGreeting(String greeting) {

        this.greeting = greeting;

    }

    public String getGreeting() {

        return this.greeting;

    }

    public void setReply(String reply) {

        this.reply = reply;

    }

    public String getReply() {

        return this.reply;

    }

}

```


39. What Is Endpoint In Mule ?

An endpoint represents the specific usage of a protocol, whether it is for listening/polling, reading from, or writing to a particular target destination. Hence it controls what underlying entities will be used with the connector they depend on. The target destination itself is defined as a URI. Depending on the connector, the URI will bear a different meaning; for example, it can represent a URL or a JMS destination.

40. What Is Transformer In Mule ?

A transformer takes care of translating the content of a message from one form to another. It is possible to chain transformers to cumulate their effects. Transformers can kick in at different stages while a message transits through a service.

41. What Is Router In Mule ?

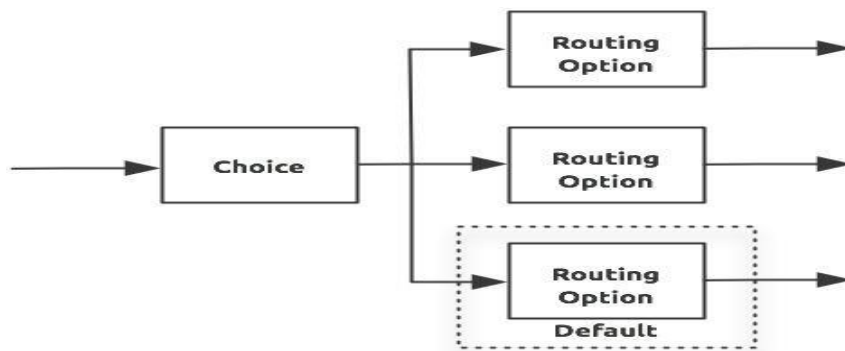
Routers play a crucial role in controlling the trajectory a message will follow when it transits in Mule. Routers are used to route data between applications

Eg are choice router, splitter , scatter gather

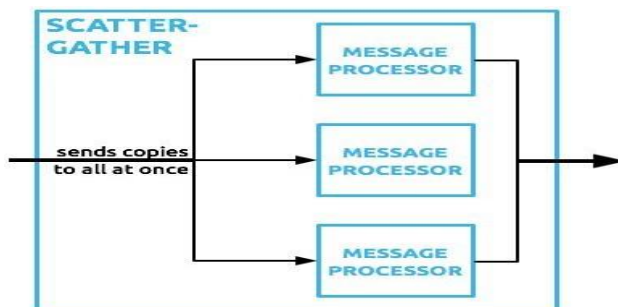
42. Choice router, scatter gather and splitter flow

Choice Router: The choice flow control dynamically routes messages based on message payload or properties. It adds conditional programming to a flow, similar to an if/then/else code block.

A choice flow control uses [expressions](#) to evaluate the content of a message, then it routes the message to one of the routing options within its scope (see image below). It directs messages to the first routing option in the scope that matches the routing configurations (evaluates to true). If none of expressions evaluate to true, the choice flow control directs the message to the default (else) route.



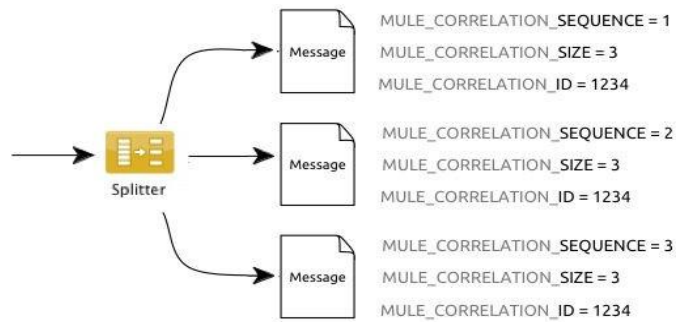
Scatter-Gather: The routing message processor Scatter-Gather sends a request message to multiple targets concurrently. It collects the responses from all routes, and aggregates them into a single message.



Splitter: The Splitter Flow Control splits a message into separate fragments, then sends these fragments one at a time to the next message processor in the flow. Segments are identified based on an expression parameter, usually written in [Mule Expression Language \(MEL\)](#), but other formats can be employed also. You can then use a Collection Aggregator Flow Control to reassemble the parts of the original message. You can also include a **Resequencer** Flow Control to put the parts back into the original sequence in case they are shuffled out of order.

Splitting and aggregating the message is especially useful when you intend to process the split parts in asynchronous

flows running on separate servers. Together, the splitter and aggregator flow controls allow you to share the workload among several servers and still be able to reassemble the message after it's processed.



43. Explain Scatter Gather. If 1 of the Scatter Gather o/p will have exception then other will be executed or not? What will be final output?

If there are no failures, Mule aggregates the results from each of the routes into a message collection (MessageCollection class). Failure in one route does not stop the Scatter-Gather from sending messages to its other configured routes, so it is possible that many, or all routes may fail concurrently.

By default, if any route fails, Scatter-Gather performs the following actions:

- sets the exception payload accordingly for each route
- throws a CompositeRoutingException, which maps each exception to its corresponding route using a sequential route ID

Catching the CompositeRoutingException allows you to gather information on all failed routes.

44. What Is Filter In Mule ?

Filters are a powerful complement to the routers. Filters provide the brains routers need to make smart decisions about what to do with messages in transit.

- Filters are used to decide on what message continues processing through a flow.
- Filtering can be done based on the message type, payload or headers.
- Payload-type-filter
 - Message passes through this filter only if the payload is of a required type
- Expression Filter
 - Is used to filter based on the result of a MEL expression
- Wildcard Filter
 - Messages that match a shell-like wildcard will be allowed to pass.
 - Is generally used for filtering text messages.
- Regular Expression Filter
 - Filtering is done based on a regular expression.
 - Is very useful for filtering text messages.

45. What Is Component In Mule ?

Components are message processors which execute business logic on messages

, Spring bean, Java bean, or script can be configured as components in a Mule application, Mule includes several scripting components that can be used in flows

eg of components are Groovy script component, Java Component, Web Service Components etc

46. Difference between Java transformer and Java component

If you want to trigger some more complex set of processes coded in Java, or your complex business logic you should better use Java component
On the other hand Java transformer can be used to create a custom transformer that can change or modify the message payload

47. Java component has 2 functions, how to invoke one as per requirement

Two methods:

1. In xml use

```
<method-entry-point-resolver>
```

```
<include-entry-point method = "xyz">
```

```
</method-entry-point-resolver>
```

2. Use invoke component : using invoke component from palette and giving object ref of which we want to invoke method and method name in method in General tab.

48. What is scope?

Mule **scopes** act as wrappers for message processors to make them function as a single unit.

- a. Async
- b. For each
- c. Transactional
- d. Until Successful

49. What is For-each scope

The **For Each** scope splits a collection into elements and processes them iteratively through the processors which are embedded in the scope, then returns the original message to the flow.

50. Until successful is used for?

The **until-successful** scope processes messages through its processors until the process succeeds. By default, until-successful processing occurs asynchronously from the main flow. After passing a message into the until-successful scope, the main flow immediately regains control of the thread. but, we can configure until-successful to run synchronously relative to the main flow.

Until-successful repeatedly retries to process a message that is attempting to complete an activity such as:

Dispatching to outbound endpoints, for example, when calling a remote web service that may have availability issues.

51. Various types of Exception Handling?

Types of Errors that can occur in Mule Applications:

- System Exceptions
- Messaging Exceptions.

Exception Handling Strategies

- Default Exception handling
- Catch Exception handling
- Rollback Exception handling
- Reference
- choice

System Exception Strategy

- System Exception Strategy is invoked when an exception is thrown at the system- level (i.e., when no message is involved).
 - System exceptions can occur during application start-up or when a connection to an external system like JMS Broker fails
 - When a system exception occurs, Mule sends an exception notification to registered listeners, logs the exception. It executes a reconnection strategy if the exception was caused by a connection failure
- System Exception Strategies are not configurable in Mule but Reconnection strategy can be configured.

Messaging Exception Strategy

- Messaging Exception occurs when an exception is thrown within a flow whenever a message is involved.
- When a message being processed through a Mule flow throws an exception, normal flow of execution stops. Mule transfers the message to the message processor sequence within the exception strategy.

For the same we have multiple exception strategies

❖ **Default Exception Strategy**

- Applied by default for all Messaging Exceptions if no other exception strategy matches for that exception
- Causes rollback of any transaction if it exists and then logs the exception

❖ **Catch Exception Strategy**

- Invokes a customized flow to handle the exception
- Consumes the inbound message and commits the transaction

Mule's catch exception strategy behavior is similar to a Java catch block, except that you cannot throw anew exception or catch another exception within a catch exception strategy

❖ **Rollback Exception Strategy**

- Inbound message is not consumed
- Causes rollback of the message. In some cases, where the message supports, causes redelivery of the message.
- If the message exceeds the redelivery attempts (defined by maxRedeliveryAttempts attribute), the flow defined by <on-redelivery-attempts-exceeded> section is invoked, failed transaction is committed and the message is consumed.

❖ **Reference Exception Strategy**

- Does not itself consume the message or cause rollback or commit
- References a global exception handler defined with a global catch, rollback or choice exception strategy.

❖ **Choice Exception Strategy**

- Does not itself consume the message or cause rollback or commit
- Routes to an exception strategy from a set of strategies based on an expression defined with the when attribute for each exception strategy.
- If a suitable exception strategy is not found then the default exception strategy is invoked.

Characteristics of Messaging Exception Strategies

- Each flow can contain only one exception strategy.
- Each exception strategy can contain any number of message processors.
- Choice exception strategies can contain one or more catch and/or rollback exception strategies.
- Rollback and catch exception strategies cannot contain other exception strategies.
- Global exception strategies can be created and reused in flows.

52. What is custom exception and global exception?

53. What u should do if you want to refer same exception strategy in all flows present?

Two ways:

1. Make it as global exception strategy and refer it through Reference exception strategy
2. Make it as default exception strategy through global element.

54. How Message In Mule Is Composed and What Is Payload In Mule ?

Answer :

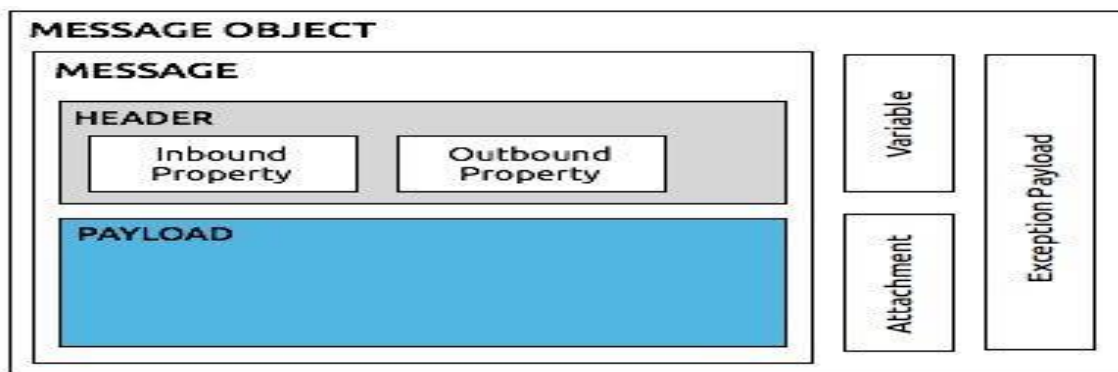
The Mule message is the data that passes through an application via one or more flows. It consists of two main parts, The message header, which contains metadata about the message, contains inbound and outbound properties, Inbound properties are immutable and outbound properties are mutable. The message payload, which contains your business-specific data.

Inbound properties are automatically generated by the message source and are immutable

- They are immutable and cannot be set or manipulated by the user.
- They contain metadata specific to the message source
- A message retains its inbound properties only for the duration of the flow
- Inbound properties are not carried forward in case a message moves from one flow to another.

Outbound properties are mutable

- **Outbound properties** are set automatically by Mule or a user can set them by using one or more transformer elements in the flow.
- They contain metadata about the message similar to that of an inbound property
- Outbound properties become inbound properties when the message passes from the outbound endpoint of one flow to the inbound endpoint of a different flow through a connector.
- Outbound properties remain outbound properties if the message moves to a new flow via a flow-ref rather than a connector.



The content of a message, also known as payload. It is wrapped in an instance of `org.mule.api.MuleMessage`, which provides different means of accessing the payload under different forms. A `MuleMessage` also contains properties, much like the header of a SOAP envelope or the properties of a JMS message, and can also have multiple named attachments.

To set message payload we use **Set Payload message processor**, can completely replace the content of the Message's payload.

55. What Are Different Type Of Messages In Mule ?

Bridge messages: Pass messages from inbound to outbound routers.

Echo and Log messages: Log messages and move them from inbound to outbound routers.

Build messages: Create messages from fixed or dynamic values

56. What are the different types of variables in mule?

Variables are name value pairs that represent data about a message.

Different kind of variables depending upon their scope

- Flow variables exist only in the flow in which they exist.
- Session variables exist across all flows within the same application.
- Record variables represent records processed as part of a batch.

57. What is message enricher

a message enricher enriches the current payload with some additional message or information and this is done without disturbing the current payload.

Enricher is used if the target system needs more information than the source system can provide. It enriches the Mule message by calling an external system or doing some transformation to the existing payload and saving it into some scope of variable, like session, and the transformation happening in the enricher scope doesn't affect the actual payload.

which method is used for it?

Set-property: Save some information extracted from payload or original payload to some invocation or flow scope variable.

how Message Enricher works.

- Enricher sends a copy of the original message into the processor.
- The original message waits.
- The copy is processed.
- The copy's response is a message.
- Part(s) of the response are added to part(s) of the original message.
- The enriched message moves forward.

58. What Are Configuration Builders In Mule ?

Answer :

Mule uses configuration builders that can translate a **human-authored** configuration file into the complex graph of objects that constitutes a running node of this ESB. The main builders are of two kinds: a Spring-driven builder, which works with XML files, and a script builder, which can accept scripting language files.

59. Why Spring-driven Configuration Builder Is Important Than Script Builder ?

Answer :

The advantages of Spring-driven configuration builder

It is the most popular — you are more likely to find examples using this syntax.

It is the most user friendly — Spring takes care of wiring together all the moving parts of the ESB, something you must do by hand with a script builder.

It is the most expressive — dedicated XML schemas define the domain-specific language of Mule, allowing you to handle higher-level concepts than the scripting approach does.

60. What Is Bridge Component In Mule ?

Answer :

A bridge component is used to pass the messages from the inbound router to the outbound one. A bridge is a neutral component: it does not perform any action or modify messages that it processes.

61. What Tags Are Used To Configure Spring Elements In Mule ?

Answer :

Tags like <spring:bean/> <spring:entry/> etc. are used to configure spring stuff.

62. Give An Example Of Stdio Connector In Mule ?

Answer :

```
<stdio:connector name="SystemStreamConnector" promptMessage="Please enter something: "
    messageDelayTime="1000"/>
```

63. Give An Example Of Http Connector In Mule ?

Answer :

```
<http:connector name="HttpConnector" proxyHostname="proxyHostname"
```

```
proxyPort="proxyPort"
proxyUsername="proxyUsername"proxyPassword="proxyPassword"/>
```

64. When Does Mule Instantiates A Connector ?

If Mule figures out that one of our endpoints needs a particular connector, it will automatically instantiate one for us, using all the default values for its different configuration parameters. **This is a perfectly viable approach if we are satisfied with the behavior of the connector when it uses its default configuration.** This is often the case for the VM or HTTP transports. **Note that Mule will name these default connectors with monikers such as connector.http.0.**

65. What Is Transport Service Descriptor In Mule ?

The connector has a technical configuration known as the Transport Service Descriptor (TSD). This hidden configuration is automatically used for each instance of the connector. It defines technical parameters such as what classes to use for the message receivers, requesters, and dispatchers; or the default transformers to use in inbound, outbound, and response routers. **Knowing these default values is essential to grasping the behavior of a transport.**

66. How Many Endpoints Are There In Mule ?

Answer :

There are two endpoints :

inbound

outbound.

You will use inbound and outbound endpoints to communicate between components and services inside Mule as well as with the outside world.

67. what is Inbound Endpoint in Mule?

Inbound endpoints are message sources which as the name suggests is where messages are created. They can be created based on external events (like an incoming HTTP request or JMS message)

68. What Is An Outbound Endpoint In Mule ?

Outbound endpoints are used to send data. An outbound endpoint is used to do things such as write to file streams, and send email messages.

Outbound endpoints are message processors that send the current message to "destinations" like a JMS queue, an HTTP server, a file, ...

69. What Is Global Endpoint In Mule ?

An endpoint destination that is shared by several routers, it is worth creating a global endpoint. A global endpoint is not typified for inbound or outbound routing, making it usable in many different places in a configuration file. It must be named so it can actually be used in a service, which will reference the global endpoint by its name.

70. Why Does An Endpoint In Mule Offer An Address Attribute ?

This allows us to configure a generic endpoint using the Mule 1.x style of URI-based destination addresses instead of the dedicated attributes of the specific endpoint element.

71. Give An Example Of File Endpoint In Mule ?

```
<file:endpoint name="tmpPoller" path="/tmp" fileAge="1000"pollingFrequency="2000"/>
```

72. What Is Streaming Property In File Connector In Mule ?

The value of this streaming property can be either true or false. If it is set to true then we are actually working on stream of file data otherwise we are working with file itself.

73. What Is Pollingfrequency Property In File Connector In Mule ?

When we want file inbound endpoints to poll their source directories for new content. This is accomplished by setting the pollingFrequency to some milliseconds value.

74. What Is Autodelete Property In File Connector In Mule ?

The default value of autoDelete is true. Therefore, a file inbound endpoint will, by default, remove the file from the source directory once it is read by the inbound endpoint. If you do not want to delete file automatically then you can set autoDelete property to false.

75. What Is Fileage Property In File Connector In Mule ?

The fileAge property specifies how long the endpoint should wait before reading the file again. For instance, a fileAge of 60000 indicates Mule should wait a minute before processing the file again.

76. How To Send Only Certain Types Of File From One Directory To Another In Mule ?

Use the below element in file inbound to filter certain types of files.

<file:filename-wildcard-filter pattern="file*.xml"/></file:inbound-endpoint> pattern indicates what pattern of file names should move from one directory to another directory.

77. What Is Multicasting Router In Mule ?

The multicasting router can send messages to multiple endpoints over different transports. The multicasting router allows you to easily move the same messages across these different endpoints.

78. What Is Mule Transformer ?

Mule transformers are used to convert the data received into transformed output

- Payload Type Transformation
Data type of the message payload is transformed from one binary form to another eg Object to String transformation
- Payload Format Transformation
Data format of message payload is transformed to another format. JSON TO XML Transformer and XML to JSON transformer
- Properties Transformation
Properties of the message are modified in this transformation

- Transformers often come in pairs. The transformer that is able to reverse the changes of another one is called a round-trip transformer.

79. What Is Mule Context ?

The Mule context is composed of references to different objects, including security credentials, the session in which the request is processed. All internals of the ESB are accessible through Mule context.

80. What is object-store

An **object store** is a facility for storing objects in or across Mule applications. Mule uses object stores to persist data for eventual retrieval. Internally, Mule uses object stores in various filters, routers, and other message processors that need to store state between messages. In most cases, Mule creates and manages object stores automatically.

81. Why we use VM transport?

You can use the Java Virtual Machine (VM) transport for intra-JVM communication between Mule [flows](#). This transport by default uses in-memory queues but can optionally be configured to use persistent queues (persistent queues are used to help prevent data loss). **Note:** VM file persistency does not work on clusters.

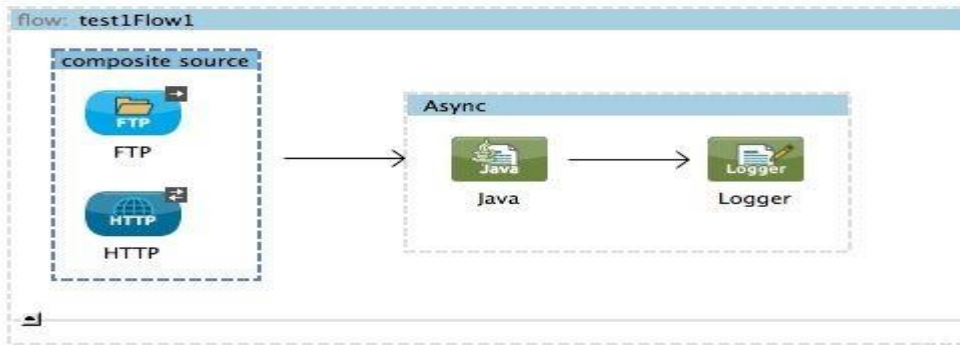
The VM transport has often been used to implement complex integrations made up of multiple applications.

VM is still useful in certain situations. Suppose, for instance, that most of the parts of your solution are local, but some need to be decoupled for testing, or because some need to be made remote.

You can use Flow References to directly reference one flow from another without a transport in the middle. use VM transports if you need redelivery of messages, this won't be possible in private flow and flow ref because for rollback exception strategy even if we define that no internal queue exists.

82. What is composite source

In the case of **Composite**, the embedded building blocks are message sources (i.e. inbound connectors) that listen in parallel on different channels for incoming messages. Whenever any of these receivers accepts a message, the **Composite** scope passes it to the first message processor in the flow, thus triggering that flow.



83. what is Munit

MUnit is a Mule application testing framework that allows us to easily build automated tests for our integrations and APIs. It provides a full suite of integration and unit test capabilities, can be used with Maven (software project management tool)

With MUnit you can: examples

- Create your Mule test by writing Mule code
- Create your Mule test by writing Java code
- Disable flow inbound endpoints
- Disable endpoint connectors

84. Custom Transformers – what class we inherit

The Mule message is the data that passes through an application via one or more flows. It consists of two main parts:

- The message header, which contains metadata about the message contains inbound and outbound properties Inbound properties are immutable and outbound properties are mutable
- The message payload, which contains your business-specific data.

AbstractTransformer – override doTransform() method

It is used when only payload needs to be access or used within Transformer

```

1 import org.mule.transformer.AbstractTransformer;
2
3 public class OrderToHtmlTransformer extends AbstractTransformer
4 {
5     public Object doTransform(Object src, String encoding) throws TransformerException
6 }

```

AbstractMessageTransformer – override transform() method

It is used when whole Mule Message needs to be access or used within Transformer

```

1 import org.mule.transformer.AbstractMessageTransformer;
2
3 public class OrderToHtmlTransformer extends AbstractMessageTransformer
4 {
5     public Object transform(MuleMessage message, String encoding) throws TransformerException
6 }

```

85. Git commands

To add your other files please follow the below commands.

git status – All the non-committed code will appear in RED.

git add *file-to-be-added* (Follow this step till your all files are added.)

git status – All added files will appear in green color

git commit

- i. Proper comment
- ii. Then press ESC :wq
- iii. Enter

git push

86. Difference between call out and service invoke?

Service Invoke: The Service Invoke is used to make a service request in either a request or response flow. The service may be Request/Response or One-Way. Multiple instances of the Service Invoke (primitive) are permitted in a flow, allowing a series of service invocations to be performed.

Callout: The Callout receives the message and calls the requested service and operation. There is a Callout node for each connected target operation in the flow.

- If the call is successful, the Callout Response node in the response flow receives the response message.
- If the call is unsuccessful, the Callout can be set to retry service invocations depending on the type of fault received.

87. How can you implement loop in flow?

Fan-in and Fan-out is used to implement the loop

88. What is the functionality of Fan-in and Fan-out?

This is used to Implement a loop in mediation

Fan-out: We can use the Fan Out to fire the output terminal once (with the input message) or to fire the output terminal *multiple* times. You can use Fan Out in isolation or as part of a Fan Out and Fan In combination.

Fan-In: Fan In is always partnered with a Fan Out in the same flow and acts as a decision point for when to continue flow execution. It receives a number of messages until a decision point is reached, at which point the last message to be received is propagated to the output terminal. The Fan In primitive can only be used in combination with Fan Out.

89. What is Shared Context?

Shared Context: Context is a temporary area which is created along with Service Message Object (SMO) in the Flows. Shared Context is a type of context. It is mainly used when we are using Aggregation process where we need to Iterate for Certain times. Shared Context maintains Aggregation data between Aggregation (FanOut and FanIn) primitives. The Content (data) which is present in the shared context BO does not persist across Request and Response flows i.e. The Data in the Shared Context which is used in Request flow can not be used again in Response flow.

90. What is Transient Context?

Transient Context: Used for passing values between connectors within the current flow —

either the requestflow or the responses flow. The transient context cannot link requests and responses and hence cannot be used across.

Used when you want to save an input message before a service invokes call (within a request or response flow). After the services invoke call, the next primitive can create another message by combining the service invoke response and the original message stored in the transient context.

91. What is the Difference between Stop and fail?

Stop: Stops a particular path in the flow, without generating an exception.

Fail: Generates an exception in the flow.

92. What is mule MMC? Why it is used?

MMC **Mule Management Console** is an management and monitoring tool designed specifically for Mule ESB instances. MMC provides a set of functionality for managing and monitoring running Mule instances, Mule clusters, applications within Mule instances, and the flows within those applications. **We can look into specific transactions also with the help of it.**

Cognizant interview questions on Mulesoft

- 1) What is Singleton pattern?
- 2) How to declare function in dataweave?
- 3) What is serialization and deserialization in Java?
- 4) What is a middleware?
- 5) What is API-led Connectivity in mulesoft?
- 6) Enlist different types of flows in mulesoft?
- 7) What are different Messaging Exception strategies available in mulesoft?
- 8) How to throw exception manually in mulesoft?
- 9) What is transport service descriptor in Mule?
- 10) What is Mule UMO?
- 11) What do you mean by Transient Context?